ENT ATLAS

CLINICAL

Neck, thyroid & salivary

12 articles · imaging key facts · November 2026

For personal study. Follow local protocols and senior advice for patients.

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· Imaging for this region

CLINICAL · NECK, THYROID & SALIVARY

Congenital neck masses

Most neck lumps in children are reactive nodes. The congenital masses that remain are explained by embryology. A midline cyst that rises when the tongue is put out is a thyroglossal cyst. A smooth lump in front of the upper third of sternomastoid is a branchial cyst, and a pit along its lower anterior border is a branchial sinus. A soft, transilluminant, multilocular swelling in a baby's posterior triangle is a lymphatic malformation. Each is cured by removing the whole tract, and each recurs if the tract is left behind.

AT A GLANCE

- Midline moving with the tongue: thyroglossal.
- Lateral: branchial.
- Treat infection first, then excise.
- Sistrunk removes the central hyoid.

1 CHAPTER ONE Embryology

Arches, clefts and pouches.

1.1 The branchial apparatus

Between the fourth and eighth weeks, five (some count six) *branchial arches* form on each side of the primitive pharynx. They are separated on the outside by **clefts** lined with ectoderm and on the inside by **pouches** lined with endoderm, and each arch contains a cartilage bar, an artery, a nerve and muscle-forming mesoderm. The second arch then overgrows the third and fourth to form the smooth contour of the neck, temporarily enclosing a space, the *cervical sinus of His*.

When a cleft, a pouch or the cervical sinus fails to obliterate, it leaves a remnant: a *cyst* with no opening, a *sinus* with one opening, inside or out, or a *fistula* with both. The skin openings always lie *anterior to the sternomastoid*, and the tract runs deep to platysma.

ARCHES, THEIR NERVES AND DERIVATIVES

ARCH	NERVE	SKELETAL AND MUSCLE DERIVATIVES	ARTERY
First (mandibular)	Trigeminal (V3)	Malleus head and neck, incus body and short process, mandible, sphenomandibular ligament; muscles of mastication, mylohyoid, anterior digastric, tensor tympani, tensor palati	Degenerates (part of maxillary)
Second (hyoid)	Facial (VII)	Manubrium of malleus, long process of incus, stapes superstructure, styloid process, stylohyoid ligament, lesser cornu and upper body of hyoid; facial muscles, stapedius, stylohyoid, posterior digastric	Stapedial (degenerates)
Third	Glossopharyngeal (IX)	Greater cornu and lower body of hyoid; stylopharyngeus	Common and internal carotid
Fourth	Superior laryngeal (X)	Thyroid and cuneiform cartilages; cricothyroid, pharyngeal constrictors, cricopharyngeus	Aortic arch (left), proximal subclavian (right)
Sixth	Recurrent laryngeal (X)	Cricoid, arytenoid and corniculate cartilages; intrinsic laryngeal muscles	Pulmonary arteries, ductus arteriosus

POUCH DERIVATIVES

POUCH	DERIVATIVES
First	Eustachian tube, middle ear, mastoid air cells
Second	Palatine tonsil and tonsillar fossa
Third	Inferior parathyroids (dorsal wing), thymus (ventral wing)
Fourth	Superior parathyroids; ultimobranchial body with the C cells

WHY THE INFERIOR PARATHYROID IS "3" AND LIES BELOW "4"

The third pouch gives rise to both the inferior parathyroid and the thymus. As the thymus descends into the chest, it drags the third-pouch parathyroid below the fourth-pouch one. That long journey explains why inferior glands are so often ectopic.

1.2 The thyroid and tongue

The thyroid begins at the *foramen caecum* at about four weeks and descends through the tongue base, passing in front of, through or behind the developing hyoid. The *thyroglossal duct* that connects it to the tongue normally disappears by the sixth to eighth week. The anterior two-thirds of the tongue come from the first-arch lingual swellings and the tuberculum impar, the posterior third from the copula of the second to fourth arches, and the muscles from the occipital somites, which is why they are supplied by the hypoglossal nerve.

2 CHAPTER TWO Midline masses

Thyroglossal and dermoid.

2.1 Thyroglossal duct cyst

The thyroglossal cyst is the commonest congenital neck mass. It usually presents in childhood but can appear at any age, as a smooth, round, cystic *midline* swelling of 2 to 4 cm, usually *at or just below the hyoid* and sometimes slightly to the left. Because it is tethered to the foramen caecum, it *rises on swallowing and on protruding the tongue*. It may become infected after an upper respiratory infection, and if it ruptures or is incised it leaves a thyroglossal sinus or "fistula". It can lie anywhere along the tract: lingual, suprahyoid, at the hyoid, subhyoid (the commonest) or suprasternal.

Two rarities matter. The cyst may contain the *only functioning thyroid tissue*, and about 1% contain carcinoma, almost always papillary. Ultrasound therefore confirms not only the cyst but also a *normal thyroid gland*, with thyroid function tests; scintigraphy is done if the gland is not seen, and solid areas in adults are sampled by FNA.

The treatment is the **Sistrunk operation**: excision of the cyst and tract *together with the central body of the hyoid bone* and a core of tongue-base muscle along the tract up to the foramen caecum. Simple cystectomy recurs in up to half of cases; the Sistrunk operation brings recurrence below 5 to 10%. Infection is first settled with aspiration or antibiotics, and the operation done afterwards.

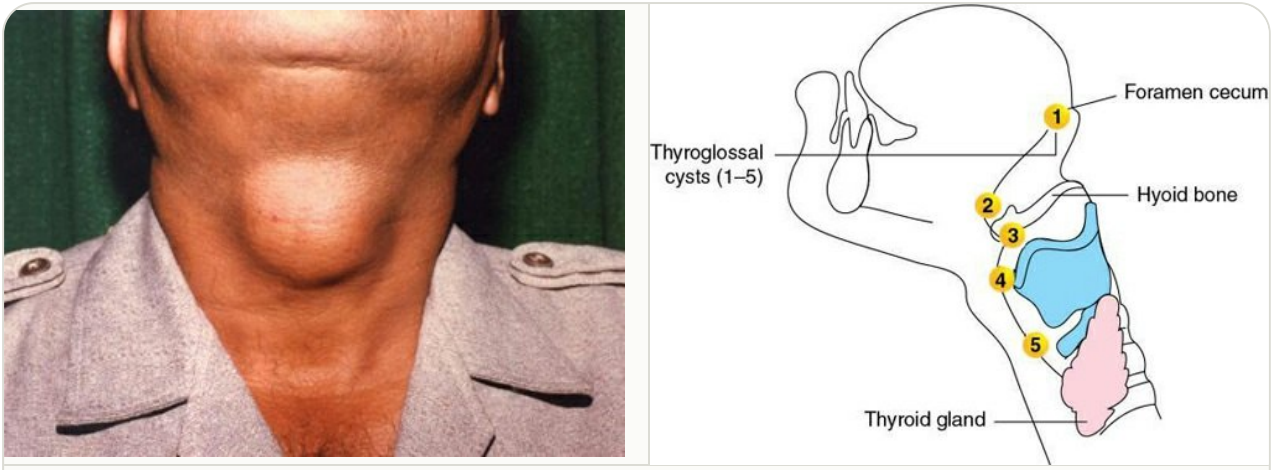
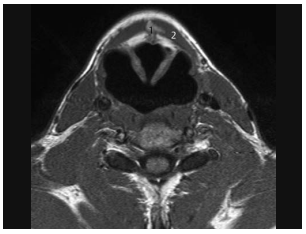


FIGURE 1 First, a thyroglossal cyst in the midline of the neck. Second, sites of thyroglossal cysts along the tract (subhyoid is the commonest).
Dhingra, figs 77.3 and 77.4

IMAGING IN THE APP 1 case

Thyroglossal duct cyst on imaging.



Thyroglossal Duct Cyst
Congenital · 11 images

OPERATIVE GUIDES IN THEATRE 1 operation

The Sistrunk operation.

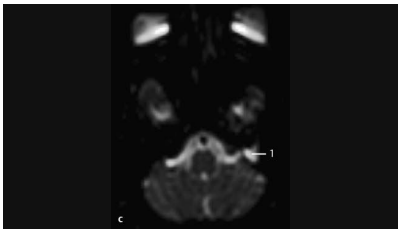
Thyroglossal Duct Cyst
Sistrunk

2.2 Dermoid cyst

A dermoid cyst is lined by skin with its adnexa, formed from ectoderm trapped at a line of fusion. It lies in the submental region or the floor of mouth (the *sublingual dermoid*), or at the suprasternal notch, and it does not move with tongue protrusion. A sublingual dermoid above the mylohyoid can push the tongue up and must be told apart from a ranula. It is excised through the mouth or the neck according to its relation to the mylohyoid.

IMAGING IN THE APP 1 case

Epidermoid and dermoid cysts.



(Epi)dermoid Cyst

Infection & inflammation · 10 images

2.3 Other midline masses

A **teratoma** in a neonate can be huge and obstruct the airway, and when it is found antenatally an EXIT (ex-utero intrapartum treatment) delivery is planned. Ectopic thyroid, a Delphian node, a bronchogenic cyst at the suprasternal notch, and a thymic cyst, usually lateral and deep to the lower third of the sternomastoid, complete the differential.

3 CHAPTER THREE Branchial anomalies

Lateral cysts, sinuses and fistulae.

3.1 Second cleft anomalies

About 90% of branchial anomalies arise from the second cleft. The **cyst** usually appears in older children and young adults as a smooth, fluctuant, non-tender swelling at the junction of the upper and middle thirds of the anterior border of the sternomastoid. It often enlarges with upper respiratory infections and may become infected, and its fluid contains cholesterol crystals.

The **sinus or fistula** is seen from infancy as a pit on the *lower third of the anterior border of the sternomastoid* that discharges mucus. Its tract runs up along the carotid sheath deep to platysma, passes *deep to the second-arch derivatives* (the external carotid, stylohyoid and posterior digastric) and *superficial to the third-arch derivatives* (the internal carotid and glossopharyngeal nerve), so that it runs between the two carotids, crosses over the hypoglossal nerve, and ends in the *tonsillar fossa*.

The anomaly is excised completely once infection has settled. A fistula is removed through an elliptical incision around the opening, with a second, higher *step-ladder* incision to follow the tract. Recurrence follows incomplete excision or surgery during infection.



FIGURE 2 A left branchial sinus discharging pus.
Dhingra, fig. 77.5

IMAGING IN THE APP 1 case

Branchial cleft cyst on imaging.



Branchial Cleft Cyst

Congenital · 12 images

THE ADULT "BRANCHIAL CYST"

A new cystic lateral neck mass in an adult over about 40 is far more often a *cystic metastasis* from HPV-related oropharyngeal carcinoma or papillary thyroid carcinoma than a branchial cyst. Examine the tonsils and tongue base, aspirate under ultrasound, test p16 and thyroglobulin, and image before excision.

3.2 First cleft anomalies

First cleft anomalies are duplications of the external ear canal. **Work type I** is ectodermal, a duplicate canal running parallel to the external canal; **Work type II** is ectodermal and mesodermal and contains cartilage. They present as a cyst, sinus or fistula near the angle of the mandible or within the parotid, draining into or near the ear canal, often as recurrent abscesses in the parotid region or as otorrhoea. Because the tract runs close to, and sometimes deep to, the facial nerve, excision needs a parotidectomy approach with the nerve identified first.

3.3 Third and fourth pouch anomalies

Third and fourth pouch anomalies open into the *pyriform sinus*, almost always on the left. They present as recurrent *left-sided neck abscesses*, as *acute suppurative thyroiditis* in children, or as cysts low in the lateral neck. A third-arch tract runs deep to the internal carotid and glossopharyngeal nerve and superficial to the hypoglossal nerve, piercing the thyrohyoid membrane above the internal laryngeal nerve. A

fourth-arch tract passes below the superior laryngeal nerve and loops round the subclavian artery on the right or the aortic arch on the left.

The diagnosis is made with a barium swallow, CT and direct laryngoscopy to see the pyriform opening. The sinus opening can be closed endoscopically by chemical or electrocautery, or the tract excised openly with a partial thyroidectomy.

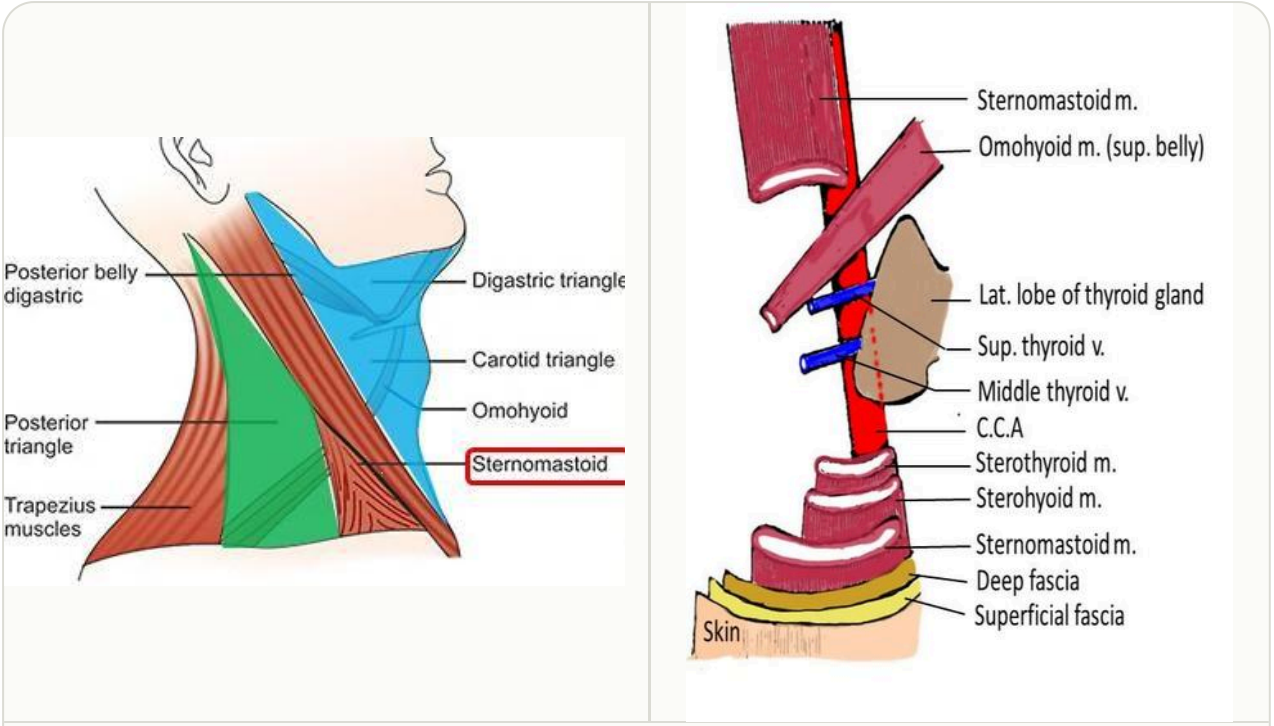


FIGURE 3 The triangles of the neck, and the common carotid artery and its relations, which guide the course of branchial tracts.

Anatomy atlas · ENT Steps p. 270, Doss p. 136

OPERATIVE GUIDES IN THEATRE 1 operation

Branchial cleft cysts and sinuses.

Branchial Cleft Cysts and Sinuses

Branchial surgery

BRANCHIAL ANOMALIES COMPARED

TYPE	EXTERNAL OPENING	INTERNAL OPENING	KEY RELATIONS
First cleft	Angle of mandible, near the ear canal	External auditory canal	Close to the facial nerve
Second cleft (commonest)	Lower third, anterior border of sternomastoid	Tonsillar fossa	Between the external and internal carotids, over XII

TYPE	EXTERNAL OPENING	INTERNAL OPENING	KEY RELATIONS
Third pouch	Lower anterior border of sternomastoid	Upper pyriform sinus	Behind the internal carotid, over XII, above the internal laryngeal nerve
Fourth pouch	Low in the neck (rare)	Apex of the pyriform sinus	Below the superior laryngeal nerve, around the subclavian or aorta

4 CHAPTER FOUR Lymphatic and vascular

Malformations and haemangiomas.

4.1 Lymphatic malformation (cystic hygroma)

A lymphatic malformation arises from sequestration of the primitive jugular lymph sacs. About half are present at birth and 90% appear by two years. They occur most often in the *posterior triangle* and extend into the floor of mouth and tongue, the parotid, the axilla and the mediastinum.

Macrocystic lesions, the classic cystic hygroma, are soft, multilocular, compressible and *brilliantly transilluminant*. **Microcystic** lesions infiltrate the tongue and face, and the overlying mucosa shows clear or blood-stained *vesicles*. Both swell suddenly with infection or bleeding into the cysts, and large ones can obstruct labour, the airway or feeding. MRI shows the extent, and when antenatal ultrasound detects a large lesion an EXIT delivery is planned.

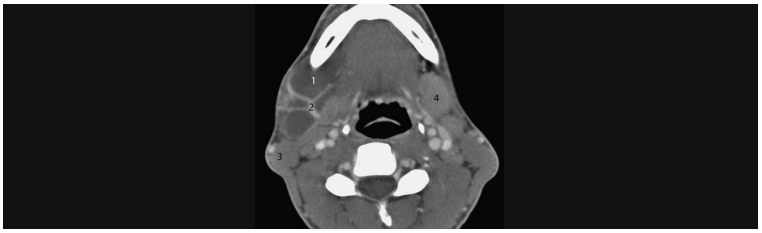
Macrocystic disease responds to *sclerotherapy* with bleomycin, doxycycline, OK-432 or ethanol. *Surgical excision* preserves nerves and vessels; recurrence is low after complete excision and about half after partial excision, and several stages may be needed. Extensive microcystic disease is treated with *sirolimus*, mucosal vesicles with laser, and airway obstruction with a tracheostomy.



FIGURE 4 A cystic hygroma in an infant.
Dhingra, fig. 77.7

IMAGING IN THE APP 1 case

Cystic hygroma (lymphatic malformation).



Lymphatic Malformation (Cystic Hygroma, Lymphangioma)
Congenital · 14 images

4.2 Infantile haemangioma and other vascular anomalies

The infantile haemangioma, the commonest tumour of infancy, appears after birth, proliferates for about a year and then slowly involutes. A lesion in a *beard distribution* means a subglottic haemangioma until proved otherwise, and a large segmental facial lesion suggests *PHACES syndrome*. Propranolol is the treatment. Venous

malformations, capillary malformations and AVMs, which grow with the child, are covered with the vascular anomalies.

5 CHAPTER FIVE Other paediatric masses

Nodes, glands, muscle.

5.1 The sternomastoid tumour and the thymus

Fibromatosis colli, the sternomastoid tumour, is a firm spindle-shaped mass within the sternomastoid of a baby two to four weeks old, often after a difficult delivery. It causes *torticollis*, with the head tilted towards the lesion and the chin turned away, and ultrasound confirms it. Stretching physiotherapy resolves most cases, and persistent cases need tenotomy to prevent plagiocephaly and facial asymmetry. A **thymic cyst** is a cervical remnant lying along the line of thymic descent from the angle of the jaw to the mediastinum, usually on the left, and it may extend into the chest.

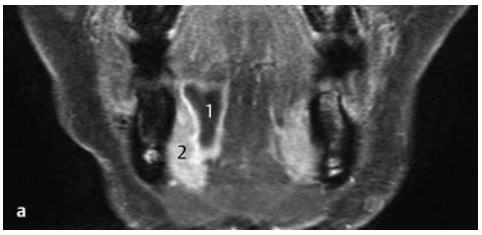
5.2 Nodes and malignancy

Reactive and infective nodes are the commonest childhood neck masses: suppurative adenitis, atypical mycobacteria and cat-scratch disease. **Kawasaki disease**, with more than five days of fever, conjunctivitis, mucosal changes, a rash and a large node, must not be missed, because coronary aneurysms follow when it goes untreated.

Malignancy is uncommon but important: lymphoma, especially Hodgkin lymphoma in adolescents, rhabdomyosarcoma, neuroblastoma, thyroid carcinoma, and nasopharyngeal carcinoma in endemic areas. A hard, fixed or matted node over 2 to 3 cm, a supraclavicular node, growth over two to six weeks despite antibiotics, and B symptoms are the warnings that call for biopsy. Salivary masses in children include parotid haemangioma, ranula and juvenile recurrent parotitis.

IMAGING IN THE APP 1 case

Ranula of the sublingual gland.



Ranula of the Sublingual Gland

Infection & inflammation · 10 images

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Embryology:** the arches with their nerves and derivatives; the pouches and what they form.
- **Thyroglossal cyst:** clinical signs; check for a normal thyroid; the Sistrunk operation and why the hyoid is removed.
- **Second branchial fistula:** its course and the step-ladder excision.
- **Third and fourth pouch sinuses:** recurrent left neck abscess and thyroiditis.
- **Cystic hygroma:** features and management.
- **The adult "branchial cyst":** a metastasis until proved otherwise.

6.2 Common pitfalls

COMMON PITFALLS

- Simple excision of a thyroglossal cyst without the hyoid and tract.
- Removing a thyroglossal "cyst" that is the patient's only thyroid tissue.
- Incising and draining a branchial cyst or atypical mycobacterial node.

- Missing a first cleft anomaly as a recurrent "parotid abscess" or discharging ear.

Written for this app from Oxford Textbook of Otolaryngology ch. 81, Oxford Specialist Handbook ch. 23, Oxford Handbook of ENT ch. 17, Dhingra ch. 77 and K. J. Lee ch. 40, 51. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · NECK, THYROID & SALIVARY

Thyroid nodules, cancer and goitre

Thyroid nodules are everywhere: one in twenty adults can feel one, and ultrasound finds them in half of older people. The task is to find the small fraction that are cancers without operating on everyone. The tools are history, ultrasound pattern, and ultrasound-guided FNA reported in Bethesda categories. Most thyroid cancers are papillary and have an excellent outlook. The surgeon's measure is two working recurrent laryngeal nerves and four living parathyroids at the end of the operation.

AT A GLANCE

- Ultrasound and FNA guide nodule care.
- Bethesda sets the next step.
- Check TSH and the vocal folds.
- Watch nerve, calcium and bleeding after surgery.
- U1–U5 and Thy1–5 grading; NIFTP is benign; low-dose iodine (HiLo) for low-risk disease.

1 CHAPTER ONE Applied anatomy

The gland and the nerves around it.

1.1 Embryology

The medial anlage starts at the *foramen caecum* of the tongue at about four weeks. It descends in front of the hyoid and larynx along the *thyroglossal tract* and reaches the root of the neck by seven weeks. Arrested descent leaves a *lingual thyroid*, often the only thyroid tissue the patient has; the lowest part of the tract persists as the *pyramidal lobe* in up to 40%; and remnants of the tract form *thyroglossal cysts*.

The lateral anlagen of the fourth and fifth pouches and the *ultimobranchial bodies* join the descending gland. The ultimobranchial bodies bring the neural-crest *C cells*, which concentrate in the upper and middle thirds of the lobes, and that is where medullary cancers arise.

1.2 The gland

The gland has two lobes joined by an isthmus over the second to fourth tracheal rings. It weighs 15 to 25 g, and each lobe measures about 5 × 3 × 2 cm. Its *true capsule* sends septa into the gland, while its *false capsule*, the perithyroidal sheath, is formed by the visceral (pretracheal) layer of the deep cervical fascia.

Two suspensory ligaments hold it. The anterior ligament runs from the isthmus to the trachea, and the posterior ligament, the **ligament of Berry**, binds the back of each lobe to the cricoid and first tracheal rings, which is why the gland rises on swallowing. The recurrent laryngeal nerve runs deep to or through Berry's ligament beside small branches of the inferior thyroid artery. This is the commonest site of nerve injury, and thyroid tissue left here explains bed uptake and measurable thyroglobulin after a "total" thyroidectomy.

The **superior thyroid artery** is the first branch of the external carotid. The **inferior thyroid artery** arises from the thyrocervical trunk, passes behind the carotid sheath and supplies the parathyroids, and a *thyroidea ima* sometimes arises from the brachiocephalic trunk or the arch. The superior and middle veins drain to the internal jugular; the middle vein has no companion artery and is the first vessel divided at surgery; and the inferior veins form a plexus draining to the brachiocephalic veins. Lymph passes first to the central compartment (level VI), to the prelaryngeal (*Delphian*), pretracheal and paratracheal nodes, and then to the superior mediastinum (level VII) and the lateral neck (levels II to IV, sometimes V).

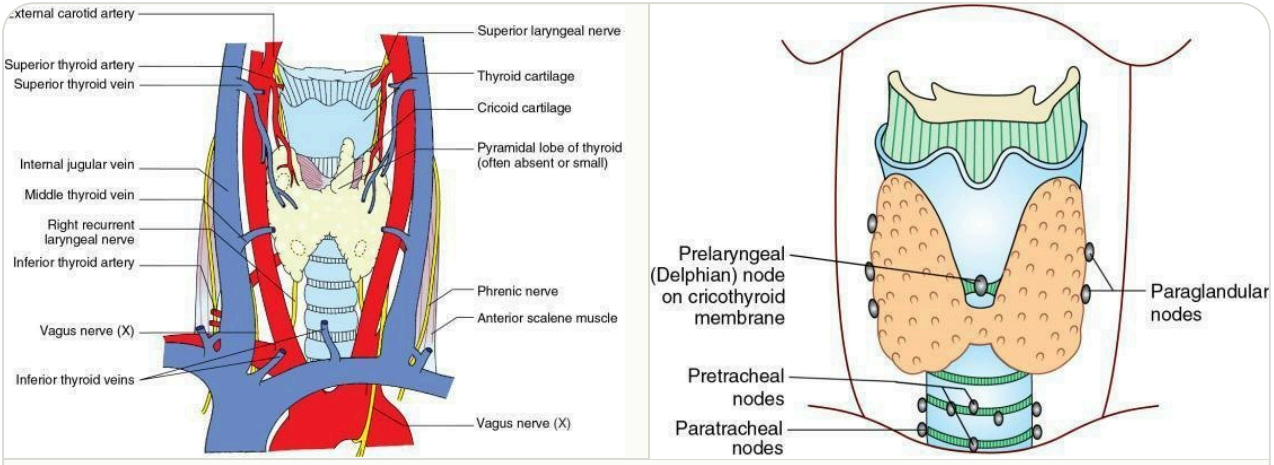


FIGURE 1 First, the thyroid and its relations. Second, lymphatic drainage of the thyroid: Delphian, pretracheal and paratracheal nodes, the jugular chain and the mediastinum.

Dhingra, figs 66.1 and 66.2

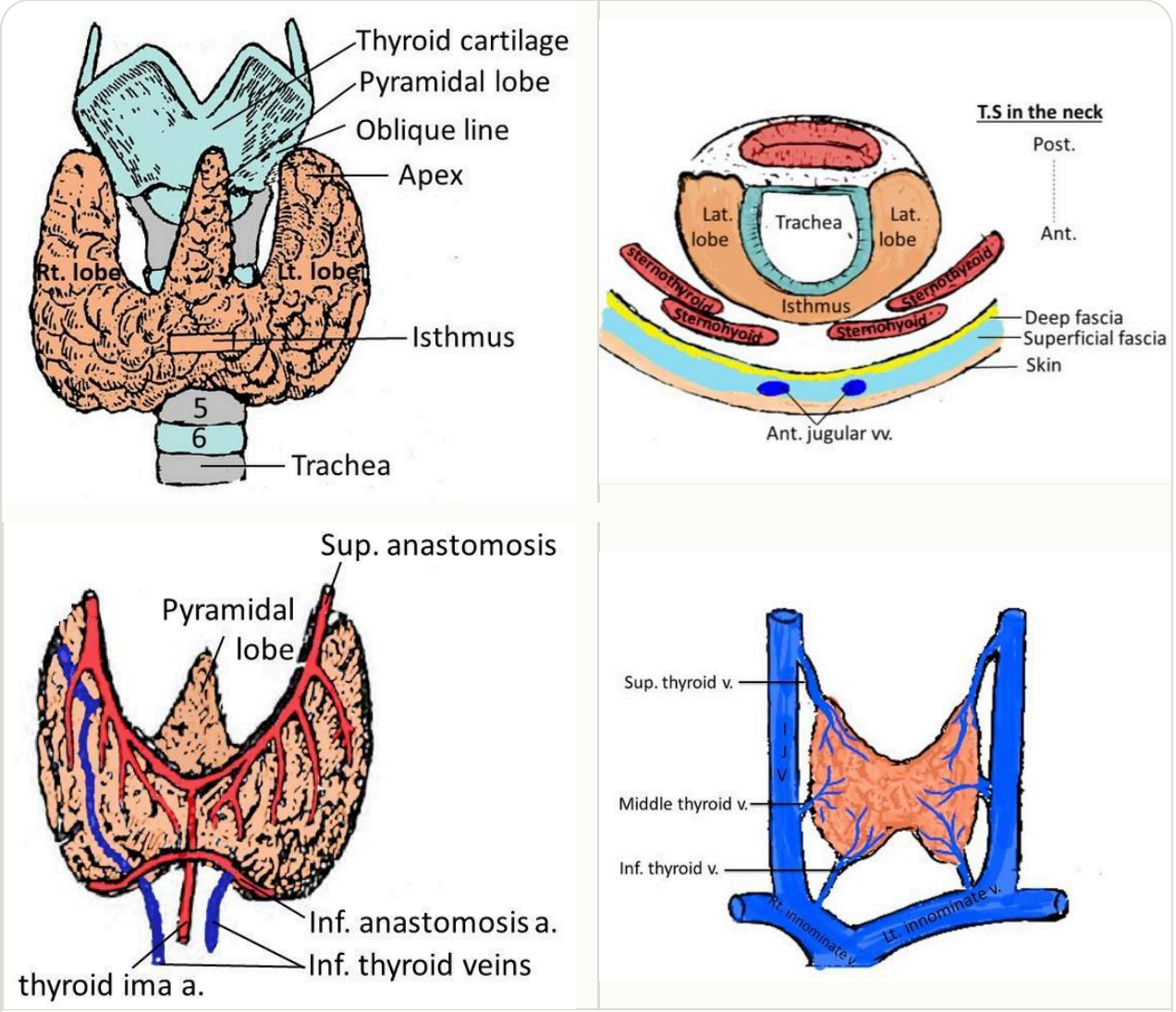


FIGURE 2 The thyroid from the front, in cross-section with its capsules, and its arteries and veins.

Anatomy atlas · Doss p. 124, Doss p. 125, Doss p. 128

1.3 Nerves at risk

The **recurrent laryngeal nerve** supplies all the intrinsic laryngeal muscles except cricothyroid and carries sensation below the folds. The *right* nerve hooks round the subclavian artery and runs obliquely from lateral to medial; the *left* hooks round the aortic arch and runs more vertically in the tracheo-oesophageal groove. Each crosses the branches of the inferior thyroid artery, superficial to them, deep to them or between them, branches before entering the larynx in about a third of people, and enters behind the inferior cornu of the thyroid cartilage. It is found in the *recurrent nerve triangle*, bounded by the trachea medially, the retracted strap muscles laterally and the lower pole above, or close to the *tubercle of Zuckerkandl*.

A **non-recurrent laryngeal nerve** occurs in about 0.5 to 1% of people on the right and almost never on the left. It runs directly from the vagus to the larynx and accompanies an aberrant retro-oesophageal right subclavian artery, the cause of *dysphagia lusoria*.

The **external branch of the superior laryngeal nerve** supplies cricothyroid, which tenses the cord for high pitch. It runs on the inferior constrictor close to the superior pole vessels and crosses them within 1 cm of the pole in about a fifth of people, so the superior vessels are ligated individually on the capsule with the gland pulled down. The **ansa cervicalis** supplies the strap muscles from below, so the straps are divided high when they must be divided.

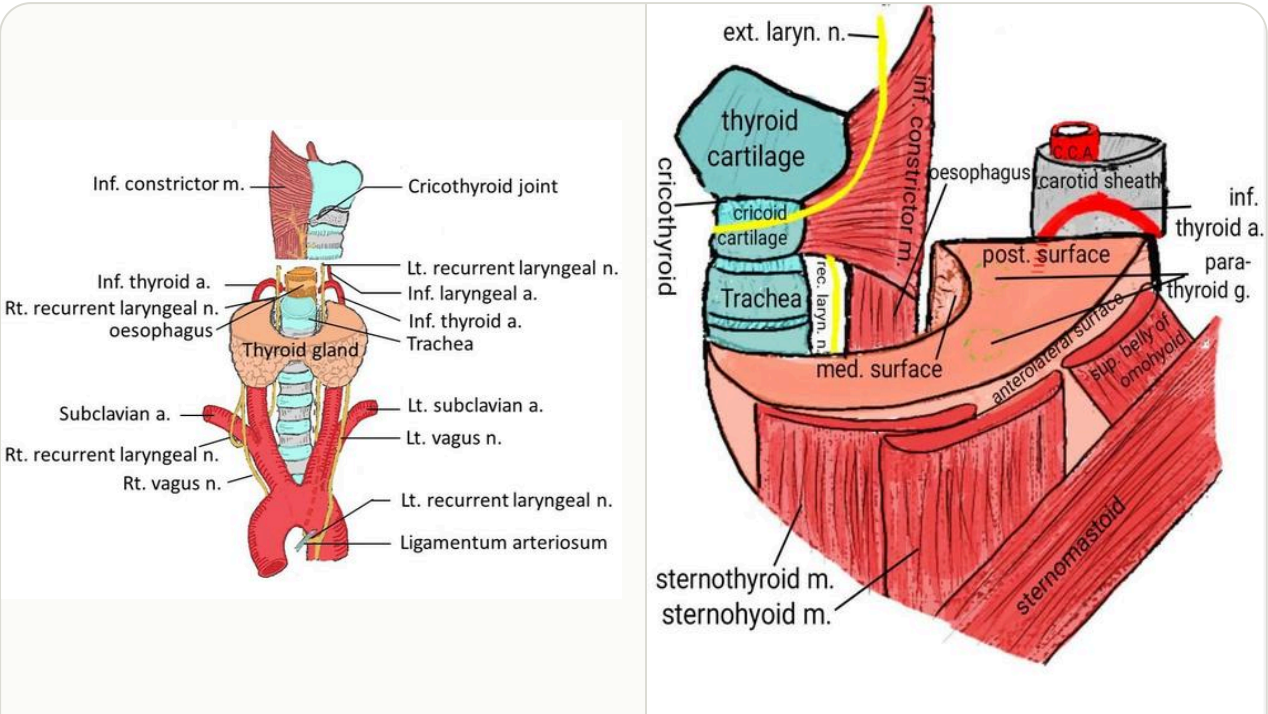


FIGURE 3 The recurrent laryngeal nerves and inferior thyroid arteries, and the relations of the lateral lobe.

Anatomy atlas · Doss p. 154, Doss p. 126

1.4 Lingual thyroid

A lingual thyroid occurs in about 1 in 3000 to 4000 patients with thyroid disease, mostly women. It is a smooth, vascular mass at the tongue base that may cause dysphagia, dysphonia, airway obstruction or bleeding, particularly at puberty or in pregnancy. Before any treatment, a scintiscan and neck ultrasound establish whether any other thyroid tissue exists. Thyroxine shrinks it and radioiodine can

ablate it; obstruction is relieved by excision through a transoral or suprahyoid transpharyngeal approach, followed by lifelong replacement if it was the only thyroid.

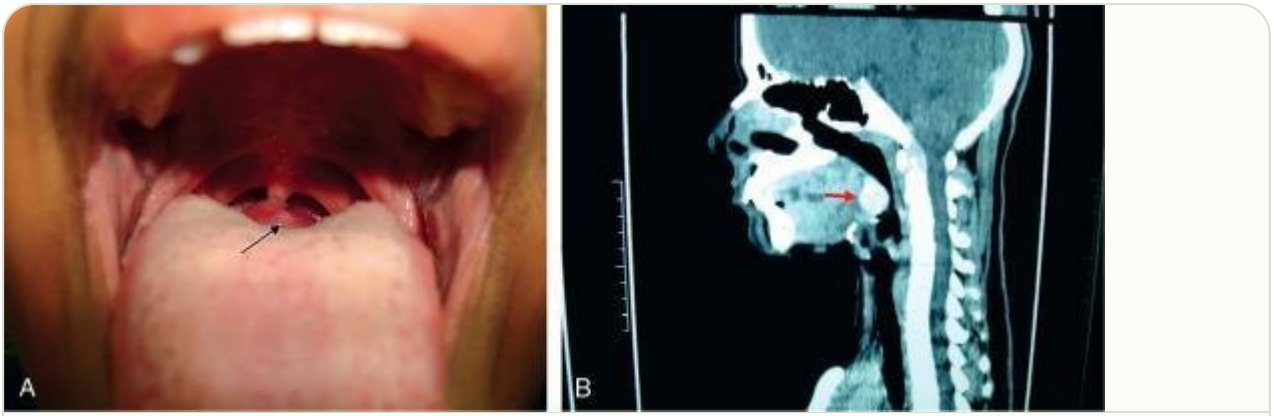


FIGURE 4 Lingual thyroid at the tongue base, with its sagittal CT.
Dhingra, fig. 66.3

2 CHAPTER TWO Physiology and function tests

Hormones and the axis.

2.1 Hormone synthesis

Follicular cells make thyroglobulin, thyroxine (T4) and tri-iodothyronine (T3), and the *parafollicular C cells* make calcitonin. Synthesis proceeds in steps:

1. Iodide is actively trapped.
2. *Thyroid peroxidase* oxidises it and attaches it to tyrosine residues on thyroglobulin, forming MIT and DIT.
3. Thyroid peroxidase couples them: DIT + DIT forms T4, and MIT + DIT forms T3.
4. The hormones are stored as colloid.
5. Colloid is taken back into the cell, thyroglobulin is cleaved, and the hormones are released.
6. Uncoupled MIT and DIT are deiodinated so that their iodine is recycled.

The gland releases mostly T4, with a half-life of about a week, and most T3, several times more potent and with a half-life of about a day, is made by peripheral deiodination. More than 99% of circulating hormone is protein-bound.

Hypothalamic TRH stimulates pituitary TSH, TSH drives hormone synthesis, growth and vascularity of the gland, and T4 and T3 feed back negatively.

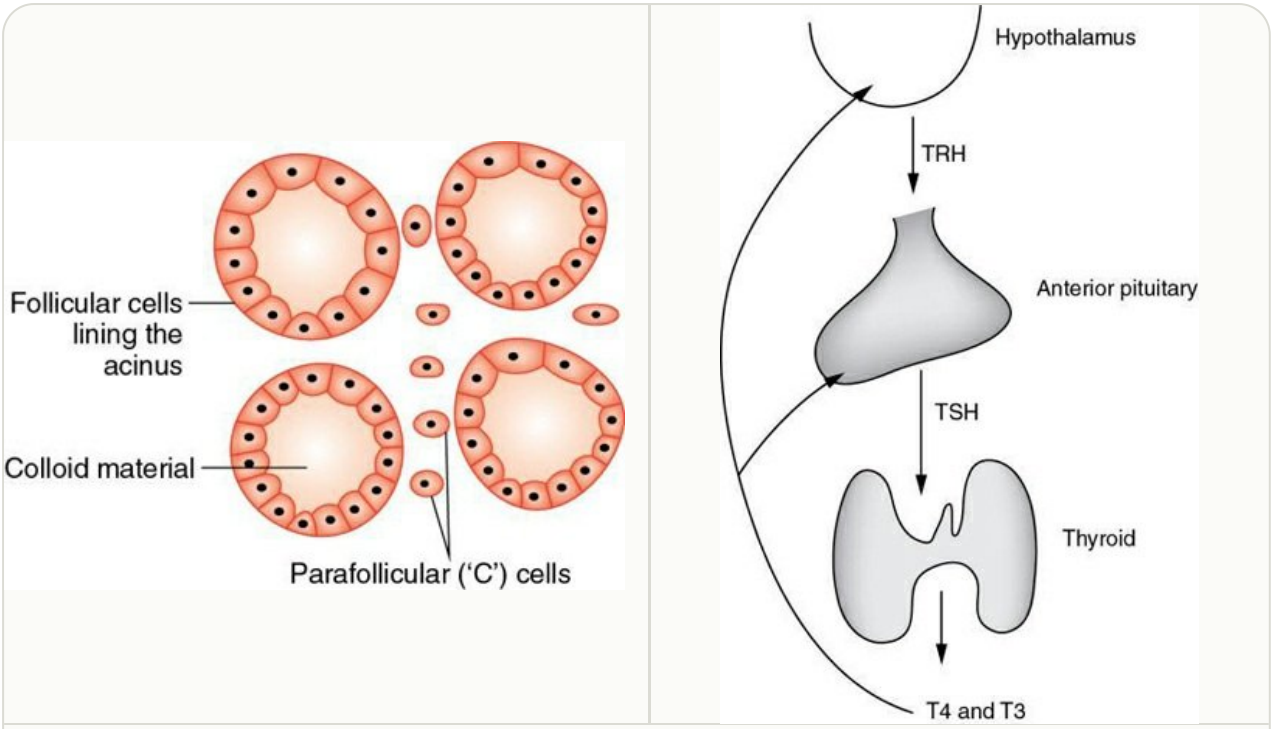


FIGURE 5 First, thyroid follicles with follicular and parafollicular C cells. Second, the hypothalamic–pituitary–thyroid axis.
Dhingra, figs 66.4 and 66.5

2.2 Tests and terms

TSH is the single best screening test: high in primary hypothyroidism and suppressed in hyperthyroidism. A high TSH with normal T4 is subclinical hypothyroidism, and a low TSH with normal T4 and T3 subclinical hyperthyroidism. **Free** T4 and T3 are measured because total levels vary with thyroid-binding globulin, which rises in pregnancy and with the pill.

TPO (antimicrosomal) antibodies are present in almost all Hashimoto disease and most Graves disease; **thyroglobulin antibodies** occur in autoimmune thyroiditis and interfere with thyroglobulin assays; and stimulating **TSH-receptor antibodies** cause Graves disease. **Thyroglobulin** is made only by thyroid tissue; it is useless for diagnosing a nodule, but after total thyroidectomy and ablation it should be undetectable, so a rising level means recurrence. **Calcitonin and CEA** are the markers of medullary carcinoma.

Ultrasound is the essential imaging study. **Scintigraphy** with technetium or I-123 shows hot and cold nodules, and **CT**, with non-iodinated contrast if radioiodine may follow, shows retrosternal extension and invasion.

3 CHAPTER THREE Benign disease

Function and inflammation.

3.1 Hypothyroidism

Iodine deficiency is the commonest cause of hypothyroidism worldwide and *Hashimoto disease* the commonest where iodine is sufficient. Thyroidectomy and radioiodine, neck irradiation (almost half of patients after laryngeal or pharyngeal radiotherapy, so TSH is checked routinely), drugs such as lithium, amiodarone and antithyroid drugs, inborn defects, and rare pituitary failure are the other causes.

The patient is tired, intolerant of cold, gains weight, has dry coarse skin and hair, constipation, menorrhagia, slow thinking, bradycardia and non-pitting *myxoedema*. In ENT terms the voice becomes hoarse and low from oedematous cords, the tongue enlarges, and sensorineural hearing loss and serous otitis occur. Congenital hypothyroidism (cretinism), with stunted growth, intellectual disability and deafness, is prevented by neonatal screening. Treatment is levothyroxine, started low in the elderly and in heart disease. Anaesthesia and sedatives can precipitate *myxoedema coma* in untreated patients.

3.2 Hyperthyroidism

Hyperthyroidism causes weight loss despite a good appetite, heat intolerance, sweating, tremor, palpitations and atrial fibrillation, diarrhoea, anxiety, proximal myopathy and oligomenorrhoea. Graves disease is the commonest cause and toxic multinodular goitre the commonest in the elderly; toxic adenoma, transient thyroiditis, iodine excess (the *Jod–Basedow phenomenon* after contrast in a nodular goitre) and exogenous hormone follow, with rare TSH-secreting pituitary tumours, struma ovarii and functioning cancers.

CAUSES OF HYPERTHYROIDISM AND THEIR SCANS

CAUSE	KEY FEATURES	ISOTOPE SCAN
Graves disease	Diffuse goitre, bruit, eye disease, pretibial myxoedema, TSH-receptor antibodies	Diffuse high uptake
Toxic multinodular goitre	Elderly, nodular goitre, no eye signs	Patchy hot areas, suppressed rest
Toxic adenoma	Single nodule, usually over 2.5 cm, often T3 toxicosis	One hot nodule, rest suppressed
Subacute (de Quervain) thyroiditis	Painful tender gland after a viral illness, raised ESR	Very low uptake
Silent and postpartum thyroiditis	Painless, transient	Very low uptake

3.3 Graves disease

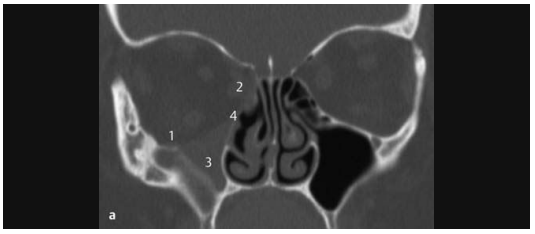
Graves disease is an autoimmune disease in which stimulating TSH-receptor antibodies drive the gland. Women are affected five to ten times more often than men, usually in the third and fourth decades. It produces a diffuse vascular goitre with a bruit, thyrotoxicosis, *ophthalmopathy* with lid retraction, proptosis, diplopia and optic neuropathy, and dermopathy as pretibial myxoedema.

Antithyroid drugs are first line: carbimazole or methimazole, or propylthiouracil in the first trimester despite its risk of liver failure. All carry a risk of agranulocytosis, so the patient is told to report a sore throat, and relapse after stopping is common. **Beta-blockers** control symptoms. **Radioiodine** is definitive but usually leads to hypothyroidism; it is avoided in pregnancy and breastfeeding, conception is deferred for at least six months, and it can worsen eye disease, especially in smokers. **Total thyroidectomy** is chosen for drug failure or side effects, a large goitre or nodules, active eye disease, pregnancy planning, or the patient's preference.

Before surgery the patient must be made euthyroid on drugs to avoid *thyroid storm*, and Lugol's iodine for about ten days reduces vascularity through the Wolff–Chaikoff effect. Severe orbitopathy is treated by endoscopic transnasal orbital decompression.

IMAGING IN THE APP 1 case

Endoscopic orbital decompression in Graves disease.



Orbital Decompression in Graves Disease

After surgery · 2 images

3.4 Thyroiditis

Hashimoto (chronic lymphocytic) thyroiditis is the commonest thyroid disease where iodine is sufficient, mostly in women. It produces a painless, firm, rubbery, symmetrical goitre with raised TPO antibodies, and the patient becomes hypothyroid over time; histology shows lymphoid follicles and Hürthle cell change. It is treated with thyroxine, but a discrete nodule still needs FNA, and a rapidly growing mass raises the possibility of *lymphoma*.

Subacute granulomatous (de Quervain) thyroiditis is the commonest painful thyroid. It follows a viral infection, with neck pain radiating to the jaw and ear, fever, a tender gland and a high ESR, and passes through a transient toxic phase with very low isotope uptake, a hypothyroid phase and recovery. It is treated with NSAIDs, steroids and beta-blockers. **Silent and postpartum thyroiditis** are painless and transient but sometimes leave permanent hypothyroidism.

Acute suppurative thyroiditis is rare: a bacterial infection with a tender red swelling and sometimes an abscess. In children it is usually *left-sided* and comes from a *pyriform sinus fistula*, a third or fourth branchial remnant; after antibiotics and drainage, the fistula is sought with barium swallow, CT or endoscopy and excised or its opening cauterised. **Riedel thyroiditis** is very rare: fibrosis spreading out of the gland, part of the IgG4-related spectrum, making it woody, fixed and non-tender, compressing the trachea and paralysing the recurrent nerve, and mimicking

anaplastic carcinoma or lymphoma. It is biopsied, often by isthmusectomy, which also relieves tracheal compression, and treated with steroids or tamoxifen.

3.5 Goitre

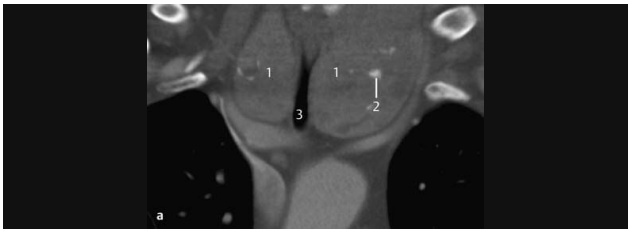
A goitre is an enlarged gland, diffuse or nodular. It results from endemic iodine deficiency, goitrogens, inherited defects of hormone synthesis (*Pendred syndrome*, with sensorineural deafness and an enlarged vestibular aqueduct), puberty and pregnancy, autoimmunity and nodular hyperplasia. Endemic goitre is prevented by iodised salt.

The **multinodular goitre** evolves slowly and may become toxic. A sudden painful enlargement means haemorrhage into a nodule, and about 5 to 10% harbour a carcinoma, usually a small papillary one. Pressure causes cough, choking, positional or nocturnal breathlessness, stridor and dysphagia, and superior vena caval obstruction gives a *positive Pemberton sign*: raising the arms brings on facial plethora, distended neck veins and breathlessness.

Assessment is by TSH, ultrasound with FNA of suspicious or dominant nodules, laryngoscopy, and CT for retrosternal extension and tracheal compression, with flow–volume loops if needed. Surgery is indicated for compression, substernal extension that can no longer be examined or sampled, suspicion of cancer, toxicity and cosmesis. A retrosternal goitre is almost always removed through the neck; sternotomy is rarely needed, mainly for a primary mediastinal goitre or one extending below the aortic arch.

IMAGING IN THE APP 1 case

Goitre on imaging, and the normal thyroid.



Hypertrophy of the Thyroid Gland, Thyroid Goiter

Infection & inflammation · 11 images

OPERATIVE GUIDES IN THEATRE 1 operation

Massive and substernal goitre.

Massive and Substernal Goiter
Substernal goitre

4 CHAPTER FOUR The thyroid nodule

Finding the cancer.

4.1 Causes

Most nodules are benign: a colloid (adenomatous) nodule in a nodular goitre, a follicular or Hürthle cell adenoma, a cyst, focal thyroiditis or a regenerative nodule, or a toxic adenoma. Carcinoma accounts for about 5 to 15%, and metastasis and lymphoma are rare. Some masses only look thyroid: a node, a parathyroid cyst, a thyroglossal cyst or a laryngocele.

CLINICAL FEATURES RAISING CONCERN FOR CANCER

LOWER CONCERN	HIGHER CONCERN
Long-standing, stable nodule	Age under 20 to 30 or over 60 years; male
Multinodular gland without a dominant nodule	Childhood neck irradiation; family history of thyroid cancer or MEN 2
Hashimoto or toxic features	Rapid growth, pain; hard, fixed nodule; size over 4 cm
Soft, mobile	Cervical lymphadenopathy; vocal fold palsy; stridor, dysphagia

EXAMINE THE LARYNX IN EVERY THYROID PATIENT

A paralysed vocal fold can coexist with a normal voice. Preoperative laryngoscopy finds unsuspected palsy, which suggests invasive cancer and changes the surgical plan. It also documents the baseline before surgery.

4.2 Investigation

TSH comes first. If it is low, a scintiscan follows, and a hot nodule is almost never malignant and is not aspirated. If it is normal or high, the patient goes on to ultrasound and FNA.

Every patient has an ultrasound of the thyroid and neck, and the pattern sets the risk. High suspicion attaches to a solid hypoechoic nodule with microcalcifications, irregular or infiltrative margins, a taller-than-wide shape or extrathyroidal extension. A solid hypoechoic nodule with smooth margins is intermediate; an iso- or hyperechoic nodule, or a partly cystic one with an eccentric solid part, is low; a spongiform nodule is very low; and a purely cystic nodule is benign. Suspicious nodes show microcalcification, cystic change, loss of the fatty hilum, a round shape and peripheral vascularity.

Ultrasound-guided FNA is done for nodules of 1 cm or more with high or intermediate suspicion, 1.5 cm or more with low suspicion, and 2 cm or more with very low suspicion (or they are observed), but not for pure cysts. Suspicious nodes are also aspirated, with thyroglobulin measured in the needle washout. **Calcitonin** is measured when there is a family history of medullary cancer or MEN 2, and **CT or MRI** is used for retrosternal extension, invasion or bulky nodal disease.

THE BETHESDA SYSTEM FOR THYROID CYTOLOGY

CATEGORY	RISK OF CANCER	USUAL MANAGEMENT
I. Non-diagnostic	5 to 10%	Repeat ultrasound-guided FNA
II. Benign	0 to 3%	Clinical and ultrasound follow-up
III. Atypia of undetermined significance (AUS/FLUS)	10 to 30%	Repeat FNA, molecular testing, or lobectomy
IV. Follicular neoplasm (including Hürthle cell)	25 to 40%	Molecular testing or diagnostic lobectomy
V. Suspicious for malignancy	50 to 75%	Lobectomy or total thyroidectomy
VI. Malignant	97 to 99%	Surgery

WHY FNA CANNOT DIAGNOSE FOLLICULAR CARCINOMA

Follicular carcinoma is diagnosed by *capsular or vascular invasion*. That means examining the whole capsule histologically, which an aspirate cannot do. A "follicular neoplasm" therefore needs a diagnostic lobectomy, and completion thyroidectomy if invasion is found. Hürthle cell neoplasms behave the same way. Molecular panels (BRAF, RAS, RET/PTC, PAX8/PPARγ, gene-expression classifiers) help sort indeterminate cytology.

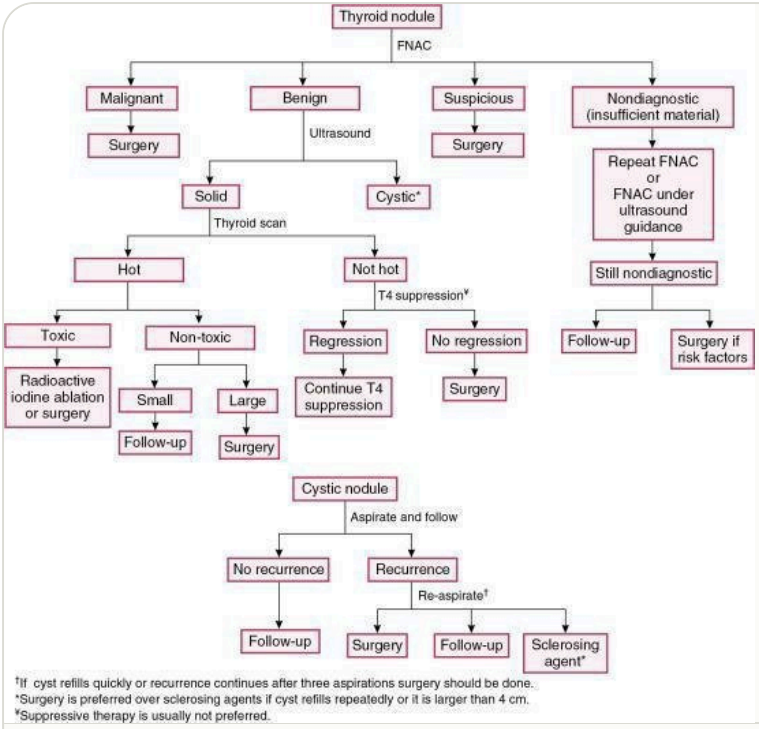


FIGURE 6 Management of a thyroid nodule.
Dhingra, fig. 66.8

4.3 Cystic nodules

About a quarter of nodules are wholly or partly cystic, and the colour of the aspirate is informative. Brown fluid is old haemorrhage into a colloid nodule. Water-clear fluid suggests a *parathyroid cyst*, confirmed by measuring PTH in the fluid. Bloody fluid that reaccumulates raises suspicion of cystic papillary carcinoma, and the solid part is aspirated under ultrasound. Cysts that recur repeatedly or exceed about 4 cm are excised, and ethanol ablation is an alternative for benign recurrent cysts.

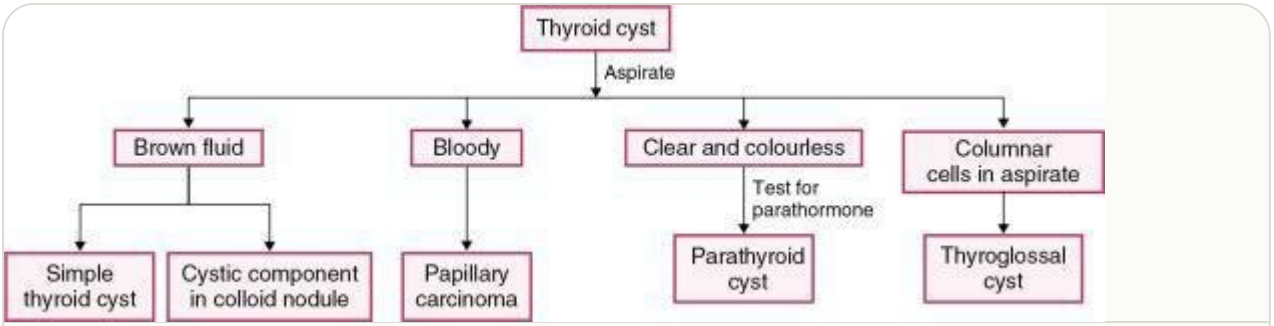


FIGURE 7 Management of a thyroid cyst according to the aspirate.
Dhingra, fig. 66.7

4.4 Benign nodules

Benign nodules are followed with examination and ultrasound, and aspirated again if they grow significantly or change pattern. TSH suppression with thyroxine is little used, because its effect is unpredictable and it harms bone and heart. A toxic adenoma is treated by lobectomy or radioiodine, and other benign nodules are removed only for compression or cosmesis, with radiofrequency or laser ablation as options for selected nodules.

5 CHAPTER FIVE Thyroid cancer

Five types.

MALIGNANT TUMOURS OF THE THYROID

TYPE	PROPORTION	CELL OF ORIGIN	SPREAD
Papillary	70 to 85%	Follicular	Lymphatic (central and lateral nodes)
Follicular, including Hürthle (oncocytic)	10 to 15%	Follicular	Blood (lung, bone)
Medullary	5%	Parafollicular C cell	Nodes early, then blood
Anaplastic	Under 5%	Follicular (dedifferentiated)	Local invasion, everywhere

TYPE	PROPORTION	CELL OF ORIGIN	SPREAD
Lymphoma	About 5%	Lymphoid (in Hashimoto)	Nodal and extranodal

5.1 Papillary carcinoma

Papillary carcinoma is the commonest type and is becoming commoner as more small tumours are detected. It usually affects women of 30 to 50, and it is also the usual childhood thyroid cancer, often with nodes and lung spread but still with an excellent prognosis. The main risk factor is *ionising radiation* in childhood, from old low-dose treatment of tonsils, acne and thymus, radiotherapy for Hodgkin disease, and Chernobyl and atomic fallout, with a latency of decades; family history, familial adenomatous polyposis, Gardner and Cowden syndromes, and adequate iodine intake also contribute.

Microscopically it forms papillae with characteristic nuclei, the ground-glass "*Orphan Annie eye*" nuclei with grooves and intranuclear inclusions, and *psammoma bodies*, and it is often multifocal. Because the diagnosis rests on nuclear features, cytology can make it. The tall cell, columnar, diffuse sclerosing and hobnail variants are aggressive, and BRAF V600E, the commonest mutation, is associated with more aggressive behaviour.

It spreads by lymphatics: clinically evident nodes are present in about 30% of adults and in many more children, and the nodal deposits may be cystic and sometimes form the presenting lump behind an occult primary. Distant spread is uncommon. Ten-year survival exceeds 90%, and prognosis worsens with older age, male sex, larger size, extrathyroidal extension and distant metastases, as captured by the *AGES* (age, grade, extent, size) and *AMES* (age, metastases, extent, size) scores. Low-risk **papillary microcarcinomas** of 1 cm or less may be watched by active surveillance or treated by lobectomy.

5.2 Follicular carcinoma

Follicular carcinoma affects an older group than papillary carcinoma, around 50, mostly women, and is commoner in iodine-deficient areas. It is usually unifocal, and the diagnosis needs capsular or vascular invasion on histology; *minimally invasive* tumours behave well and *widely invasive* ones do not. It spreads by *blood* to lung and

bone, with distant metastases in up to 15% at diagnosis, while nodal spread is uncommon. RAS mutations and the PAX8/PPARγ rearrangement are characteristic, and the prognosis is worse with age over 50, size over 4 cm and distant disease.

5.3 Hürthle cell (oncocytic) carcinoma

Hürthle cell carcinoma consists of large eosinophilic cells packed with mitochondria. It is more aggressive than follicular carcinoma, more often multifocal and with nodal and distant spread, and because it takes up *radioiodine poorly*, surgery is all-important.

5.4 Medullary carcinoma

Medullary carcinoma arises from the C cells, secretes *calcitonin* and CEA, which are used for diagnosis and follow-up, and shows amyloid stroma on histology. It is equally common in both sexes. About 75 to 80% are **sporadic**, unifocal tumours in the fifth decade. The other 20 to 25% are **familial**, caused by autosomal dominant germline *RET* mutations; they are preceded by C-cell hyperplasia, are bilateral and multifocal, and occur in MEN 2A, MEN 2B and familial medullary carcinoma, now counted as a variant of MEN 2A.

It spreads early to central and lateral nodes, then to lung, liver and bone, and its hormones cause diarrhoea and flushing. The work-up measures calcitonin and CEA, tests *every* patient for RET, excludes a *phaeochromocytoma* with plasma or urinary metanephrines before any surgery (removing it first to avoid a hypertensive crisis), and checks serum calcium for hyperparathyroidism.

Treatment is total thyroidectomy with central (level VI) neck dissection, adding a lateral dissection for nodal disease or according to the calcitonin level. Radioiodine does not work, and RET and multikinase inhibitors such as selpercatinib, vandetanib and cabozantinib treat progressive metastatic disease. RET carriers have a prophylactic thyroidectomy *within the first year* for the highest-risk (MEN 2B) mutations, *by about five years* for high-risk MEN 2A mutations, and later, guided by calcitonin, for moderate-risk mutations.

MULTIPLE ENDOCRINE NEOPLASIA

SYNDROME	GENE	COMPONENTS
MEN 1 (Wermer)	MEN1	Parathyroid hyperplasia, pituitary tumours, pancreatic islet tumours
MEN 2A (Sipple)	RET	Medullary carcinoma (nearly all), phaeochromocytoma (about 50%), hyperparathyroidism (20 to 30%), cutaneous lichen amyloidosis, Hirschsprung disease
MEN 2B	RET (M918T)	Medullary carcinoma (early, aggressive), phaeochromocytoma, mucosal neuromas of lips and tongue, marfanoid habitus, intestinal ganglioneuromatosis; no hyperparathyroidism

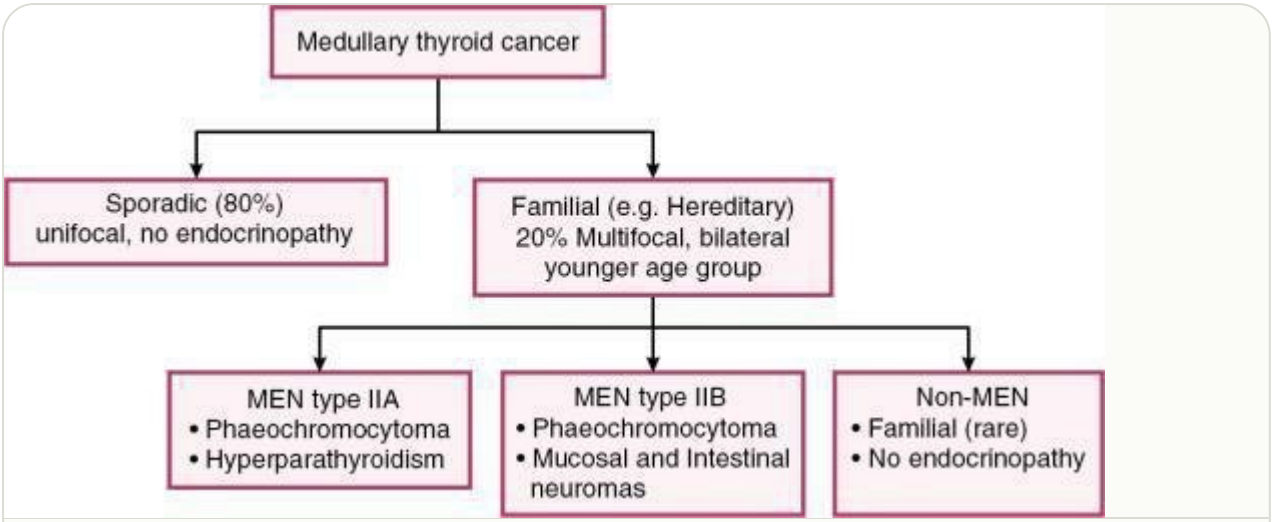


FIGURE 8 Types of medullary carcinoma: sporadic (about 80%) and familial (MEN 2A, MEN 2B, and familial non-MEN).
Dhingra, fig. 66.6

5.5 Anaplastic carcinoma

Anaplastic carcinoma is one of the most lethal cancers, with a median survival of about six months. It arises in old people, usually over 60, often by dedifferentiation of a long-standing differentiated cancer, and is always stage IV. It presents as a rapidly enlarging, hard, often painful mass fixed to its surroundings, with hoarseness, stridor, dysphagia and nodes, and half the patients have distant metastases.

A core or open biopsy must provide enough tissue to exclude *lymphoma*, which is curable. The tumour is resected only if resection can be complete; otherwise it is treated with radiotherapy and chemotherapy, and BRAF-mutant tumours respond to

dabrafenib with trametinib. Palliating the airway, including the hard decision about tracheostomy through tumour, and supportive and palliative care are central.

5.6 Lymphoma

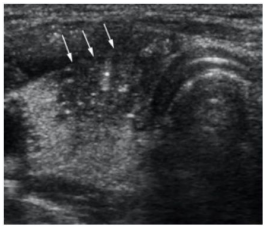
Thyroid lymphoma usually affects older women and almost always arises in *Hashimoto thyroiditis*. It is mostly a B-cell non-Hodgkin lymphoma, diffuse large B-cell or MALT type, presenting as a rapidly enlarging, painless, firm goitre with compression, often in a hypothyroid patient. It is diagnosed by core or open biopsy with immunohistochemistry and flow cytometry, and treated with chemotherapy and radiotherapy, surgery being limited to biopsy or relief of the airway. The outlook is good.

5.7 Metastases to the thyroid

Metastases reach the thyroid from the kidney, breast, lung, melanoma and head and neck squamous carcinoma.

IMAGING IN THE APP 1 case

Thyroid carcinoma on imaging.



Thyroid cancer

Tumours · 10 images

6 CHAPTER SIX Treating differentiated cancer

Surgery, iodine, suppression.

6.1 Extent of thyroidectomy

Lobectomy with isthmusectomy is enough for low-risk, unifocal, intrathyroidal tumours up to about 4 cm without nodes or extrathyroidal extension. **Total thyroidectomy** is needed for tumours over 4 cm, gross extrathyroidal extension, nodal or distant metastases, bilateral disease, previous irradiation, and whenever radioiodine is planned. **Completion thyroidectomy** follows when lobectomy histology shows a higher-risk tumour.

6.2 The neck

Clinically involved central nodes are removed by a **therapeutic central (level VI) dissection**. A **prophylactic central dissection** is debated; it is considered for T3 and T4 tumours and when lateral nodes are involved, at the cost of more hypoparathyroidism. Lateral nodes proven on FNA are treated by a compartmental selective neck dissection of levels II to IV, with V as needed, and never by "berry picking".

OPERATIVE GUIDES IN THEATRE 1 operation

Central compartment neck dissection.

Central Compartment Neck Dissection
Central dissection

6.3 Invasive disease

A functioning recurrent nerve with tumour on its surface is shaved and kept, while a paralysed or encased nerve is resected. Involved strap muscles are removed en bloc, and invasion of the trachea, larynx or oesophagus is treated by shave or full-thickness window or sleeve resection according to its depth.

OPERATIVE GUIDES IN THEATRE 1 operation

Invasive thyroid cancer.

Invasive Thyroid Cancer
Invasive cancer

6.4 After surgery

Radioactive iodine (I-131) ablates the thyroid remnant and treats microscopic or metastatic disease in intermediate- and high-risk patients. The patient is prepared by thyroxine withdrawal or recombinant TSH and a low-iodine diet, and iodinated contrast is avoided beforehand. Its side effects are sialadenitis, xerostomia, taste change, nasolacrimal duct obstruction and second cancers.

TSH suppression with levothyroxine aims for a TSH below 0.1 mU/L in high-risk disease and a low-normal level in low-risk disease. **Surveillance** follows thyroglobulin, with its antibodies, against the baseline, together with neck ultrasound, diagnostic radioiodine scans, and PET-CT for thyroglobulin-positive, iodine-negative disease. External beam radiotherapy treats unresectable or gross residual disease and tracheal invasion in older patients, and iodine-refractory progressive disease is treated with kinase inhibitors such as lenvatinib and sorafenib, and targeted agents for RET, NTRK and BRAF changes.

7 CHAPTER SEVEN Thyroidectomy

The operation and its complications.

7.1 Types

OPERATIONS ON THE THYROID

OPERATION	WHAT IS REMOVED	TYPICAL USE
Lobectomy (hemithyroidectomy)	One lobe with the isthmus	Benign nodule, follicular neoplasm, low-risk cancer, toxic adenoma
Isthmusectomy	The isthmus	Isthmic nodule; biopsy and airway relief in anaplastic carcinoma or lymphoma
Subtotal thyroidectomy	Both lobes leaving a small posterior remnant	Formerly Graves and multinodular goitre

OPERATION	WHAT IS REMOVED	TYPICAL USE
Near-total thyroidectomy	One lobe totally, the other leaving under 1 g near the nerve and parathyroid	Cancer, Graves, goitre
Total thyroidectomy	Both lobes and isthmus	Cancer, Graves, large multinodular goitre
Completion thyroidectomy	The remaining lobe after lobectomy	Lobectomy histology shows cancer needing total thyroidectomy

7.2 Indications and preparation

The indications are the four Cs plus one: cancer, suspicion of cancer, compression, cosmesis, and hyperthyroidism. Before operating, the patient should be euthyroid; calcium and vitamin D are measured, the larynx is examined, the thyroid and neck are scanned by ultrasound with FNA, CT is obtained for retrosternal extension, calcitonin is measured when medullary cancer is suspected, and the usual anaesthetic assessment is made.

7.3 Technique

1. **Incision:** a collar incision in a skin crease about two fingerbreadths above the sternal notch.
2. **Flaps:** subplatysmal flaps raised.
3. **Strap muscles:** separated in the midline and lifted off the gland.
4. **Lateral mobilisation:** the middle vein divided, and the gland rotated medially.
5. **Superior pole:** the vessels divided individually on the capsule, in the avascular cricothyroid (Reeve) space, to spare the external branch of the superior laryngeal nerve.
6. **The recurrent nerve:** identified and traced to its laryngeal entry. Nerve monitoring confirms and protects it, and loss of signal on the first side may stop a planned bilateral operation.
7. **Parathyroids:** each gland identified and kept on its blood supply by dividing inferior thyroid artery branches on the capsule; a devascularised gland is confirmed by frozen section, minced and reimplanted into sternomastoid.

8. **Berry's ligament:** divided carefully; cautery near the nerve is dangerous here.
9. **Closure:** haemostasis checked with a Valsalva manoeuvre, then straps and platysma closed. Drains are optional; they do not prevent haematoma.
10. **Minimal-access approaches:** video-assisted and remote-access robotic routes for selected patients.

OPERATIVE GUIDES IN THEATRE 3 operations

Open, video-assisted and robotic thyroidectomy.

Open Thyroidectomy
Thyroidectomy
Minimally Invasive Video-Assisted Thyroidectomy
Video-assisted
Robotic-Assisted Thyroidectomy
Robotic

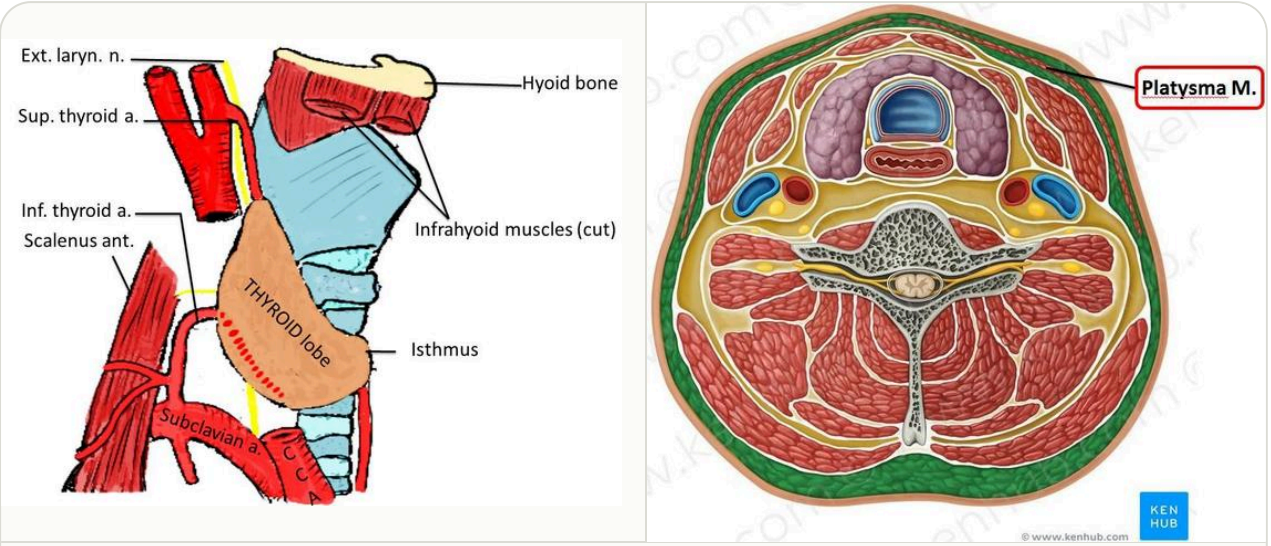


FIGURE 9 The thyroid with the strap muscles reflected, and a cross-section of the neck at the level of the thyroid.

Anatomy atlas · Doss p. 127, ENT Steps p. 273

7.4 Complications

Haematoma complicates under 2% of thyroidectomies, usually within hours, and is the complication that kills if it is not acted on at once.

NECK HAEMATOMA AFTER THYROIDECTOMY

Swelling, tight neck, stridor or distress:

1. Open the skin and the strap muscles at the bedside.

2. Evacuate the clot.
3. Secure the airway.
4. Return to theatre.

Venous and lymphatic congestion makes the larynx oedematous, so intubation is difficult. Release of pressure comes first.

Recurrent laryngeal nerve injury is temporary in up to about 5 to 10% and permanent in about 1%, with a higher risk in reoperation, cancer and retrosternal goitre. Unilateral injury causes hoarseness and aspiration. *Bilateral* injury may leave a near-normal voice but causes stridor on extubation, often needing reintubation and tracheostomy. Injury to the **external branch of the superior laryngeal nerve** costs high pitch and vocal projection, which matters greatly to singers.

Hypocalcaemia from hypoparathyroidism is common and transient after total thyroidectomy and permanent in about 1 to 3% in expert hands, more after central dissection. It appears at 24 to 72 hours with perioral and fingertip tingling, cramps, Chvostek's and Trousseau's signs, tetany, laryngospasm and a long QT. Calcium and PTH are measured after surgery, and hypocalcaemia is treated with oral calcium and calcitriol, or intravenous calcium gluconate when severe. Thyroid storm, rare tracheomalacia, expected hypothyroidism after total thyroidectomy, wound infection, seroma, keloid, pneumothorax, and chyle leak after lateral dissection complete the list.

WHERE THE BOOKS DIFFER · SIZE CUT-OFFS AND LOBECTOMY

Dhingra	Lee
Microcarcinomas under 1.5 cm and intrathyroidal tumours without nodes may have lobectomy; follicular neoplasms over 4 cm in the elderly go straight to total thyroidectomy; no elective neck dissection.	Following the 2015 ATA guidelines, lobectomy suffices for low-risk intrathyroidal tumours up to 4 cm; total thyroidectomy for larger, extrathyroidal or metastatic disease or when radioiodine is planned; prophylactic central dissection is debated.

Current practice follows the ATA approach and uses risk stratification rather than size alone.

BRITISH PRACTICE: U AND THY GRADES, NIFTP AND LOW-DOSE IODINE (OXFORD TEXTBOOK)

Thyroid cancer incidence has risen sharply over three decades while mortality has risen much less, a sign that imaging is finding many indolent cancers. British practice grades nodules on ultrasound from **U1** (normal) and **U2** (benign) to **U3** (indeterminate), **U4** (suspicious) and **U5** (malignant), and requires ultrasound-guided FNA for U3 to U5. Cytology is graded **Thy1** (non-diagnostic), **Thy2** (non-neoplastic), **Thy3a** (atypia), **Thy3f** (follicular neoplasm), **Thy4** (suspicious) and **Thy5** (malignant), roughly equivalent to the Bethesda categories. Thy3 and above, and any discordance between imaging and cytology, go to the thyroid multidisciplinary team. Thy3f and Thy4 lesions usually have a **diagnostic hemithyroidectomy**.

New entities include tumours of uncertain malignant potential and **NIFTP** (non-invasive follicular thyroid neoplasm with papillary-like nuclear features), which is now regarded as benign and needs no further treatment after lobectomy. **Completion thyroidectomy** is advised for tumours over 4 cm, multifocal or bilateral disease, extrathyroidal spread, familial disease, nodal or distant metastases. A solitary **papillary microcarcinoma** under 1 cm without adverse features needs only hemithyroidectomy without TSH suppression, and active surveillance is an option in selected patients.

Radioactive iodine is given for high-risk disease (tumour over 4 cm, extrathyroidal extension, distant metastases) and can usually be omitted in low-risk disease. Recombinant human TSH has largely replaced thyroid hormone withdrawal before treatment, and the **HiLo** and **ESTIMABL** trials showed that a low ablation dose (1.1 GBq) is as effective as a high dose (3.7 GBq) for low- and intermediate-risk patients. At 9 to 12 months, **dynamic risk stratification** using thyroglobulin, ultrasound and imaging re-grades each patient's risk and guides how hard TSH is suppressed and how closely they are followed.

CHAPTER EIGHT

8 For the examination

What the examiner wants to hear.

8.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Surgical anatomy:** Berry's ligament; the course of the RLN on each side, and the non-recurrent nerve; the external branch of the superior laryngeal nerve; the parathyroids and their blood supply.

- **Lingual thyroid:** scan before excision.
- **Work-up of a nodule:** TSH, ultrasound pattern, FNA, Bethesda categories; why FNA cannot diagnose follicular carcinoma.
- **The five cancers:** their spread, markers, MEN syndromes, RET testing, and phaeochromocytoma first.
- **Extent of surgery,** radioiodine, TSH suppression and thyroglobulin follow-up.
- **Complications:** haematoma (open at the bedside), nerve palsies, hypocalcaemia.

8.2 Common pitfalls

COMMON PITFALLS

- Operating for medullary carcinoma without excluding phaeochromocytoma.
- Calling a cystic lateral neck node in a young adult a branchial cyst without considering metastatic papillary carcinoma.
- Excising a lingual thyroid without checking for other thyroid tissue.
- Giving iodinated contrast just before planned radioiodine.
- Failing to check the cords before and after surgery.

Written for this app from Oxford Textbook of Otolaryngology ch. 59, Oxford Specialist Handbook ch. 11, Oxford Handbook of ENT ch. 15, Dhingra ch. 66 and K. J. Lee ch. 38, 51. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · NECK, THYROID & SALIVARY

Hyperparathyroidism

Primary hyperparathyroidism used to be the disease of "bones, stones, abdominal groans and psychic moans". Today most patients are found by a raised calcium on a routine blood test. Usually one gland is at fault, a single adenoma, and a localised, focused operation cures it. The art is to confirm the biochemistry, localise the gland, and remember the exceptions: multigland disease, the ectopic gland and familial hypocalciuric hypercalcaemia.

AT A GLANCE

- High calcium plus high PTH.
- Confirm, then localise.
- Focused or four-gland surgery.
- Check PTH during surgery.

1 CHAPTER ONE Applied anatomy

Four small glands, two journeys.

1.1 Embryology

The two pairs of glands have different journeys, and their journeys explain where they are found. The **superior parathyroids** arise from the *fourth pharyngeal pouch* and travel only a short distance with the lateral thyroid anlage, so their position is fairly constant. The **inferior parathyroids** arise from the *third pouch* together with the thymus and descend much further, so they are more variable, lying anywhere from the angle of the jaw to the mediastinum and often in the *thyrothymic ligament* or the thymus itself.

1.2 The normal glands

A normal gland measures about 5 × 3 × 1 mm and weighs about 40 mg. It is flat and bean- or leaf-shaped, caramel, tan or mahogany in colour, darker than the paler, brighter yellow fat around it, and it "glides" within that fat when gently pushed (the

gliding sign). Most people have four glands and about 5% have supernumerary ones. The glands are usually mirror images of each other, which helps in finding the opposite one.

The **superior gland** lies at the level of the cricothyroid joint on the back of the upper thyroid lobe, about 1 cm above the crossing of the recurrent laryngeal nerve and inferior thyroid artery, and *deep (dorsal) to the nerve*. The **inferior gland** lies within 1 to 2 cm of the lower pole and *superficial (ventral) to the nerve*. Both draw most of their blood from the inferior thyroid artery, which is why its branches are divided on the thyroid capsule at thyroidectomy rather than in the trunk.

Adenomas can lie in ectopic sites: retro-oesophageal and retrotracheal, in the anterior mediastinum within the thymus, inside the thyroid, in the carotid sheath, or undescended high in the neck.

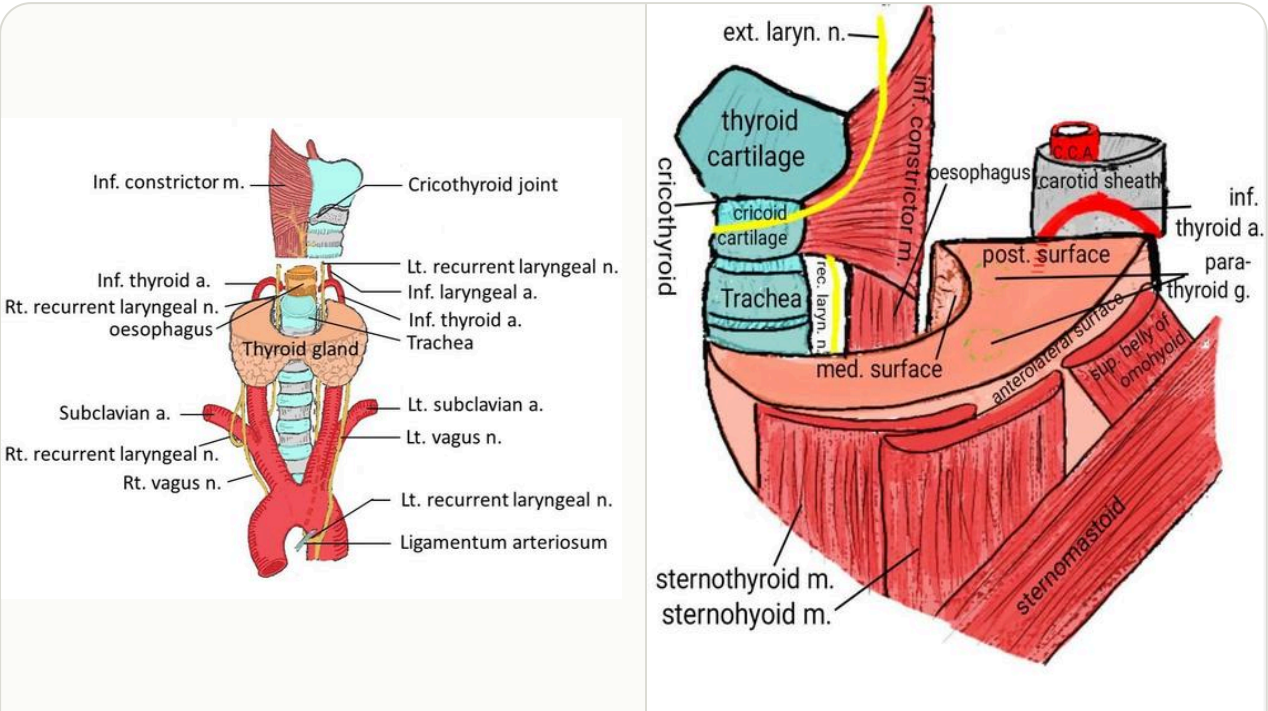


FIGURE 1 The inferior thyroid arteries and recurrent laryngeal nerves, landmarks for the parathyroids; and the relations of the thyroid lobe.

Anatomy atlas · Doss p. 154, Doss p. 126

2 CHAPTER TWO Calcium physiology

PTH and vitamin D.

2.1 Parathyroid hormone

Parathyroid hormone raises serum calcium in four ways: it mobilises calcium from bone through osteoclast activity; it increases renal tubular reabsorption of calcium while increasing excretion of phosphate; it stimulates renal 1α-hydroxylation of vitamin D; and, through vitamin D, it increases absorption of calcium from the gut. Calcitonin from the thyroid C cells has only a minor opposing role in humans. PTH has a half-life of only a few minutes, which is what makes *intraoperative PTH monitoring* possible.

2.2 Measuring calcium

Total calcium varies with albumin and is corrected by adding about 0.8 mg/dL (0.2 mmol/L) for each 1 g/dL fall in albumin below normal. When albumin is abnormal, ionised calcium is measured instead.

3 CHAPTER THREE Hyperparathyroidism

Primary, secondary, tertiary.

3.1 Primary hyperparathyroidism

Primary hyperparathyroidism affects about 1 in 500 women and 1 in 2000 men, mostly after the menopause. About 85% of cases are due to *a single adenoma*, 10 to 15% to *four-gland hyperplasia* (sporadic, familial, or part of MEN 1 or MEN 2A), 2 to 5% to *double adenomas*, mostly in the elderly, and under 1% to *carcinoma*. Neck irradiation and lithium are risk factors.

The biochemistry is high calcium with a raised or *inappropriately normal* PTH, a low phosphate, and a high or high-normal urinary calcium. Vitamin D is measured and

replaced if low, because deficiency both confuses the picture and worsens bone disease.

3.2 Clinical features

MANIFESTATIONS OF HYPERCALCAEMIA

SYSTEM	FEATURES
Bones	Pain, osteoporosis, fractures; osteitis fibrosa cystica, brown tumours (rare now)
Kidneys	Stones, nephrocalcinosis, polyuria and polydipsia, renal impairment
Gut	Constipation, nausea, peptic ulcer, pancreatitis
Mind and muscle	Fatigue, weakness, depression, poor memory, confusion
Other	Hypertension, short QT, band keratopathy, gout and pseudogout

A **hypercalcaemic crisis**, with dehydration, confusion and arrhythmia at very high calcium levels, is treated with intravenous saline, a bisphosphonate or denosumab, and calcitonin, followed by urgent surgery.

3.3 Differential diagnosis of hypercalcaemia

CAUSES OF HYPERCALCAEMIA

PTH RAISED OR NORMAL	PTH SUPPRESSED
Primary hyperparathyroidism	Malignancy (PTH-related peptide from lung, breast, kidney cancers; bone metastases; myeloma)
Tertiary hyperparathyroidism	Granulomas: sarcoidosis, tuberculosis (excess vitamin D activation)
Familial hypocalciuric hypercalcaemia (low urinary calcium)	Vitamin D or A excess, milk-alkali syndrome
Lithium	Thyrotoxicosis, adrenal insufficiency, immobilisation, Paget disease; thiazides

FAMILIAL HYPOCALCIURIC HYPERCALCAEMIA (FHH)

- **Cause:** an autosomal dominant mutation of the calcium-sensing receptor.
- **Biochemistry:** lifelong mild hypercalcaemia with normal or slightly raised PTH, and a *low urinary calcium* (calcium/creatinine clearance ratio under 0.01).
- **Course:** benign.
- **Surgery:** does not help. Always check the 24-hour urinary calcium before operating.

3.4 Secondary and tertiary hyperparathyroidism

Secondary hyperparathyroidism is a hyperplastic response of all the glands to low calcium, mostly in *chronic kidney disease*, where phosphate is retained and active vitamin D is lacking, but also in vitamin D deficiency and malabsorption. PTH is high and calcium low or normal. It is treated medically with phosphate binders, active vitamin D and *calcimimetics* such as cinacalcet. Refractory disease with bone pain, pruritus or calciphylaxis needs *subtotal (three-and-a-half gland) parathyroidectomy*, or total parathyroidectomy with autotransplantation into the forearm, together with thymectomy.

Tertiary hyperparathyroidism is autonomous hypersecretion that persists after the cause has been corrected, typically after renal transplantation, and produces hypercalcaemia.

3.5 Parathyroid carcinoma

Carcinoma is suspected when calcium and PTH are very high (calcium often over 14 mg/dL), when there is a palpable neck mass or a recurrent laryngeal nerve palsy, and when bone and renal disease are present together. At surgery the gland is hard, grey-white and adherent, invading the thyroid or muscle. It must be resected *en bloc* with the thyroid lobe and adjacent tissue without rupturing the capsule. Recurrence is common, and death usually comes from uncontrollable hypercalcaemia.

3.6 Indications for surgery

Parathyroidectomy is the only cure, and it is recommended for every symptomatic patient. In asymptomatic patients it is recommended when the serum calcium is more than 1 mg/dL (0.25 mmol/L) above the upper limit of normal, when the patient is under 50, when there is osteoporosis (a T-score of −2.5 or less) or a vertebral fracture, when creatinine clearance is under 60 mL/min or there are kidney stones, nephrocalcinosis or high urinary calcium, and when the patient cannot be followed up. Many surgeons also offer surgery to fit asymptomatic patients, because of the benefits to bone and cognition.

CHAPTER FOUR

4 Localisation

Before a focused operation.

4.1 The tests

Localisation does not make the diagnosis; the biochemistry does. Its purpose is to plan a focused operation, and it is essential before any reoperation.

Ultrasound is cheap but operator-dependent. It shows a hypoechoic oval nodule behind the thyroid, often with a polar feeding vessel, but is poor for retrotracheal and mediastinal glands. **Sestamibi (technetium-99m MIBI) scintigraphy** relies on the fact that both thyroid and parathyroid take up the tracer but the mitochondria-rich adenoma *retains* it after the thyroid has washed out. It detects about 85 to 90% of single adenomas, less reliably small adenomas, double adenomas and hyperplasia, and SPECT or SPECT-CT gives a three-dimensional position.

Four-dimensional CT shows the typical behaviour of an adenoma, with arterial enhancement and early washout, and is useful when the other scans disagree and before reoperation. Choline PET-CT is increasingly used, and MRI and selective venous sampling for PTH help in difficult reoperations.

CONCORDANCE MATTERS

A *concordant* ultrasound and sestamibi scan localise a single adenoma in most patients and permit a focused approach. *Discordant* or negative scans suggest multigland

disease or an ectopic gland. These patients need bilateral exploration by an experienced surgeon.

5 CHAPTER FIVE Surgery

Focused or bilateral.

5.1 Approaches

In **focused (minimally invasive) parathyroidectomy**, a small incision over the localised gland allows the adenoma to be removed by controlling its vascular pedicle while keeping clear of the recurrent nerve. *Intraoperative PTH* is measured before and 10 minutes after excision, and a fall of more than 50% from the highest value into the normal range (the *Miami criterion*) confirms cure.

Bilateral neck exploration identifies all four glands. It is used when the scans do not localise, in multigland disease, in MEN and familial disease, and when the PTH fails to fall after a focused excision.

When a gland is missing, the search follows embryology. A missing superior gland is sought in the tracheo-oesophageal groove and retro-oesophageal space along the inferior thyroid artery; a missing inferior gland in the thymus and thyrothymic ligament, by cervical thymectomy, and in the carotid sheath. Intrathyroidal ultrasound or a lobectomy on that side follows, but a blind sternotomy is not performed at the first operation.

Hyperplasia is treated by subtotal parathyroidectomy, leaving a well-vascularised remnant of about 50 mg marked with a clip, or by total parathyroidectomy with autotransplantation; thymectomy is added in MEN 1.

OPERATIVE GUIDES IN THEATRE 4 operations

Parathyroidectomy for primary and secondary disease, intraoperative PTH, and reoperation.

Parathyroidectomy for Primary Hyperparathyroidism and Nonlocalizing Hyperparathyroidism
Primary

Intraoperative Parathyroid Hormone Monitoring
IOPTH
Parathyroidectomy for Secondary Hyperparathyroidism
Secondary
Reoperative Parathyroidectomy
Reoperative

5.2 Complications

Temporary hypocalcaemia is common. *Hungry bone syndrome*, a severe and prolonged hypocalcaemia with low phosphate and magnesium, follows removal of a long-standing adenoma in a patient with bone disease, as the starved skeleton takes up calcium, and needs large doses of calcium and calcitriol. Permanent hypoparathyroidism follows about 1 to 5% of adenoma operations and more after surgery for hyperplasia or secondary disease. Recurrent laryngeal nerve palsy occurs in 0.4 to 4%, and haematoma can occur.

Persistent or recurrent hyperparathyroidism has a short list of causes: a missed or second adenoma, unrecognised hyperplasia, a supernumerary or ectopic gland, parathyromatosis from rupture of the capsule, carcinoma, and a wrong diagnosis such as FHH.

6 CHAPTER SIX Hypoparathyroidism and hypocalcaemia

After thyroid and parathyroid surgery.

6.1 Causes and features

Hypoparathyroidism usually follows removal or devascularisation of the parathyroids during thyroidectomy or central neck dissection. Rarer causes are autoimmune disease, DiGeorge syndrome (absent third and fourth pouch derivatives), magnesium deficiency and infiltration.

Symptoms usually appear 24 to 72 hours after surgery, starting with perioral and fingertip tingling and cramps and progressing to carpopedal spasm and tetany, laryngospasm with stridor, seizures, confusion, and a prolonged QT with arrhythmia. *Chvostek's sign* is twitching of the face when the facial nerve is tapped in front of the

ear; because 5 to 10% of normal people have it, it should be recorded before surgery. *Trousseau's sign* is carpal spasm when a blood-pressure cuff is inflated above systolic pressure for three minutes.

6.2 Management

Calcium is measured after total thyroidectomy, and a PTH level a few hours after surgery predicts who will become hypocalcaemic and who can safely go home. Mild symptoms are treated with oral calcium carbonate and calcitriol or alfacalcidol. Severe symptoms or tetany need intravenous 10% calcium gluconate, 10 to 20 mL over ten minutes followed by an infusion, with cardiac monitoring and correction of magnesium.

Hypoparathyroidism lasting beyond six months is permanent and needs lifelong calcium and active vitamin D, aiming for a low-normal calcium to avoid hypercalciuria, with recombinant PTH in selected cases. It is prevented by capsular dissection, by identifying and preserving the glands with their blood supply, by autotransplanting any gland that is devascularised or removed by mistake, and by near-infrared autofluorescence to identify the glands.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Embryology and anatomy:** fourth pouch superior glands deep to the RLN, third pouch inferior glands superficial to it, and the ectopic sites.
- **Biochemistry:** high calcium with non-suppressed PTH. Exclude FHH with urinary calcium.
- **Adenoma versus hyperplasia;** the MEN syndromes.
- **Localisation:** ultrasound, sestamibi, 4D CT; why localisation is not diagnosis.
- **Focused parathyroidectomy** and the intraoperative PTH fall.

- **Post-thyroidectomy hypocalcaemia:** signs and treatment.

7.2 Common pitfalls

COMMON PITFALLS

- Operating on familial hypocalciuric hypercalcaemia.
- Treating negative localisation scans as a reason not to operate on proven disease.
- Stopping after one adenoma in MEN 1, where all glands are abnormal.
- Rupturing the capsule of a suspected carcinoma or adenoma, which seeds parathyromatosis.

Written for this app from Oxford Textbook of Otolaryngology ch. 63, Oxford Specialist Handbook ch. 11, Oxford Handbook of ENT ch. 15, Dhingra ch. 66 and K. J. Lee ch. 38, 51. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · NECK, THYROID & SALIVARY

Salivary gland infections, stones and swellings

Most salivary swellings are not tumours. A child with painful, swollen cheeks has mumps; an old, dehydrated patient after surgery with a hot, tender parotid and pus at the duct has suppurative parotitis; a man whose submandibular gland swells every time he eats has a stone. Know the glands and their ducts, recognise the patterns of swelling (acute or recurrent, painful or painless, one gland or all), and remember that a persistent, firm lump in one gland is a tumour until proved otherwise.

AT A GLANCE

- Mealtime pain suggests a stone.
- Slow lump suggests tumour.
- Palsy with a lump suggests cancer.
- Ultrasound, FNA and careful surgery.
- Drooling: separate anterior from posterior; avoid duct relocation in children who aspirate.

1 CHAPTER ONE Applied anatomy

Three pairs of glands and hundreds of small ones.

1.1 Parotid gland

The largest, lying on the masseter and ramus, wrapped round the back of the mandible, from the zygomatic arch to below the angle (the tail), in front of the ear and sternomastoid; its deep surface reaches the prestyloid parapharyngeal space. It is enclosed in the parotidomasseteric fascia (part of the investing layer), whose unyielding nature makes parotitis so painful.

- **Facial nerve:** leaves the stylomastoid foramen, enters the gland and divides at the *pes anserinus* into temporofacial and cervicofacial divisions and five branches; it separates the larger *superficial lobe* (about 80%) from the *deep lobe*, the plane of parotid surgery. Landmarks for the trunk: the *tragal pointer* (the nerve lies about 1 cm deep and inferior to it), the *tympanomastoid suture*, and the *posterior belly of digastric* (the nerve is at its depth, above it).

- **Other contents:** the retromandibular vein (superficial to the external carotid), the external carotid artery dividing into maxillary and superficial temporal arteries, and intraparotid lymph nodes (about 20), the only salivary gland with intraglandular nodes.
- **Stensen's duct:** runs forward over the masseter, turns through the buccinator and opens opposite the upper second molar.
- **Nerves:** parasympathetic secretomotor fibres from the glossopharyngeal nerve via Jacobson's nerve, the lesser petrosal nerve, the otic ganglion and the auriculotemporal nerve; sensation of the overlying skin by the great auricular nerve (C2, C3).

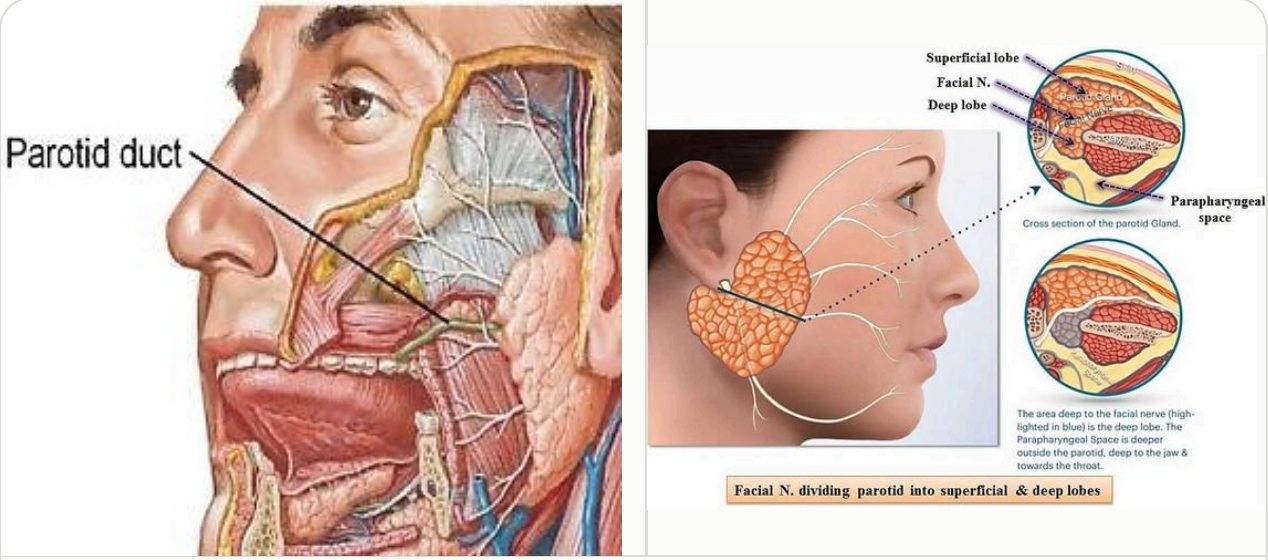
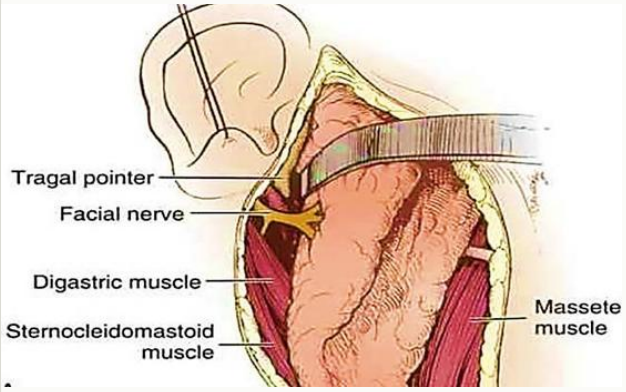


FIGURE 1 The parotid gland and Stensen's duct, and the facial nerve dividing it into superficial and deep lobes.

Anatomy atlas · ENT Steps p. 181



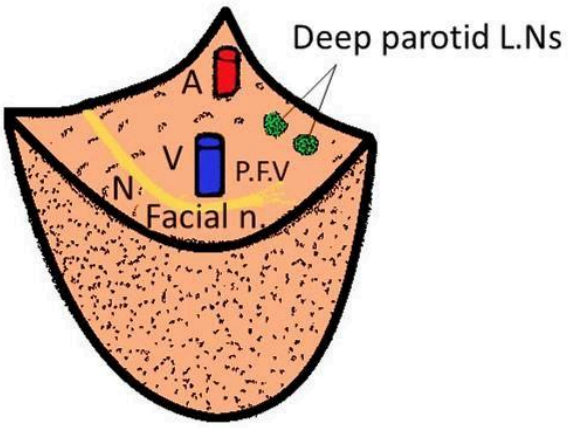
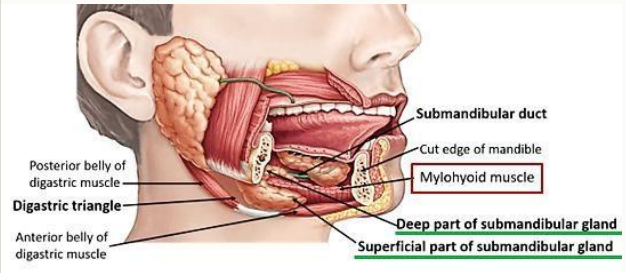


FIGURE 2 Landmarks for the facial nerve trunk, and the structures within the parotid.
Anatomy atlas · ENT Steps p. 185, Doss p. 74

1.2 Submandibular gland

In the digastric triangle, wrapped round the back of the mylohyoid: a large superficial part below it and a deep part above it. *Wharton's duct* runs forwards between the mylohyoid and hyoglossus, crossed by the *lingual nerve* (which loops beneath it), to open beside the frenulum. The facial artery grooves the deep part, the facial vein lies superficial to it, and the *marginal mandibular nerve* runs over the gland within its capsule; the hypoglossal nerve lies deep. Secretomotor supply: chorda tympani via the submandibular ganglion.



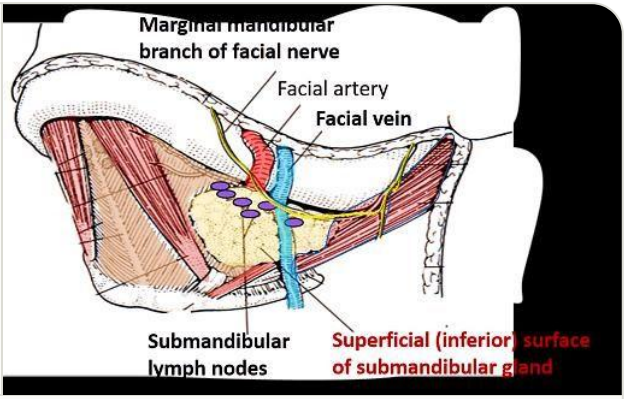


FIGURE 3 The submandibular gland around the mylohyoid, and the marginal mandibular nerve.
Anatomy atlas · ENT Steps p. 189, ENT Steps p. 190

1.3 Sublingual and minor glands

The sublingual glands lie under the floor of mouth mucosa above the mylohyoid, draining by many small ducts (of Rivinus), some into Wharton's duct (Bartholin's duct). Six hundred to a thousand minor salivary glands line the mouth, palate (most), pharynx, larynx and sinuses.

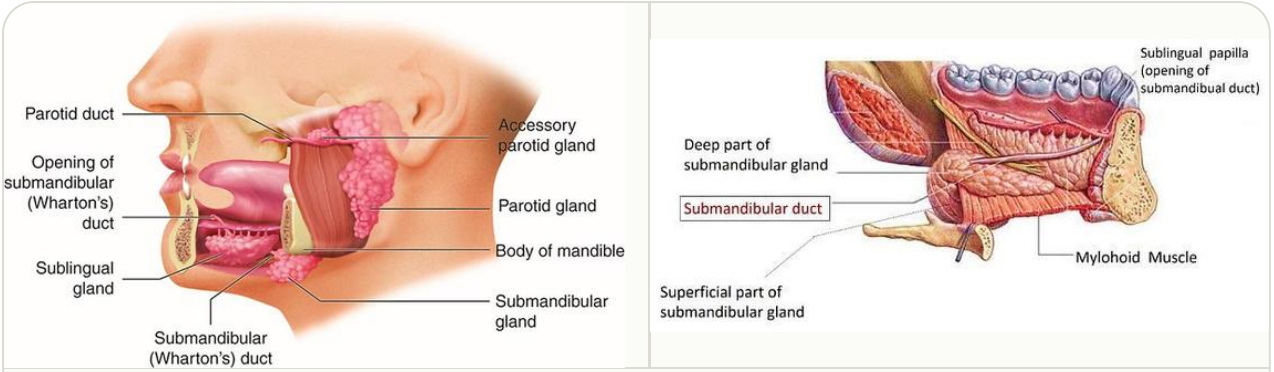


FIGURE 4 The major salivary glands and their ducts, and the sublingual and submandibular glands with their ducts.

Anatomy atlas · ENT Steps p. 176, ENT Steps p. 191

1.4 Saliva

About a litre to a litre and a half a day. At rest, most comes from the submandibular glands (more mucinous, richer in calcium); when stimulated by taste or smell, the parotids (serous, watery) supply most. Saliva lubricates, begins starch digestion (amylase), buffers acid (bicarbonate), protects teeth and kills microbes (IgA, lysozyme, lactoferrin). Flow is driven mainly by parasympathetic stimulation; dehydration, anticholinergic drugs, radiotherapy and age reduce it.

2 CHAPTER TWO Assessment

History, examination, imaging.

2.1 History and examination

Which gland, one or both sides; acute or recurrent; pain and swelling with meals (obstruction); fever and pus (infection); dry eyes and mouth (Sjögren); drugs,

alcohol, diabetes; and red flags for tumour (persistent mass, facial weakness, pain, skin fixation, nodes). Inspect and milk the duct orifices, and palpate *bimanually* the submandibular gland and floor of mouth (stones are often felt along the duct). Test the facial nerve.

2.2 Imaging

- **Ultrasound:** first line for stones, ductal dilatation, abscess and masses, and for guided aspiration.
- **Plain radiographs:** about 80% of submandibular stones (fewer parotid stones) are radio-opaque; an occlusal film shows duct stones.
- **Non-contrast CT:** detects stones reliably; contrast CT for abscess.
- **Sialography** (conventional or MR sialography): ducts, strictures, sialectasis and radiolucent stones; contraindicated in acute infection.
- **MRI:** the best study for masses.
- **Sialendoscopy:** both diagnosis and treatment.
- **Scintigraphy (technetium pertechnetate):** gland function; uptake in Warthin tumour and oncocytoma.

3 CHAPTER THREE Infections

Viral and bacterial.

3.1 Mumps (epidemic parotitis)

A paramyxovirus spread by droplets, saliva and urine; mostly children of four to six years in unvaccinated populations; incubation two to three weeks, infectious from before the swelling to about a week after.

- **Features:** fever, malaise and muscle pains, then painful swelling of one parotid and usually, within days, the other; sometimes the submandibular glands; swelling settles in about a week.
- **Complications:** orchitis (painful testis, usually one-sided; sterility rare), oophoritis, pancreatitis, aseptic meningitis and encephalitis, *unilateral sensorineural deafness*, sometimes sudden, thyroiditis, myocarditis, nephritis and arthritis.

- **Diagnosis:** clinical; raised serum and urinary amylase; IgM or rising IgG serology, or PCR.
- **Treatment:** supportive: fluids, rest, analgesics, warm or cold compresses, avoiding sour foods; support for orchitis.
- **Prevention:** MMR vaccine.

3.2 Acute suppurative sialadenitis

Usually the parotid, in the elderly, dehydrated and debilitated, after abdominal or hip surgery, and with a dry mouth from drugs: stasis of saliva lets mouth bacteria ascend Stensen's duct. *Staphylococcus aureus* (including MRSA), streptococci and anaerobes.

- Sudden, severe pain and swelling, worse on chewing, fever and toxicity; the duct orifice is red and swollen and pus can be milked from it.
- Treatment: rehydration, intravenous antibiotics covering staphylococci and anaerobes, sialogogues (lemon drops), massage from back to front, oral hygiene. Ultrasound or CT if it does not settle; an abscess (often hard to feel through the fascia) is drained through a parotidectomy incision with radial incisions in the line of the facial nerve branches.



FIGURE 5 Acute suppurative parotitis on the left in a 58-year-old woman.
Dhingra, fig. 45.1

3.3 Neonatal and childhood parotitis

Neonatal suppurative parotitis in premature babies (staphylococcal). *Juvenile recurrent parotitis*: recurrent swelling, often one-sided, in children aged three to ten without pus, with punctate sialectasis on imaging; usually remits at puberty; antibiotics during attacks, and sialendoscopic lavage with steroid.

3.4 Chronic recurrent sialadenitis and sialectasis

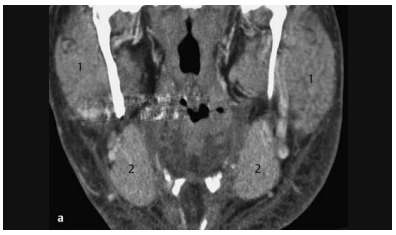
Recurrent painful swelling with pus during attacks and a firm, slightly enlarged gland between them, from duct obstruction, strictures, stones and reduced flow. Sialography shows ductal dilatation (punctate, globular or cavitary *sialectasis*) or strictures. Treat attacks with antibiotics; between them, hydration, oral hygiene, sialogogues and avoiding drying drugs; sialendoscopy to dilate strictures and wash out plugs; gland excision for persistent symptoms. The *Küttner tumour* (chronic sclerosing sialadenitis, part of IgG4-related disease) can mimic a submandibular neoplasm.

3.5 Granulomatous infections

- **Tuberculosis:** of the parotid (often its intraglandular nodes) or submandibular gland, as an acute sialadenitis or a chronic mass mimicking tumour, sometimes with skin changes or sinuses. Diagnosis by FNA (granulomas, acid-fast bacilli, culture) and tuberculin testing; antituberculous treatment.
- **Atypical mycobacteria:** in children of one to three years, a painless mass with a violaceous skin and sinus formation, a negative chest film and weak tuberculin reaction; excision or curettage.
- **Actinomycosis:** after dental infection; a woody swelling with sinuses discharging sulphur granules; prolonged penicillin.
- **Cat-scratch disease** (*Bartonella*) and **toxoplasmosis** cause periparotid and cervical lymphadenitis.

IMAGING IN THE APP 1 case

Sialadenitis on imaging.



Sialadenitis

Infection & inflammation · 11 images

4 CHAPTER FOUR Sialolithiasis

Stones.

4.1 Pathology

Calcium phosphate (hydroxyapatite) deposited on a nidus of mucus or debris. About 80 to 90% occur in the *submandibular* system, because its saliva is more alkaline, viscous and calcium-rich and its duct is long and runs uphill; about 10% in the parotid (often in the gland), rarely the sublingual.

4.2 Features

Pain and swelling of the gland at mealtimes (*salivary colic*), subsiding over an hour or two; recurrent infection with pus at the orifice; a stone may be seen at the papilla or felt along the duct bimanually.

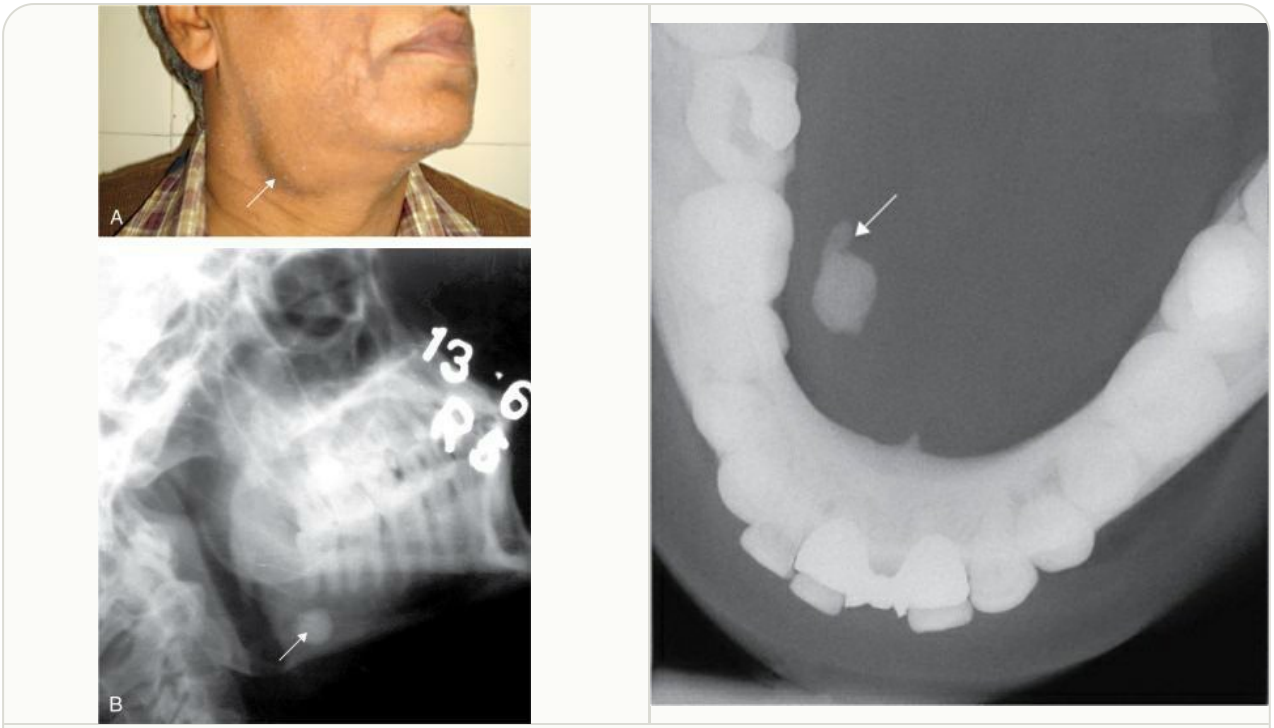


FIGURE 6 First, (A) an adult with submandibular sialadenitis (arrow) and (B) a radio-opaque stone in the submandibular gland on a lateral radiograph. Second, a radio-opaque stone in the right submandibular duct on an occlusal film.

Dhingra, figs 45.2 and 45.3

4.3 Diagnosis

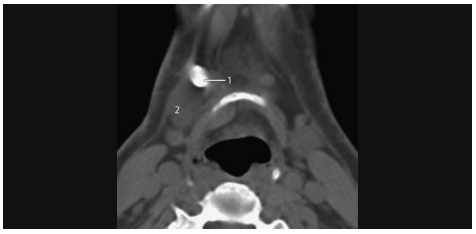
Plain and occlusal films (radio-opaque stones), ultrasound, non-contrast CT, and sialography or MR sialography for radiolucent stones and strictures.

4.4 Treatment

- **Small stones near the orifice:** conservative (hydration, sialogogues, massage) may pass them.
- **Stones in the anterior or middle duct:** intraoral *sialolithotomy*: a stay suture behind the stone to stop it slipping back, an incision over it along the duct, protecting the lingual nerve, and marsupialisation.
- **Sialendoscopy:** small stones (under about 5 mm) retrieved with a basket; larger ones fragmented by laser or extracorporeal lithotripsy; combined endoscopic and transoral approaches remove many hilar stones while keeping the gland.
- **Gland excision:** for stones in the hilum or parenchyma that cannot be reached, and for a gland destroyed by recurrent infection.

IMAGING IN THE APP 1 case

A submandibular sialolith on imaging.



Sialolith of the Submandibular Gland

Infection & inflammation · 5 images

OPERATIVE GUIDES IN THEATRE 2 operations

Sialendoscopy, and excision of the submandibular gland.

Salivary Endoscopy

Sialendoscopy

Excision of the Submandibular Gland

Submandibular gland

5 CHAPTER FIVE Systemic and autoimmune

Bilateral swelling and a dry mouth.

5.1 Sjögren syndrome

An autoimmune exocrinopathy with lymphocytic destruction of the salivary and lacrimal glands, mostly in women.

- **Primary:** dry mouth (xerostomia) and dry eyes (keratoconjunctivitis sicca), with parotid enlargement.
- **Secondary:** sicca features with another connective tissue disease, usually rheumatoid arthritis or lupus.
- **Diagnosis:** Schirmer test, anti-Ro (SS-A) and anti-La (SS-B) antibodies, rheumatoid factor and ANA, and *labial minor salivary gland biopsy* showing focal lymphocytic infiltrates; the gland shows a benign lymphoepithelial lesion (formerly Mikulicz disease).

- **Treatment:** saliva substitutes, frequent sips, sialogogues (pilocarpine, cevimeline), dental care, artificial tears.

LYMPHOMA IN SJÖGREN SYNDROME

The risk of non-Hodgkin lymphoma (usually MALT lymphoma of the parotid) is greatly increased. A new, firm or enlarging mass in a Sjögren gland is biopsied.

5.2 Sarcoidosis

Bilateral painless parotid swelling; *Heerfordt syndrome* (uveoparotid fever) combines parotitis, uveitis, fever and facial palsy. Steroids.

5.3 HIV

Bilateral parotid enlargement from *benign lymphoepithelial cysts* and lymphoid hyperplasia, sometimes the presenting sign of HIV; a sicca picture (diffuse infiltrative lymphocytosis syndrome). Antiretroviral treatment, aspiration or sclerotherapy for cosmetically troublesome cysts; a solid mass needs biopsy because of the risk of lymphoma.

5.4 Sialadenosis (sialosis)

Painless, symmetrical, non-inflammatory enlargement of the parotids from acinar hypertrophy, with diabetes, alcoholism, malnutrition, bulimia, obesity and drugs. Distinguish from other causes of bilateral swelling by FNA or biopsy; treat the cause.

5.5 Necrotising sialometaplasia

An ulcer or swelling of the hard palate (minor salivary glands) after ischaemic injury, mostly in men in their forties, whose pseudoepitheliomatous hyperplasia mimics squamous or mucoepidermoid carcinoma; it heals spontaneously in five to six weeks. Biopsy and wait.

CAUSES OF BILATERAL PAROTID SWELLING

GROUP	EXAMPLES
Infective	Mumps, HIV, recurrent parotitis of childhood
Autoimmune and granulomatous	Sjögren syndrome, sarcoidosis, IgG4-related disease
Metabolic	Sialadenosis (diabetes, alcohol, malnutrition, bulimia)
Neoplastic	Warthin tumour (bilateral in some), lymphoma, acinic cell carcinoma (rarely)
Drugs	Iodides, phenylbutazone and others

6 CHAPTER SIX Dry and wet mouths

Xerostomia and sialorrhoea.

6.1 Xerostomia

From drugs (anticholinergics, antidepressants, antihistamines, diuretics), radiotherapy (irreversible damage above about 25 Gy), Sjögren syndrome, dehydration, diabetes and age. Effects: difficulty chewing, swallowing and speaking, caries, candida and altered taste. Treat with water, saliva substitutes, sugar-free gum, pilocarpine, fluoride and dental care; prevent radiation xerostomia with parotid-sparing IMRT and submandibular gland transfer.

6.2 Sialorrhoea (drooling)

Usually poor oral control rather than excess saliva, in children with cerebral palsy and adults with neurological disease. Positioning and therapy, anticholinergics (glycopyrrolate, hyoscine patches), botulinum toxin into the submandibular and parotid glands, and surgery (submandibular duct relocation or gland excision with parotid duct ligation).

DROOLING IN CHILDREN (OXFORD TEXTBOOK)

Drooling is normal up to about four years and usually stops by then in children without neuromuscular disease. Beyond that it is usually caused by poor oral and pharyngeal control, open lips and poor posture, as in cerebral palsy, rather than by too much saliva. At rest about 70% of saliva comes from the submandibular glands, which is why they are the main target of treatment. Separate **anterior drooling** (saliva spilling from the lips, a social and skin problem) from **posterior drooling** (saliva pooling in the pharynx and aspirated, causing cough, night choking and chest infections), because the choice of surgery depends on it. Severity and progress are measured with scales such as the Teacher's Drooling Scale or by counting bib changes.

Management is stepwise and multidisciplinary: oral motor training, posture and behavioural work; then **glycopyrronium** orally or a **hyoscine patch**; then ultrasound-guided **botulinum toxin** into the submandibular and parotid glands, which works for several months but can cause dysphagia if it spreads. Surgery is considered for persisting problems. **Duct ligation** (two, three or four ducts) is a simple day-case option but can recur. **Submandibular duct relocation** to the tonsillar fossa, often with sublingual gland excision to prevent a ranula, suits anterior drooling but can worsen aspiration and must be avoided in children who aspirate. **Bilateral submandibular gland excision**, with or without parotid duct ligation, is preferred for posterior drooling and aspiration.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Parotid anatomy:** the facial nerve and its landmarks; the course of Stensen's and Wharton's ducts.
- **Mumps** and its complications, including deafness.
- **Acute suppurative parotitis:** who gets it and why.
- **Why stones form in the submandibular gland,** and their management.
- **Sjögren syndrome:** diagnosis and the lymphoma risk.

- **Causes of bilateral parotid swelling.**

7.2 Common pitfalls

COMMON PITFALLS

- Performing sialography during acute infection.
- Calling a firm, persistent parotid lump "chronic parotitis" without imaging and FNA.
- Injuring the lingual nerve when removing a duct stone.

Written for this app from Oxford Textbook of Otolaryngology ch. 82, Oxford Specialist Handbook ch. 7, Oxford Handbook of ENT ch. 11, Dhingra ch. 45 and K. J. Lee ch. 35. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · NECK, THYROID & SALIVARY

Salivary gland tumours

Salivary tumours follow a simple rule of size: the bigger the gland, the more likely its tumour is benign. Four out of five parotid tumours are benign, about half of submandibular tumours, and only a minority of minor salivary and sublingual tumours. Most parotid lumps are pleomorphic adenomas, slow and painless, that must be removed with a cuff of normal gland because they leave seedlings behind if shelled out. Pain, rapid growth, fixity, nodes and above all facial weakness mean malignancy.

AT A GLANCE

- 80% of parotid tumours are benign; most minor gland tumours are malignant.
- Pleomorphic adenoma: pseudopodia, so never enucleate.
- Warthin: tail of parotid, older male smokers, bilateral in some.
- Adenoid cystic carcinoma spreads along nerves and to the lungs.
- Permanent facial weakness about 7% after parotidectomy; Frey syndrome about 20%; Milan system for cytology.

1 CHAPTER ONE Overview

Classification and clues.

1.1 Proportions

- **Parotid:** about 80% of all salivary tumours; about 80% benign.
- **Submandibular:** about 10%; 50 to 60% benign.
- **Minor salivary and sublingual:** the remainder; most minor salivary tumours (and nearly all sublingual tumours) are malignant.

TUMOURS OF THE SALIVARY GLANDS (SIMPLIFIED WHO)

BENIGN	MALIGNANT
Pleomorphic adenoma	Mucoepidermoid carcinoma (low, intermediate, high grade)
Warthin tumour (papillary cystadenoma lymphomatosum)	Adenoid cystic carcinoma

BENIGN	MALIGNANT
Oncocytoma	Acinic cell carcinoma
Basal cell, canalicular adenoma; myoepithelioma	Carcinoma ex pleomorphic adenoma
Haemangioma, lymphangioma (children)	Salivary duct carcinoma, adenocarcinoma, polymorphous adenocarcinoma
Lipoma, neurofibroma, schwannoma	Epithelial–myoepithelial, secretory carcinoma
	Squamous and undifferentiated carcinoma (often metastatic), lymphoma, sarcoma

1.2 Features of malignancy

Rapid growth, pain, fixation to skin or deep structures, ulceration, *facial nerve weakness*, trismus, and cervical nodes. Many malignant tumours, however, present as a painless mobile lump exactly like an adenoma.

1.3 Evaluation

- **Ultrasound** with **fine-needle aspiration** (or core biopsy): about 90% accurate in separating benign from malignant; incisional biopsy of a parotid mass is avoided because it seeds tumour and endangers the nerve.
- **MRI:** the best study for extent, deep lobe and parapharyngeal extension, and perineural spread (especially of adenoid cystic carcinoma along the facial and trigeminal nerves); **CT** for bone erosion; **PET-CT** for high-grade cancers.
- **Facial nerve** function recorded before surgery.

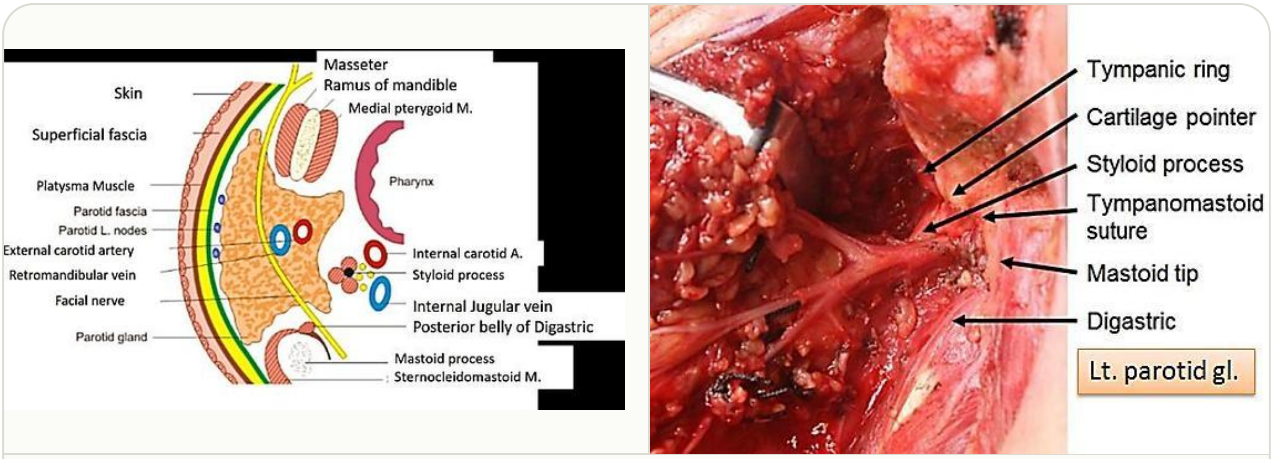


FIGURE 1 The parotid in cross-section with its relations, and the surgical landmarks of the facial nerve at parotidectomy.

Anatomy atlas · ENT Steps p. 182, ENT Steps p. 187

2 CHAPTER TWO Benign tumours

Adenomas and others.

2.1 Pleomorphic adenoma

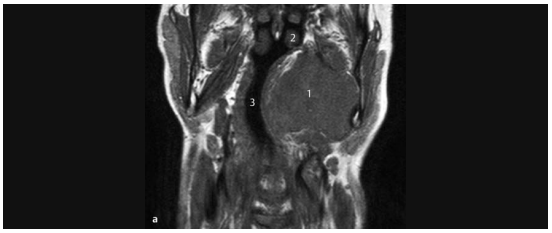
The commonest salivary tumour in adults and children; about 85% in the parotid, usually the superficial lobe or tail; also the submandibular gland and palate. Women more than men, usually in the third to fifth decades. A slow, painless, firm, mobile, lobulated lump that may grow very large; from the deep lobe it can bulge into the parapharyngeal space and push the tonsil and soft palate medially.

- **Pathology:** a *mixed tumour* of epithelial and myoepithelial cells in a myxoid, chondroid or fibrous stroma, within an *incomplete capsule* from which finger-like pseudopodia protrude.
- **Treatment:** *superficial parotidectomy* (or partial superficial parotidectomy) with identification and preservation of the facial nerve, removing the tumour with a cuff of normal tissue; extracapsular dissection is used by some for small, superficial, mobile tumours. *Enucleation* leaves pseudopodia and gives high, often multinodular recurrence; spillage at surgery also seeds recurrence. Deep lobe and parapharyngeal tumours are removed through the neck (occasionally with mandibulotomy), not through the mouth.

- **Malignant change:** *carcinoma ex pleomorphic adenoma* in long-standing or recurrent tumours (risk rising with time, several per cent at ten to fifteen years): sudden growth, pain or facial weakness in an old lump.

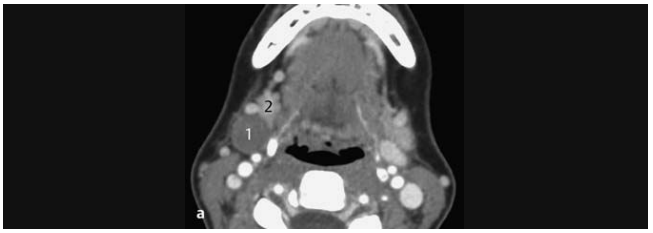
IMAGING IN THE APP 2 cases

Pleomorphic adenoma of the parotid and submandibular glands.



Pleomorphic Adenoma of the Parotid Gland

Tumours · 11 images



Pleomorphic Adenoma of the Submandibular Gland

Tumours · 2 images

2.2 Warthin tumour

Papillary cystadenoma lymphomatosum: the second commonest benign parotid tumour, almost always in the tail of the parotid, in older men who smoke; up to a fifth multifocal and about 5 to 10% bilateral. A soft or cystic, sometimes fluctuating, lump that may become inflamed and painful. Histologically oncocytic epithelium in papillary cysts over a lymphoid stroma; it takes up technetium pertechnetate (a "hot" nodule). Not a true neoplasm and very rarely malignant: removal by partial parotidectomy or extracapsular excision, or observation in the elderly once FNA is confident.

2.3 Oncocytoma

Rare (about 1%), from oncocytes rich in mitochondria, in the elderly, in the superficial parotid, usually under 5 cm; hot on pertechnetate scanning. Superficial parotidectomy.

2.4 Other benign tumours

Basal cell adenoma (can mimic adenoid cystic carcinoma), canalicular adenoma (upper lip), myoepithelioma, lipoma (characteristic on imaging), and neural tumours.

2.5 Tumours in children

Vascular lesions are the commonest. *Haemangioma* of the parotid in infants, usually girls: soft, bluish, enlarging with crying, growing in the first months then involuting by about five to seven years; half have skin haemangiomas; propranolol, surgery only if it fails. *Lymphangioma* does not involute and is excised or sclerosed. Among solid tumours in children, pleomorphic adenoma is the commonest benign one, but up to half are malignant, mostly *mucoepidermoid carcinoma*.

3 CHAPTER THREE Malignant tumours

By type.

3.1 Mucoepidermoid carcinoma

The commonest salivary malignancy in adults and children, usually in the parotid: a mixture of mucous, epidermoid and intermediate cells.

- **Low grade:** more mucous cells, cystic, slow, behaving almost like a benign tumour; five-year survival about 90%.
- **High grade:** more epidermoid cells, solid, aggressive, with nodal spread; survival about 30 to 50%.

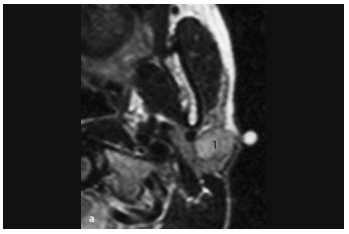
Low-grade tumours: parotidectomy (superficial or total according to site) preserving the nerve. High-grade: total parotidectomy, sacrificing only nerve that is invaded, with neck dissection and postoperative radiotherapy.

3.2 Adenoid cystic carcinoma (cylindroma)

About 10% of salivary cancers, the commonest in the submandibular gland, sublingual gland and minor glands (especially the palate). Slow but relentless: it infiltrates widely along tissue planes and especially *nerves* (perineural spread, causing pain, facial palsy and skip lesions to the skull base), recurs locally decades later, and metastasises to the *lungs* (often with long survival) more than to nodes. Histology: cribriform ("Swiss cheese", best prognosis), tubular, and solid (worst). Treatment: wide excision with clear nerve margins and postoperative radiotherapy for almost all; survival keeps falling beyond five and ten years, so follow-up is lifelong.

IMAGING IN THE APP 1 case

Adenoid cystic carcinoma on imaging.



Adenoid Cystic Carcinoma

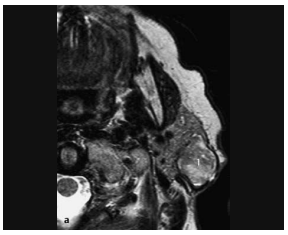
Tumours · 3 images

3.3 Acinic cell carcinoma

Usually in the parotid, sometimes bilateral, mostly in women: a small, firm, mobile lump resembling an adenoma; usually low grade. Parotidectomy; recurrence more often local than regional.

IMAGING IN THE APP 1 case

Acinic cell carcinoma on imaging.



Acinic Cell Carcinoma

Tumours · 3 images

3.4 Carcinoma ex pleomorphic adenoma

A carcinoma (often salivary duct carcinoma) arising in a long-standing or recurrent mixed tumour, up to 10% of salivary cancers: a sudden phase of rapid growth, pain or palsy. Radical surgery and radiotherapy; poor prognosis. (Rarer forms: true carcinosarcoma, and the metastasising pleomorphic adenoma.)

3.5 Salivary duct carcinoma

A high-grade cancer resembling breast ductal carcinoma, in older men, with early nodal and distant spread; often HER2 or androgen receptor positive, which allows targeted treatment of recurrence.

3.6 Others

Polymorphous adenocarcinoma (palate; good prognosis), epithelial–myoepithelial carcinoma, secretory carcinoma (ETV6–NTRK3 fusion, treatable with TRK inhibitors), adenocarcinoma NOS, and undifferentiated carcinoma (rapid, painful, fixed, with facial palsy and nodes).

3.7 Squamous cell carcinoma

Primary salivary squamous carcinoma is rare; most squamous cancer in the parotid is *metastatic* to intraparotid nodes from skin cancer of the scalp, ear, temple and face, which spreads to the parotid in about 5% of high-risk cases. Rapid growth, pain,

ulceration, nodes. Radical parotidectomy with excision of skin, muscle, mandible or temporal bone as needed, neck dissection, and postoperative radiotherapy.



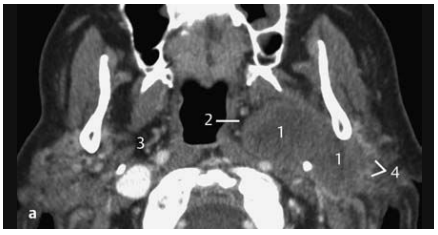
FIGURE 2 Squamous cell carcinoma of the right parotid: (A) a parotid swelling and (B) facial palsy.
Dhingra, fig. 46.1

3.8 Lymphoma and metastases

Primary MALT lymphoma of the parotid (especially in Sjögren syndrome) and secondary nodal lymphoma: biopsy and treat as lymphoma. Melanoma from the scalp and face also metastasises to the parotid.

IMAGING IN THE APP 1 case

Malignancies of the parotid gland.



Malignancies of the Parotid Gland
Tumours · 10 images

4 CHAPTER FOUR Treatment principles

Surgery first.

- **Primary tumour:** complete excision. In the parotid, superficial, total or radical parotidectomy according to site and extent; in the submandibular gland, excision of the gland with the contents of the triangle; minor gland cancers by wide local excision (often partial maxillectomy for the palate).
- **Facial nerve:** preserved whenever it can be separated from tumour, even with malignancy, provided it functions before surgery; a nerve that is paralysed or encased is resected and grafted immediately (great auricular or sural nerve), with mastoidectomy if needed to reach a clear proximal margin.
- **Neck:** therapeutic dissection (levels I to V) for clinical nodes; elective dissection for high-grade tumours, T3 and T4 tumours, extraglandular extension and facial nerve involvement.
- **Postoperative radiotherapy:** high-grade histology, close or positive margins, perineural invasion, extraglandular extension, nodes, and recurrent tumours; neutrons and protons have special roles.
- **Systemic treatment:** for metastatic disease; targeted agents for HER2, androgen receptor and NTRK-driven tumours.

T STAGING OF MAJOR SALIVARY GLAND CANCER (AJCC)

CATEGORY	DEFINITION
T1	2 cm or less, no extraparenchymal extension
T2	Over 2 cm up to 4 cm, no extraparenchymal extension
T3	Over 4 cm, or extraparenchymal extension
T4a	Skin, mandible, ear canal or facial nerve
T4b	Skull base, pterygoid plates, or encasing the carotid

5 CHAPTER FIVE Parotidectomy

The operation and its sequelae.

5.1 Technique

A modified Blair (lazy S, preauricular-cervical) incision or a facelift incision; a skin flap raised superficial to the parotid fascia; the great auricular nerve identified (its posterior branch preserved where possible); the gland mobilised off the ear cartilage and sternomastoid; the *facial nerve trunk* found using the tragal pointer, tympanomastoid suture and posterior belly of digastric, then followed forwards along its branches, lifting the superficial lobe off them. Nerve monitoring is widely used. Retrograde dissection from a peripheral branch helps in revision surgery.

OPERATIVE GUIDES IN THEATRE 2 operations

Superficial and deep lobe parotidectomy.

Superficial Parotidectomy
Superficial parotidectomy

Deep Lobe Parotidectomy
Deep lobe parotidectomy

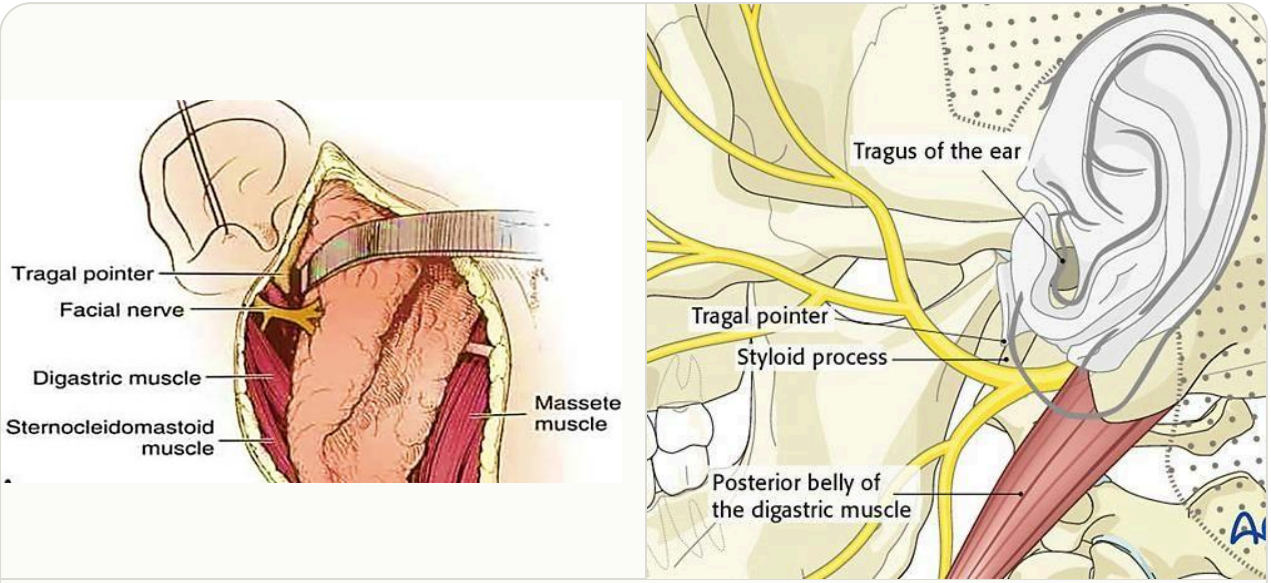


FIGURE 3 Landmarks for the facial nerve trunk: tragal pointer, styloid process and posterior belly of digastric.

Anatomy atlas · ENT Steps p. 185, ENT Steps p. 186

5.2 Complications

- **Facial nerve weakness:** temporary in a fifth to a third (especially the marginal mandibular branch), permanent in under 1 to 2% in benign surgery.
- **Numbness** of the ear lobe (great auricular nerve).

- **Frey syndrome (gustatory sweating):** sweating and flushing of the cheek skin while eating, months after surgery, because cut parasympathetic secretomotor fibres to the parotid regenerate into the sympathetic nerves of the sweat glands. Confirmed by the starch–iodine (Minor) test. Most need only reassurance; botulinum toxin injected into the skin is effective; an interposed barrier (SMAS flap, sternomastoid flap, acellular dermis, fat) reduces it; tympanic neurectomy is an old remedy.
- **Sialocele and salivary fistula** (usually settle with pressure, aspiration or botulinum toxin), haematoma, infection, a hollow contour, and tumour recurrence.

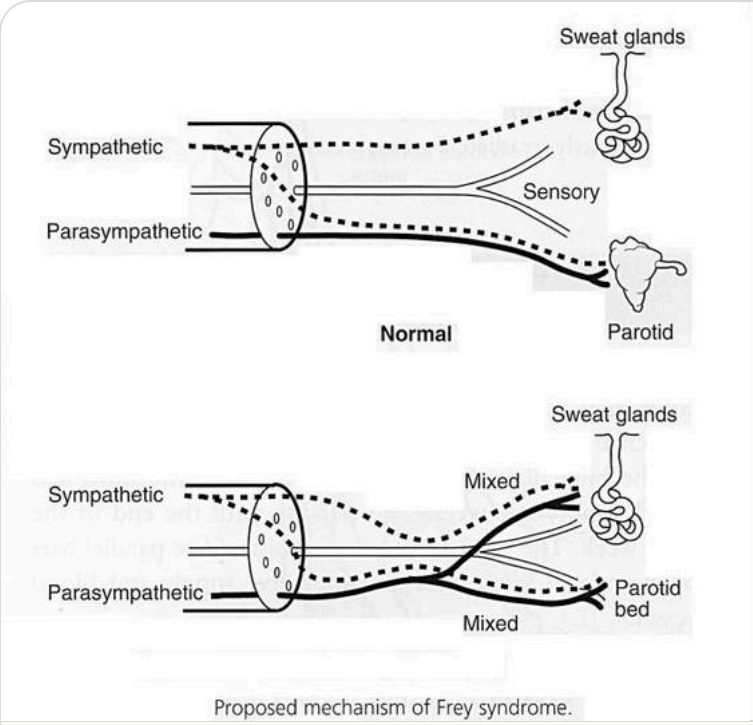


FIGURE 4 The mechanism of Frey syndrome.

Anatomy atlas · ENT Steps p. 188

NUMBERS FOR CONSENT (OXFORD TEXTBOOK)

The parotid holds 70 to 85% of salivary tumours, of which about 20% are malignant; the submandibular gland 8 to 15% (about 40% malignant); and minor and sublingual glands the rest, with about 80% malignant. Malignancies are graded low, intermediate or high, and high grade predicts local recurrence and distant spread. Fine-needle aspiration has a sensitivity and specificity approaching 90%, but up to 30% of results are indeterminate, which is where the **Milan system** for reporting risk, and core biopsy, help.

After parotidectomy, temporary facial weakness occurs in up to about a third (lower, about 22%, with continuous nerve monitoring) and permanent weakness in about 7%, which monitoring does not change. **Frey syndrome** (gustatory sweating and flushing from misdirected parasympathetic fibres) follows about 20% of superficial

parotidectomies; **first bite syndrome**, a sharp pain with the first mouthful of each meal from loss of sympathetic supply, follows deep lobe and parapharyngeal surgery. For the submandibular gland, the incision is placed 3 to 4 cm below the mandible, and the marginal mandibular nerve is protected by raising the fascia over the gland (the Hayes Martin manoeuvre) or by finding the nerve directly. A modified Blair incision is used whenever a neck dissection may be needed.

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The rule of glands:** larger gland, more likely benign.
- **Pleomorphic adenoma:** pathology, why not to enucleate, superficial parotidectomy, malignant change.
- **Warthin tumour:** who, where, bilaterality, pertechnetate.
- **Mucoepidermoid carcinoma** grades; **adenoid cystic carcinoma** and perineural spread.
- **Features of malignancy;** FNA versus open biopsy.
- **Parotidectomy:** facial nerve landmarks, complications, Frey syndrome.

6.2 Common pitfalls

COMMON PITFALLS

- Enucleating a pleomorphic adenoma.
- Doing an open biopsy of a parotid lump.
- Removing a parapharyngeal deep lobe tumour through the mouth.
- Sacrificing a working facial nerve that could be dissected free.

Written for this app from Oxford Textbook of Otolaryngology ch. 58, Oxford Specialist Handbook ch. 7, Oxford Handbook of ENT ch. 11, Dhingra ch. 46 and K. J. Lee ch. 35. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · NECK, THYROID & SALIVARY

Paragangliomas, neural tumours and vascular anomalies

A slow-growing, painless lump high in the neck that moves from side to side but not up and down, sometimes with a bruit, is a carotid body tumour until the scan says otherwise. Never needle it. Paragangliomas are often hereditary: every patient is tested for SDH mutations and screened for catecholamine secretion. In children, the vascular anomalies are sorted by one question. A haemangioma is a tumour that grows after birth and then involutes. A malformation is present at birth and grows with the child.

AT A GLANCE

- Carotid body tumour splays the internal and external carotids (lyre sign).
- Up to 40% of paragangliomas are hereditary (SDH).
- Vagal tumours push the carotids forward without splaying.
- Haemangiomas involute; malformations grow with the child.

1 CHAPTER ONE Paragangliomas

Tumours of the paraganglia.

1.1 Biology

Paragangliomas arise from the neural-crest *paraganglia*, the chemoreceptor tissue of the carotid body, the vagal ganglia, the jugular bulb and the middle ear. Histologically they are nests of *chief (type I) cells* surrounded by *sustentacular (type II) cells*, the pattern called *Zellballen* ("balls of cells"), in a highly vascular stroma. The carotid body is the commonest site in the head and neck, accounting for about half to 60%, followed by the jugular foramen (glomus jugulare), the middle ear (glomus tympanicum) and the vagus (glomus vagale, about 5 to 10%).

Up to 40% are hereditary, mostly from mutations in the *succinate dehydrogenase* genes SDHB, SDHC, SDHD and others. Hereditary tumours appear at a younger age, are more often multiple and bilateral (about a third), and come with phaeochromocytomas; SDHB carriers have the highest risk of malignancy. *Genetic*

testing is therefore offered to every patient. Chronic high-altitude hypoxia, by contrast, causes carotid body hyperplasia.

About 5 to 10% of carotid tumours, and more vagal tumours, are malignant, and malignancy is defined only by metastasis to nodes or distant sites, not by histology or local invasion. Only 1 to 3% secrete catecholamines, but everyone is screened with *plasma free metanephrines* or 24-hour urinary fractionated metanephrines, both to find a functioning tumour and to find a phaeochromocytoma before anaesthesia.

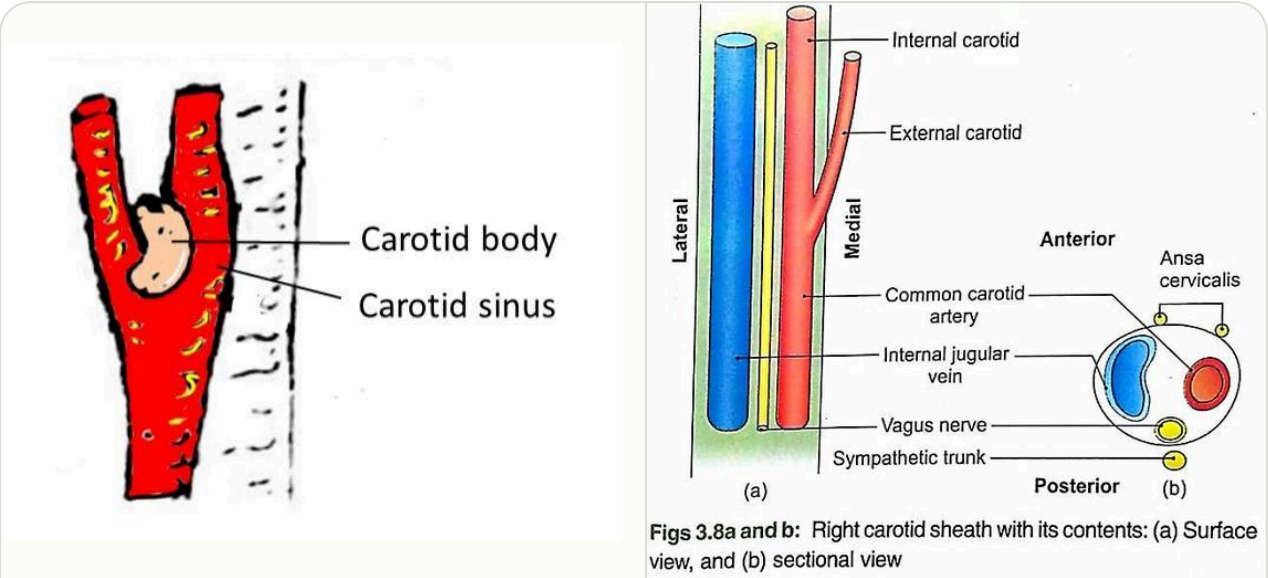


FIGURE 1 The carotid body at the bifurcation, and the carotid sheath and its contents.
Anatomy atlas · Doss p. 137, ENT Steps p. 266

1.2 Carotid body tumour

The carotid body tumour is a painless, slowly enlarging mass at the *carotid bifurcation*, at the anterior border of the sternomastoid at the level of the hyoid. It grows about 1 mm a year, doubling in four to five years. It moves from side to side but *not vertically* (*Fontaine's sign*), transmits or has its own pulsation, may have a bruit, and may bulge into the oropharynx. Cranial nerve palsies of X, XII or the sympathetic chain are uncommon before surgery, and pain or rapid growth suggests malignancy.

Contrast CT or MRI shows an intensely enhancing mass *splaying the internal and external carotids* apart. On MRI it has a "*salt-and-pepper*" pattern, haemorrhage providing the salt and flow voids the pepper, and angiography shows the *lyre sign* of splayed carotids around a vascular blush. Functional imaging with Ga-68 DOTATATE

PET looks for multiple tumours, and if carotid sacrifice may be needed, the circle of Willis and collateral flow are assessed by balloon occlusion testing.

NO NEEDLE, NO BIOPSY

FNA of a suspected carotid body tumour is avoided, and open or core biopsy condemned. They bleed, they rarely give a diagnosis, and imaging is characteristic.

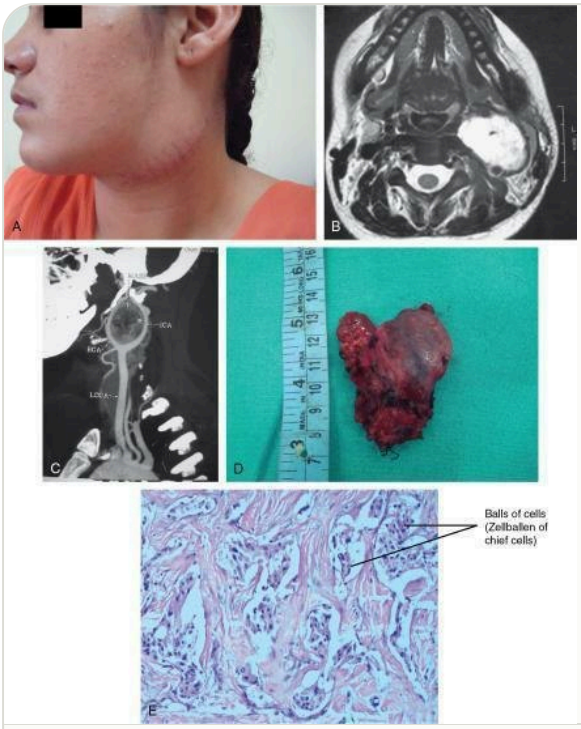


FIGURE 2 Carotid body tumour: (A) clinical swelling; (B) MRI; (C, D) angiography showing splaying of the internal and external carotids (lyre sign) with tumour blush; (E) the excised specimen. Dhingra, fig. 77.6

SHAMBLIN CLASSIFICATION

GROUP	RELATION TO THE CAROTIDS	SURGICAL MEANING
I	Small, localised, easily separated from the vessels	Low risk
II	Adherent to or partly surrounding the vessels	Careful subadventitial dissection; nerve injury more likely
III	Completely encasing the carotids	May need carotid resection and grafting; highest risk of stroke and nerve injury

Surgery is the only cure and is preferred for younger, fit patients and for tumours that are growing, secreting or causing symptoms. Older or frail patients with small,

stable tumours can be observed with serial MRI. Conventional or stereotactic radiotherapy halts growth in about 90% of unresectable tumours, unfit patients and some bilateral cases, though the tumour does not disappear.

1. Transcervical approach, beginning with a level II to III dissection.
2. Control of the common, internal and external carotids.
3. Identification of the vagus, hypoglossal and superior laryngeal nerves and the sympathetic chain.
4. Dissection in the periadventitial plane (the "white line"), working from below upwards.
5. The external carotid may be ligated.
6. A vascular surgeon should be available for reconstruction.

Some surgeons embolise large tumours beforehand, accepting a risk of stroke and nerve palsy. After surgery, cranial nerve deficits of the vagus, hypoglossal and sympathetic chain are common with large tumours, though many recover, and stroke is the feared complication. *First bite syndrome*, a sharp parotid pain with the first bites of a meal, follows loss of the sympathetic supply to the parotid. Resection of both carotid bodies causes *baroreflex failure* with labile blood pressure and tachycardia, so bilateral tumours are staged, or one is observed.

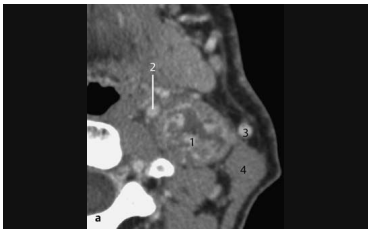
OPERATIVE GUIDES IN THEATRE 1 operation

Excision of carotid body tumours.

Excision of Carotid Body Tumors
Carotid body tumour

IMAGING IN THE APP 1 case

Carotid body paraganglioma.



Carotid Body Paraganglioma
Tumours · 13 images

1.3 Vagal paraganglioma

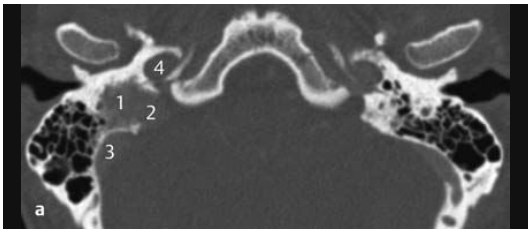
The vagal paraganglioma usually arises from the inferior (nodose) ganglion, high in the parapharyngeal space. It displaces both carotids *anteromedially* without splaying them, separates the internal jugular vein from the carotid, and may extend to the jugular foramen and intracranially. About a third of patients already have vagal weakness and hoarseness, and malignancy is commoner than with carotid tumours. Because surgery almost always sacrifices the vagus, small tumours with normal nerve function are observed or irradiated, and surgery is kept for growing tumours and for those that have already paralysed the nerve.

1.4 Jugular and tympanic paragangliomas

Glomus jugulare and tympanicum tumours present with *pulsatile tinnitus*, conductive hearing loss and a red mass behind the drum, the *rising sun* sign, which blanches on pneumatic pressure (*Brown's sign*), and jugulare tumours may cause lower cranial nerve palsies, the *jugular foramen syndrome*. They are classified by Fisch or by Glasscock–Jackson and treated by surgery through transcanal, mastoid or infratemporal approaches, by radiotherapy, or by observation.

IMAGING IN THE APP 1 case

Glomus jugulare paraganglioma.



Glomus Jugulare Tumor (Paraganglioma)

Tumours · 11 images

OPERATIVE GUIDES IN THEATRE 1 operation

Glomus tumours.

Glomus Tumors

Glomus tumours

2 CHAPTER TWO Neural tumours

Schwannomas and neurofibromas.

2.1 Schwannoma (neurilemmoma)

The schwannoma is a benign, encapsulated tumour of Schwann cells that grows *eccentrically* on its nerve, so the fascicles can often be separated from it. Histology shows cellular *Antoni A* areas with palisading *Verocay bodies* and loose *Antoni B* areas.

Most head and neck schwannomas arise in the *parapharyngeal space*, where they are the second commonest tumour after salivary tumours, usually from the vagus or the cervical sympathetic chain, but also from IX, XI, XII and the cervical and brachial plexuses. They present as a slow, painless lateral neck or parapharyngeal mass that moves from side to side but not vertically, and pressure may send pain along the nerve. The nerve of origin usually still works.

On imaging they enhance less intensely than a paraganglioma, have *no flow voids*, and show a target sign on T2. A *vagal* schwannoma pushes the carotids forward and separates the carotid from the jugular vein, while a *sympathetic* schwannoma pushes both carotids forward without separating the vessels. They are removed by intracapsular enucleation preserving the fascicles, but deficits remain common afterwards: hoarseness after vagal tumours, and *Horner syndrome* and first bite syndrome after sympathetic tumours. Small tumours in older patients can be observed.

2.2 Neurofibroma

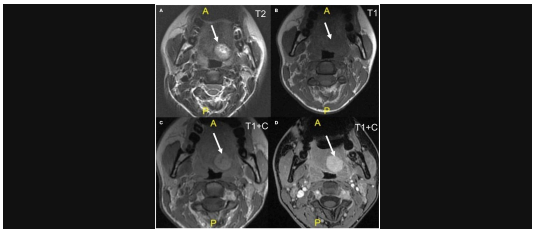
The neurofibroma is unencapsulated and intertwined with its nerve, so the nerve has to be sacrificed to remove it. Multiple and plexiform neurofibromas are the hallmark of neurofibromatosis type 1, and rapid growth or pain suggests malignant change to a *malignant peripheral nerve sheath tumour*.

DISTINGUISHING PARAPHARYNGEAL AND CAROTID SPACE MASSES

FEATURE	CAROTID BODY TUMOUR	VAGAL PARAGANGLIOMA	VAGAL SCHWANNOMA	SYMPATHETIC SCHWANNOMA	PLEOMORPHIC ADENOMA (PRESTYLOID)
Position of the carotids	Splayed apart at the bifurcation	Pushed anteromedially, not splayed	Pushed anteriorly	Pushed anteriorly, vein not separated	Pushed posteriorly
Flow voids	Yes	Yes	No	No	No
Nerve deficit before surgery	Rare	Vagal in about a third	Uncommon	Horner occasionally	None

IMAGING IN THE APP 1 case

Schwannomas of the oral cavity and neck.



Schwannomas of the oral cavity and neck

After surgery · 3 images

OPERATIVE GUIDES IN THEATRE 2 operations

Excision of neural tumours, and tumours of the prestyloid parapharyngeal space.

Excision of Neural Tumors of the Parapharyngeal Space

Neural tumours

Management of Tumors in the Prestyloid Parapharyngeal Space

Parapharyngeal space

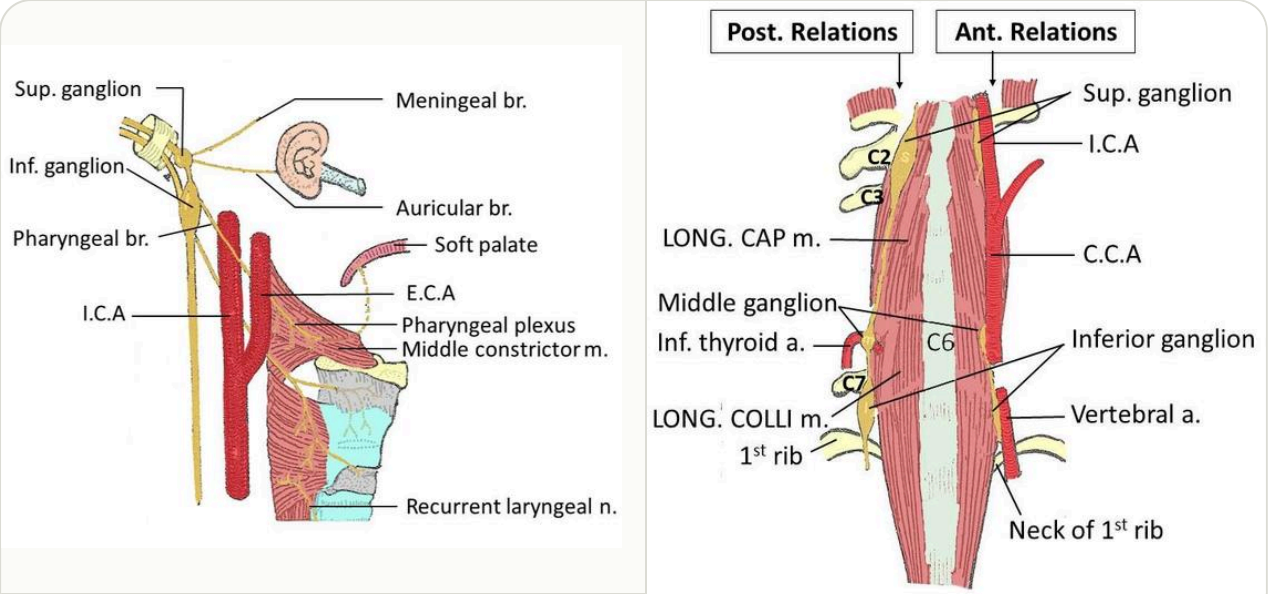


FIGURE 3 The vagus nerve and the cervical sympathetic chain in the neck.
Anatomy atlas · Doss p. 153, Doss p. 160

3 CHAPTER THREE Vascular anomalies

Tumours and malformations.

3.1 Classification (ISSVA)

HAEMANGIOMA VERSUS VASCULAR MALFORMATION

FEATURE	INFANTILE HAEMANGIOMA	VASCULAR MALFORMATION
Nature	True tumour (endothelial proliferation)	Error of vessel development
At birth	Absent or a faint precursor mark	Present (may not be visible)
Course	Proliferates for about a year, then involutes over years	Grows in proportion with the child; never involutes; growth spurts at puberty, pregnancy, infection, trauma
Marker	GLUT-1 positive	GLUT-1 negative

FEATURE	INFANTILE HAEMANGIOMA	VASCULAR MALFORMATION
Treatment	Observation; propranolol	Laser, sclerotherapy, embolisation, surgery according to type

3.2 Infantile haemangioma

The infantile haemangioma is the commonest tumour of infancy, affecting about 4 to 5% of infants, more often girls and premature or low-birth-weight babies. It appears in the first weeks, *proliferates* rapidly to about 9 to 12 months, and then *involutes* slowly, largely by five to ten years, often leaving fibrofatty tissue and telangiectasia. It may be superficial (the bright red "strawberry"), deep and bluish, or mixed, and focal, multifocal or segmental. **Congenital haemangiomas** are fully grown at birth and involute rapidly (RICH), partially (PICH) or not at all (NICH).

Several associations matter. A haemangioma in a *beard distribution* is accompanied by a subglottic haemangioma in about two-thirds of children, with biphasic stridor appearing at about two to three months. A *segmental facial lesion* raises the possibility of PHACES syndrome (posterior fossa anomalies, haemangioma, arterial anomalies, cardiac and aortic anomalies, eye anomalies and sternal clefts), assessed with MRI and echocardiography. *Five or more skin lesions* prompt a search for hepatic haemangiomas. The *Kasabach–Merritt phenomenon* of platelet trapping and coagulopathy belongs to kaposiform haemangioendothelioma and tufted angioma, not to the common haemangioma.

Most haemangiomas are left to involute. Treatment is needed when the airway is threatened, when a periocular lesion threatens vision through amblyopia, for ulceration and bleeding, for lesions of the lip, nose and ear that will deform, for large facial lesions, and for high-output heart failure. *Propranolol*, at about 2 mg/kg/day with monitoring for hypoglycaemia, bradycardia and bronchospasm, is first line; topical timolol suits small superficial lesions; steroids are now second line; the pulsed-dye laser treats ulceration and residual telangiectasia; and surgery is used for residual deformity and for selected subglottic and localised lesions.

3.3 Vascular malformations

The **capillary malformation**, or port-wine stain, is a flat pink to purple patch present at birth, often in a trigeminal distribution, which darkens and thickens into "cobblestones" with age. Involvement of V1 raises the question of *Sturge–Weber syndrome*, with leptomeningeal angiomatosis, seizures and glaucoma. It is treated with the pulsed-dye laser.

The **venous malformation** is a soft, bluish, compressible mass that swells when dependent or with a Valsalva manoeuvre. Imaging shows *phleboliths*, thrombosis makes it painful, and large lesions raise the D-dimer. It is treated by sclerotherapy with ethanol, sodium tetradecyl sulfate or bleomycin, by the Nd:YAG laser, or by excision of localised lesions, while compression and head elevation ease symptoms. The **lymphatic malformation**, macrocystic or microcystic, is described with the congenital neck masses.

The **arteriovenous malformation** is a high-flow lesion with a *nidus*: warm and pulsatile, with a thrill and bruit. It is quiescent in childhood and expands at puberty, in pregnancy or after trauma, causing ulceration, bleeding, bone destruction and heart failure, and is staged by *Schobinger* as quiescence, expansion, destruction and decompensation. Treatment is superselective embolisation followed within days by complete surgical excision of the nidus with reconstruction. *The feeding vessels must never be ligated proximally*: ligation does not cure, it recruits collaterals, and it blocks future embolisation. An **arteriovenous fistula**, usually traumatic with one or two connections, is closed endovascularly or surgically.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Carotid body tumour:** Fontaine sign and bruit; splaying, the lyre sign and salt-and-pepper MRI; no biopsy; metanephrines and SDH genetics; Shamblin groups;

complications including first bite syndrome and baroreflex failure.

- **Vagal paraganglioma versus schwannoma:** vessel displacement and flow voids.
- **Haemangioma versus malformation:** natural history and GLUT-1; propranolol; subglottic haemangioma with beard-distribution lesions.
- **AVM:** embolise and excise; never ligate proximally.

4.2 Common pitfalls

COMMON PITFALLS

- Needling a pulsatile mass at the bifurcation.
- Operating on a functioning paraganglioma, or on a patient with an undiagnosed phaeochromocytoma, without alpha blockade.
- Resecting both carotid body tumours at one sitting.
- Calling a venous malformation a "haemangioma" and waiting for it to involute.

Written for this app from Oxford Textbook of Otolaryngology ch. 67, Oxford Specialist Handbook ch. 12, Dhingra ch. 77 and K. J. Lee ch. 40. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · NECK, THYROID & SALIVARY

Radiotherapy and chemotherapy for head and neck cancer

Most patients with head and neck cancer receive radiotherapy at some point: alone for early tumours, with cisplatin for advanced ones, after surgery when the pathology shows risk, or for palliation. The surgeon needs to know enough radiation biology and chemotherapy to choose the right treatment, prepare the patient (teeth first), recognise the toxicities that come back to the ENT clinic, and know when a dry mouth, a stiff jaw or an exposed bone is the radiotherapy and when it is the cancer returning.

AT A GLANCE

- Conventional 2 Gy a day; 70 Gy for gross disease.
- Positive margins or extranodal extension: add cisplatin.
- Cisplatin is ototoxic: baseline audiogram.
- Osteoradionecrosis: prevent by dental care before treatment.

1 CHAPTER ONE Radiation basics

Physics and biology.

1.1 Radiation and dose

Cancer is treated with *ionising* radiation of short wavelength and high energy: photons, either X-rays from machines or gamma rays from isotopes such as cobalt-60, and particles such as electrons, protons, neutrons and carbon ions. Dose is measured in **grays**; 1 Gy is 1 joule absorbed per kilogram and equals 100 rad.

Most treatment uses megavoltage photons of 4 to 25 MV from a *linear accelerator*, which spare the skin because the maximum dose builds up below the surface. **Electrons** deposit their dose near the surface and fall off sharply, which suits the skin, lip and superficial nodes overlying the spinal cord. **Protons** deposit most of their energy at a set depth, the *Bragg peak*, with almost no exit dose, which suits the skull base, orbit and brainstem, re-irradiation and children. **Neutrons and carbon**

ions have a high linear energy transfer and a role in salivary and radioresistant tumours.

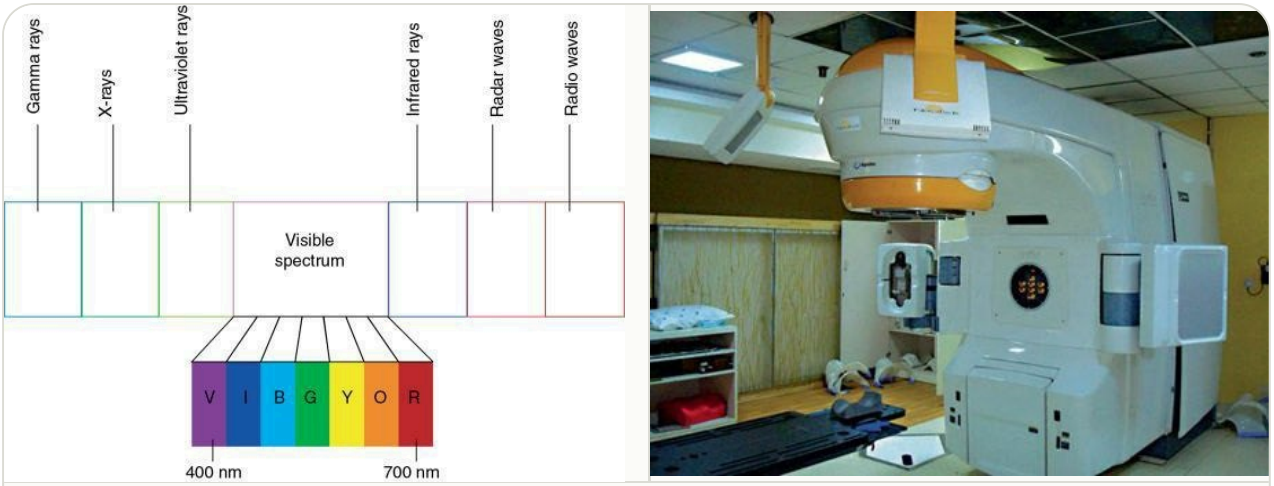


FIGURE 1 First, the electromagnetic spectrum. Second, a modern linear accelerator. Dhingra, figs 73.1 and 73.2

1.2 How radiation kills

Radiation damages DNA, mostly indirectly, by ionising water into free radicals that cause *double-strand breaks*; cells then die when they try to divide, or by apoptosis. Oxygen "fixes" the damage, so hypoxic cells are two to three times more resistant. That is why anaemia is corrected to a haemoglobin of about 12 g/dL, smoking is stopped, and hypoxic sensitisers such as nimorazole have been tried.

Fractionation works because of the **four R's**. Between fractions, normal tissues *repair* sublethal damage better than tumour cells; tumour cells *reoxygenate*; and cells *redistribute* into sensitive phases of the cycle. Against this, tumour *repopulation* accelerates after about four weeks, which is why gaps in treatment are harmful and overall treatment time matters.

The biological effect depends on dose per fraction as well as total dose, as described by the linear-quadratic model, $BED = nd(1 + d/(\alpha/\beta))$. Late-reacting tissues, with a low α/β of about 3, are spared by small fractions, whereas tumours and acutely reacting tissues, with an α/β of about 10, are not.

1.3 Fractionation

Conventional fractionation gives 1.8 to 2 Gy a day, five days a week. About 50 Gy sterilises microscopic disease such as the elective neck, 60 Gy is given after surgery, 66 Gy for close or positive margins, and 70 Gy in 35 fractions over seven weeks for gross tumour.

Hyperfractionation gives smaller fractions of 1.1 to 1.2 Gy twice daily to a higher total of 74 to 80 Gy, improving local control with fewer late effects but more acute toxicity. **Accelerated fractionation**, with six fractions a week or a concomitant boost, shortens overall time to counter repopulation, and helps mainly patients treated without chemotherapy. **Hypofractionation**, with fractions of 2.5 Gy or more, is used for palliation, frail patients and stereotactic treatment.

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2 CHAPTER TWO Delivery

From mask to beam.

2.1 Planning

1. **Immobilisation:** a thermoplastic mask, with a mouth bite to displace the tongue.
2. **Planning CT,** fused with MRI or PET.
3. **Contouring:** the gross tumour volume (GTV); the clinical target volume (CTV), adding microscopic spread and nodal levels; the planning target volume (PTV), a margin for set-up error; and the *organs at risk*: spinal cord, brainstem, parotids, optic apparatus, cochlea, larynx, mandible and constrictors.
4. **Dose calculation** on a planning system.
5. **Daily image verification.**

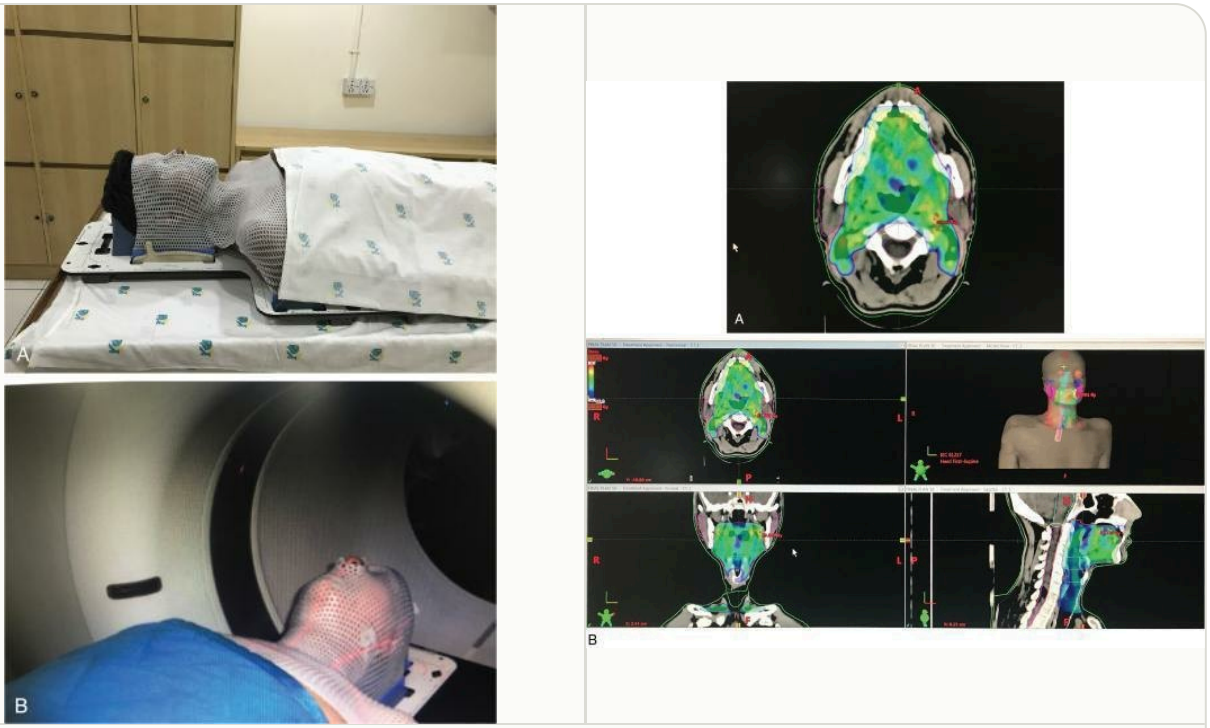


FIGURE 2 First, a thermoplastic immobilisation mask. Second, contouring of target volumes and organs at risk on the planning scan.

Dhingra, figs 73.3 and 73.4

2.2 Techniques

Conventional two-dimensional treatment with parallel opposed lateral fields is now historical, replaced first by CT-based **three-dimensional conformal radiotherapy** with several shaped beams and then by **intensity-modulated radiotherapy (IMRT)**, the current standard. IMRT uses inverse planning with many beams or arcs (VMAT), whose intensity is modulated by moving multileaf collimators, to "curve" the dose around the spinal cord and parotids and to give different doses to different targets at the same time, the *simultaneous integrated boost*. *Parotid-sparing IMRT* reduces permanent xerostomia.

Image-guided radiotherapy images the patient on the machine each day with cone-beam CT, correcting set-up errors and allowing smaller margins, and **adaptive radiotherapy** re-plans during treatment as the tumour shrinks and the patient loses weight. **Stereotactic radiosurgery and body radiotherapy** deliver very precise high doses in one to five sessions, for vestibular schwannoma (12 to 13 Gy in one session), paraganglioma, small recurrences and re-irradiation.

2.3 Brachytherapy

Brachytherapy places a sealed source in or against the tumour, interstitially or within a cavity, giving a high local dose that falls off rapidly. Iridium-192 is used with high-dose-rate remote afterloading, replacing caesium-137, radium and gold seeds. It treats cancers of the lip, oral tongue and floor of mouth and the nasopharynx, as a boost or for recurrence, and boosts external beam treatment.



FIGURE 3 First, interstitial brachytherapy needles for cancer of the lip. Second, brachytherapy of the tongue (A–C).
Dhingra, figs 73.5 and 73.6

3 CHAPTER THREE Radiotherapy in practice

When and why.

3.1 Roles

Definitive radiotherapy preserves the organ. Alone, it treats early T1 to T2 N0 tumours of the larynx (with control around 90% for T1 glottic cancer), oropharynx, nasopharynx and hypopharynx; with concurrent chemotherapy, it treats advanced tumours.

Postoperative radiotherapy starts four to six weeks after surgery, and the whole interval from surgery to the end of radiotherapy should be under about 11 weeks. It is indicated for close (under 5 mm) or positive margins, extranodal extension, two or more positive nodes or a node over 3 cm, pT3 and pT4 tumours, perineural or lymphovascular invasion, nodes in levels IV or V from oral or oropharyngeal cancer, and poor differentiation. *Positive margins and extranodal extension* are the indications for adding *concurrent cisplatin*.

Preoperative radiotherapy is now rarely used, because it increases wound complications and fistulae, and **palliative radiotherapy** uses short courses for pain, bleeding, fungation and obstruction.

PRE- VERSUS POSTOPERATIVE RADIOTHERAPY

PREOPERATIVE	POSTOPERATIVE
Shrinks borderline tumours and sterilises the edges	Surgery done on unirradiated tissue: easier, better healing
Oxygenated, undisturbed tumour bed	Dose tailored to pathology (margins, nodes); a higher dose possible
Smaller fields	Larger fields; scarred, hypoxic tumour bed
More fistulae, flap failures, carotid blow-out	Delay from surgical complications allows regrowth

3.2 Site by site

In the **larynx**, early glottic cancer is treated with radiotherapy alone or transoral laser surgery, and advanced cancer with concurrent cisplatin chemoradiation for larynx preservation, as established by RTOG 91-11; total laryngectomy is reserved for T4a disease with cartilage destruction or a non-functioning larynx. In the **hypopharynx**, larynx preservation is possible but less often successful.

In the **nasopharynx**, radiotherapy is the primary treatment, covering levels II and V and the retropharyngeal nodes; T1N0 disease is treated with radiotherapy alone and more advanced disease with cisplatin chemoradiation, often with induction or adjuvant chemotherapy, and plasma EBV DNA guides follow-up. In the **oropharynx**, radiotherapy or chemoradiation competes with transoral surgery followed by adjuvant treatment. HPV-positive disease has a much better outlook, with three-

year survival of 82% against 57% in RTOG 0129, but de-escalation is done only within trials.

Oral cavity cancer is primarily surgical, with adjuvant radiotherapy according to the pathology. **Salivary cancers** are operated on and then irradiated for high grade, positive margins, nodes or perineural spread; for adenoid cystic carcinoma the named nerve is covered to the skull base.

After chemoradiation, the **neck** is assessed with a PET-CT at 12 weeks, and dissection is done only for PET-positive residual disease, as shown by the PET-NECK trial. For **recurrence**, salvage surgery is preferred; re-irradiation, often hyperfractionated with chemotherapy or delivered by SBRT or protons, cures a minority at a high risk of toxicity, and the best candidates have a second primary many years after the first treatment.

TYPICAL DOSES

SETTING	DOSE
Elective neck (microscopic disease)	50 to 54 Gy
Postoperative, clear margins	60 Gy in 30 fractions
Postoperative, close or positive margins, extranodal extension	66 Gy in 33 fractions
Gross disease (definitive)	70 Gy in 35 fractions over 7 weeks
Early glottic cancer (T1)	63 Gy in 28 fractions, or 66 Gy in 33
Vestibular schwannoma (radiosurgery)	12 to 13 Gy single fraction

4 CHAPTER FOUR Chemotherapy and drugs

Partners to radiation.

4.1 Goals

Chemotherapy is given to sensitise the tumour to radiation and so improve locoregional control, to treat micrometastases, to preserve organs, to palliate recurrent or metastatic disease, and to debulk tumours before surgery or radiotherapy. On its own, it does not cure squamous carcinoma of the head and neck.

4.2 How it is given

Concurrent chemoradiation is the standard for locally advanced disease. The MACH-NC meta-analysis showed an absolute survival gain of about 6.5% at five years, all of it from concurrent treatment. *Cisplatin*, at 100 mg/m² every three weeks or 40 mg/m² weekly, is the drug of choice, and adding other drugs adds toxicity rather than cure.

Induction chemotherapy with *TPF* (docetaxel, cisplatin and 5-FU), which is better than PF, is used to select patients for larynx preservation and for bulky nodal disease, but it has not consistently beaten chemoradiation alone (DeCIDE, PARADIGM). **Adjuvant chemotherapy** after definitive treatment brings no benefit, except in nasopharyngeal cancer. **Postoperative chemoradiation** is given for positive margins and extranodal extension, on the evidence of the EORTC 22931 and RTOG 9501 trials.

4.3 Targeted therapy and immunotherapy

Cetuximab is a monoclonal antibody against EGFR, which is overexpressed in over 90% of these tumours. Combined with radiotherapy it improved survival over radiotherapy alone in the Bonner trial, at the cost of an acneiform rash and infusion reactions, and it is now used for patients unfit for cisplatin. With platinum and 5-FU it was the first-line treatment for metastatic disease in the EXTREME regimen.

The **PD-1 checkpoint inhibitors** nivolumab and pembrolizumab improved survival in recurrent or metastatic disease after platinum (CheckMate 141, KEYNOTE-040), and pembrolizumab, alone for PD-L1-positive tumours or with platinum and 5-FU, is now first-line treatment for recurrent or metastatic disease (KEYNOTE-048). Their immune-related side effects include thyroiditis, pneumonitis, colitis, hepatitis, adrenal insufficiency and rash.

CETUXIMAB IS NOT A SUBSTITUTE FOR CISPLATIN

In HPV-positive oropharyngeal cancer, radiotherapy with cetuximab gave worse survival and control than radiotherapy with cisplatin, without less toxicity (RTOG 1016, De-ESCALaTE HPV). Cisplatin remains the standard partner. Cetuximab is reserved for patients who cannot receive platinum.

DRUGS AND THEIR CHARACTERISTIC TOXICITIES

DRUG	USE	KEY TOXICITY
Cisplatin	Concurrent, induction, palliative	Nephrotoxicity (hydrate), high-frequency sensorineural hearing loss and tinnitus, neuropathy, severe nausea, marrow
Carboplatin	Cisplatin-unfit, weekly concurrent	Marrow (thrombocytopenia)
5-Fluorouracil	Induction (PF, TPF), palliative	Mucositis, diarrhoea, hand–foot syndrome, coronary spasm
Docetaxel, paclitaxel	Induction (TPF), palliative	Neutropenia, neuropathy, alopecia, hypersensitivity
Methotrexate	Palliative	Mucositis, marrow, renal and liver toxicity
Bleomycin	Formerly squamous and lymphoma	Pulmonary fibrosis, fever
Doxorubicin	Sarcomas, salivary cancers	Cardiomyopathy
Cetuximab	With radiotherapy if cisplatin-unfit; palliative	Acneiform rash, infusion reaction, hypomagnesaemia
Pembrolizumab, nivolumab	Recurrent and metastatic disease	Immune-related: thyroid, lung, gut, liver, adrenal

BEFORE CISPLATIN

- **Baseline audiogram:** cisplatin ototoxicity is cumulative and permanent, starting at 4 to 8 kHz.
- **Renal function:** creatinine clearance.
- **Blood count and electrolytes, including magnesium.**
- **Fitness for hydration:** check cardiac function.

Children and patients with existing hearing loss deserve monitoring during treatment. Sodium thiosulfate protects hearing in children.

5 CHAPTER FIVE Toxicity

Early and late.

5.1 Acute effects

Acute effects appear from the second week and settle within weeks of finishing. **Mucositis** causes pain, odynophagia and weight loss, and is managed with salt and bicarbonate mouth rinses, analgesia including opioids, treatment of candida, and nutritional support from a dietitian with a nasogastric tube or gastrostomy when needed. The **skin** reddens and then desquamates, dry and later moist, and is washed gently without rubbing or unprescribed creams. Taste is lost, saliva becomes sticky, the larynx swells, fatigue sets in, and chemotherapy suppresses the marrow.

5.2 Late effects

Late effects are permanent and depend on dose per fraction and total dose. **Xerostomia** follows a mean parotid dose above about 26 Gy and brings caries, candida, dysphagia and taste loss; it is reduced by parotid-sparing IMRT and treated with pilocarpine, saliva substitutes and, prophylactically, submandibular gland transfer. **Radiation caries** is prevented by daily fluoride trays and lifelong dental care, and **osteoradionecrosis** is described below.

Fibrosis of the soft tissues causes **trismus**, managed with jaw exercises and devices, and fibrosis of the constrictors causes **dysphagia**, stricture and aspiration, which is why swallowing exercises continue throughout treatment ("use it or lose it"). The **larynx** may develop oedema, chondronecrosis and a fixed cord. **Hypothyroidism** affects up to half of patients after neck radiotherapy, so TSH is checked every 6 to 12 months.

The **ear** suffers serous otitis media, sensorineural hearing loss that is worse with cisplatin, and radionecrosis of the canal and temporal bone. The **eye and nervous system** are at risk of cataract, retinopathy, *optic neuropathy*, brainstem and temporal lobe necrosis, and *radiation myelitis*, which is avoided by keeping the cord dose under about 45 to 50 Gy; Lhermitte's sign is an early, reversible form. Radiation accelerates **carotid atherosclerosis** and stroke, sets the scene for *carotid blow-out* after salvage surgery, and occasionally causes **second malignancies** such as sarcomas and thyroid cancer.

EARLY AND LATE COMPLICATIONS OF RADIOTHERAPY

EARLY	LATE
Mucositis, odynophagia	Permanent xerostomia, radiation caries
Skin erythema and desquamation	Skin atrophy, fibrosis, telangiectasia
Loss of taste, thick saliva	Osteoradionecrosis, chondronecrosis
Laryngeal oedema	Trismus, dysphagia, stricture
Candidiasis	Hypothyroidism, hypopituitarism
Marrow suppression	Serous otitis, sensorineural loss
Fatigue, nausea	Cataract, optic neuropathy, myelitis, brain necrosis; second cancers

5.3 Osteoradionecrosis (ORN)

Osteoradionecrosis is exposed, devitalised irradiated bone that fails to heal over three months in the absence of tumour. It almost always affects the posterior body of the *mandible*, where radiation leaves the bone hypovascular, hypocellular and hypoxic, with radiation-induced fibrosis. The risk rises with doses above 60 Gy,

dental extraction after radiotherapy, poor dentition, surgery on irradiated bone, smoking and alcohol.

It presents with pain and exposed bone, trismus, an orocutaneous fistula or a pathological fracture, and recurrence must always be excluded by biopsy and imaging. Prevention is the key: a dental assessment *before* radiotherapy, extraction of unsalvageable teeth at least two weeks before starting, fluoride, and atraumatic dentistry afterwards. Established disease is treated conservatively with irrigation, antibiotics and removal of loose sequestra; medically with pentoxifylline and tocopherol, with or without clodronate ("PENTOCLO"), and hyperbaric oxygen in selected cases; and, when advanced, by resection and reconstruction with vascularised bone such as a fibula free flap.

OPERATIVE GUIDES IN THEATRE 2 operations

Segmental mandibulectomy, and microvascular reconstruction.

Segmental Mandibulectomy
Segmental mandibulectomy

Microvascular Reconstruction of the Head and Neck
Free flap reconstruction

5.4 Care during treatment

Before treatment the mouth is cleared of unsalvageable teeth, smoking is stopped, anaemia is corrected, and the dietitian, speech and swallowing therapist and nurse specialist see the patient, with a prophylactic gastrostomy for selected high-risk patients. Each week the weight, blood count (and renal function with cisplatin), mucositis, skin care and analgesia are reviewed, with psychological support and help to stop drinking. Admission is needed for dehydration, neutropenic sepsis, aspiration or uncontrolled pain.

6 CHAPTER SIX
Other tumours treated with radiation

Beyond squamous carcinoma.

6.1 Benign and unusual tumours

Stereotactic radiosurgery controls more than 95% of small and residual **vestibular schwannomas**, and radiotherapy or radiosurgery halts the growth of about 90% of **paragangliomas**. Unresectable intracranial residual **juvenile angiofibroma** is irradiated and regresses slowly over years. Radiotherapy is central in **sinonasal NK/T-cell lymphoma**, combined with asparaginase-based chemotherapy, and is part of combined treatment for **lymphoma of Waldeyer's ring** and chloroma. **Chordoma and chondrosarcoma** of the skull base receive high-dose proton therapy after maximal safe resection, and **olfactory neuroblastoma** receives postoperative radiotherapy after craniofacial or endoscopic resection.

7 CHAPTER SEVEN
For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The gray;** the four R's; why fractionate; why gaps and anaemia are bad.
- **IMRT and IGRT;** what parotid sparing achieves; proton Bragg peak.
- **Indications for postoperative radiotherapy,** and when to add cisplatin (margins, extranodal extension).
- **Chemoradiation:** cisplatin and its ototoxicity; the roles of TPF induction, cetuximab and immunotherapy.
- **Early and late toxicities;** osteoradionecrosis and its prevention.
- **PET-CT at 12 weeks** to decide on neck dissection after chemoradiation.

7.2 Common pitfalls

COMMON PITFALLS

- Starting radiotherapy before the teeth are assessed.
- Extracting teeth in an irradiated mandible without precautions.
- Long breaks in radiotherapy.
- Starting postoperative radiotherapy more than six weeks after surgery.
- Forgetting the TSH check and audiometry in follow-up.

Written for this app from Oxford Textbook of Otolaryngology ch. 65, Oxford Specialist Handbook ch. 12, Dhingra ch. 73, 74 and K. J. Lee ch. 46, 47. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · NECK, THYROID & SALIVARY

HIV and ENT

Up to half of people with HIV have head and neck problems, and an ENT finding is often the first clue to the diagnosis. Examples are bilateral parotid cysts, oral hairy leukoplakia, persistent thrush, necrotising gingivitis, generalised nodes, and a purple lesion on the palate. Effective antiretroviral therapy has made the dramatic opportunistic diseases less common, but the ENT surgeon still needs to recognise them, test for HIV, and protect the theatre team with universal precautions and post-exposure prophylaxis.

AT A GLANCE

- CD4 below 200 defines AIDS.
- Oral hairy leukoplakia is EBV-driven and predicts progression.
- Bilateral parotid lymphoepithelial cysts regress on ART.
- PEP within 72 hours, ideally within hours, for 28 days.

1 CHAPTER ONE The virus and the disease

Biology, course, diagnosis.

1.1 HIV

HIV is an enveloped, single-stranded RNA retrovirus of the lentivirus genus. **HIV-1** causes most infections worldwide and is the more pathogenic; **HIV-2** is mainly West African and less virulent.

The virion is a lipid bilayer carrying the envelope glycoproteins *gp120*, which binds CD4, and *gp41*, which mediates fusion. Inside lie a matrix (p17), a conical capsid (*p24*) and two copies of RNA, together with the three enzymes *reverse transcriptase*, *integrase* and *protease*. Each enzyme is a drug target.

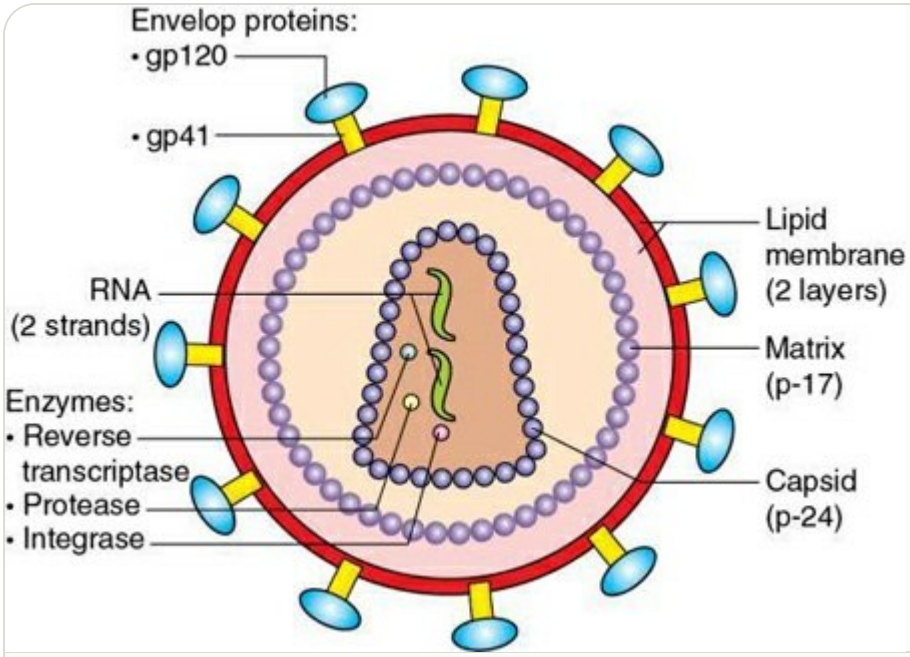


FIGURE 1 Structure of the HIV-1 virion.
Dhingra, fig. 75.1

1.2 Life cycle

1. gp120 binds CD4 and a co-receptor (CCR5 or CXCR4) on helper T cells, macrophages and dendritic cells.
2. gp41 fuses the envelope with the cell membrane.
3. Reverse transcriptase copies the RNA into DNA.
4. Integrase inserts it into the host genome as a *provirus*.
5. The provirus directs new viral RNA and polyproteins, which protease cleaves into functional proteins.
6. New virions bud off.

The progressive loss of *CD4 helper T cells* is what destroys cell-mediated immunity, and it explains the pattern of disease that follows.

1.3 Transmission

HIV spreads by sexual contact, through blood and blood products, by shared needles, and from mother to child during pregnancy, delivery or breastfeeding. The infectious fluids are blood, semen, vaginal fluid and breast milk, together with the

deep body fluids: CSF and pleural, peritoneal, pericardial and amniotic fluid. Saliva, tears, sweat, urine and faeces carry a negligible risk unless they are visibly bloody.

1.4 Course

Primary (acute) infection appears two to four weeks after exposure as a glandular-fever-like illness, with fever, pharyngitis, cervical nodes, a rash, mouth and genital ulcers, and sometimes meningoencephalitis. Viral load is very high at this stage and the patient is highly infectious.

A **clinical latency** of about ten years on average follows. The patient is well, sometimes with *persistent generalised lymphadenopathy*, but the virus keeps replicating in lymphoid tissue, including the tonsils and adenoids, while CD4 counts fall.

AIDS is defined by a CD4 count below 200 cells/mm³ (normal 500 to 1500) or by an AIDS-defining illness. These include Kaposi sarcoma, *Pneumocystis* pneumonia, oesophageal candidiasis, extrapulmonary tuberculosis, cryptococcosis, cerebral toxoplasmosis and certain lymphomas.

1.5 Diagnosis

The first-line test is a **fourth-generation combined antigen and antibody immunoassay**, which detects p24 antigen and antibodies and becomes positive at about two to six weeks. A positive result is confirmed with a second, differentiating assay; the Western blot has largely been replaced. **HIV RNA** by PCR (the viral load) diagnoses acute infection and monitors treatment, and the **CD4 count** measures immune damage and predicts opportunistic disease. Patients are staged clinically by the WHO stages 1 to 4, or by the CDC stages based on the CD4 count.

1.6 Treatment

Antiretroviral therapy (ART), formerly called HAART, combines usually three drugs, often in a single daily tablet. It is started *at diagnosis, whatever the CD4 count*, because immediate treatment halves serious illness and death (the START and TEMPRANO trials). A person with an undetectable viral load does not transmit HIV sexually: *undetectable equals untransmittable*.

For prevention, *pre-exposure prophylaxis (PrEP)* with tenofovir and emtricitabine protects people at high risk, and *post-exposure prophylaxis (PEP)* is described below.

CLASSES OF ANTIRETROVIRAL DRUGS

CLASS	ACTION	EXAMPLES
Nucleoside and nucleotide reverse transcriptase inhibitors (NRTI)	Chain termination of viral DNA	Tenofovir, emtricitabine, lamivudine, abacavir, zidovudine
Non-nucleoside reverse transcriptase inhibitors (NNRTI)	Bind and block reverse transcriptase	Efavirenz, nevirapine, rilpivirine
Integrase inhibitors	Block integration into the host genome	Dolutegravir, bictegravir, raltegravir
Protease inhibitors	Block cleavage of polyproteins	Darunavir, atazanavir (with ritonavir boosting)
Entry and fusion inhibitors	Block CCR5 or gp41	Maraviroc, enfuvirtide

2 CHAPTER TWO ENT manifestations

Head to neck.

2.1 Categories of disease

Four kinds of disease account for almost everything the ENT surgeon sees. **Infections**, both opportunistic and ordinary, become more severe and more often complicated as the CD4 count falls; they may be bacterial, viral, fungal, mycobacterial or protozoal. **Malignancies** are either *AIDS-defining*, namely Kaposi sarcoma and non-Hodgkin lymphoma, or *non-AIDS-defining*, such as Hodgkin lymphoma, skin cancers and HPV-related oropharyngeal cancer. **Neurological disease** includes HIV encephalopathy, neuropathies and cranial nerve palsies, especially of the facial nerve. Finally there are **inflammatory and lymphoproliferative conditions**, chiefly lymphoepithelial cysts and diffuse infiltrative lymphocytosis.

2.2 Ear

In the **external ear**, seborrhoeic dermatitis and otitis externa are common, including *necrotising (malignant) otitis externa* from *Pseudomonas* or *Aspergillus*, and Kaposi sarcoma may involve the pinna and canal.

In the **middle ear**, serous otitis media follows adenoid hyperplasia or a nasopharyngeal mass. A unilateral effusion in an adult with HIV therefore needs a nasopharyngeal biopsy to exclude lymphoma or Kaposi sarcoma. Recurrent and chronic otitis media are frequent, especially in children, and *Pneumocystis* or mycobacterial otitis occurs.

The **inner ear and nerves** suffer sensorineural hearing loss from HIV itself, from opportunistic infections (CMV, syphilis, cryptococcal meningitis) and from ototoxic drugs. Herpes zoster oticus (*Ramsay Hunt syndrome*) and facial palsy are commoner than in the general population.

2.3 Nose and sinuses

Rhinosinusitis is common. While the CD4 count is above 200 the usual organisms are found, but at low counts infections become atypical and complicated, with *Pseudomonas*, CMV and *invasive fungal sinusitis* from *Aspergillus* or mucor. Herpes simplex can ulcerate the nose, and the tumours to remember are Kaposi sarcoma and B-cell, Burkitt and plasmablastic lymphoma.

2.4 Mouth and pharynx

Oral candidiasis is the commonest oral lesion. It takes four forms: *pseudomembranous* thrush, with white plaques that wipe off; *erythematous* (atrophic) candidiasis; *angular cheilitis*; and *hyperplastic* candidiasis, whose plaques do not wipe off. Topical nystatin or clotrimazole suffices when the CD4 count is above 200; systemic fluconazole is used when it is lower or when odynophagia suggests oesophageal disease. Resistance develops with repeated courses.

Oral hairy leukoplakia produces white, vertically corrugated, hair-like plaques on the *lateral border of the tongue* that cannot be wiped off. It is caused by *EBV*, is asymptomatic and is not premalignant, but it is a strong marker of HIV that predicts

progression to AIDS and should prompt testing. Biopsy it if in doubt, because the differential includes leukoplakia, candida, lichen planus and carcinoma.

Periodontal disease ranges from *linear gingival erythema* to *necrotising ulcerative gingivitis and periodontitis*, which is painful, foul and bleeding, with loss of the interdental papillae and bone. It is treated with chlorhexidine, debridement, metronidazole and dental care.

Ulcers need particular care. *Giant aphthous ulcers* are large, deep and painful; biopsy them to exclude herpes, CMV, tuberculosis, histoplasmosis and lymphoma before giving topical or intralesional steroids or thalidomide. Chronic herpes simplex ulcers lasting more than a month are AIDS-defining, as is oral histoplasmosis with its granulomatous ulcers.

In adults, **adenotonsillar hypertrophy** causes nasal obstruction and middle ear effusion and must be biopsied to exclude lymphoma. Kaposi sarcoma and lymphoma also arise on the palate, gingiva, tongue and tonsil.

2.5 Larynx

The larynx may be infected by candida, herpes, CMV or tuberculosis, and epiglottitis occurs. Kaposi sarcoma and lymphoma here can obstruct the airway.

2.6 Salivary glands

Benign lymphoepithelial cysts (BLEC) make up about three-quarters of HIV parotid lesions. They present as painless, soft, often *bilateral* parotid swellings (up to 80% are bilateral on imaging) with cervical lymphadenopathy, and they may be the presenting sign of HIV, especially in children. They are thought to arise from intraparotid nodes. Ultrasound or CT shows multiple thin-walled cysts, and FNA confirms the diagnosis. The cysts regress on ART; serial aspiration, *doxycycline or alcohol sclerotherapy* and low-dose radiotherapy are alternatives, and parotidectomy is reserved for disfiguring lesions or suspected malignancy.

Diffuse infiltrative lymphocytosis syndrome (DILS) resembles Sjögren's syndrome, with bilateral parotid enlargement and sicca symptoms, but the infiltrate is of CD8 lymphocytes, Sjögren antibodies are negative, and lymphocytic interstitial pneumonitis often coexists. It is treated with ART and symptomatic measures.

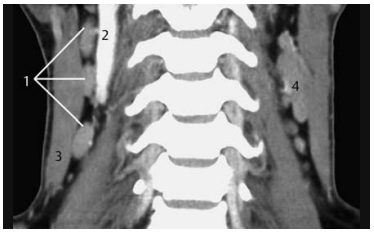
Xerostomia (often drug-related), parotitis and lymphoma complete the salivary picture.

2.7 Neck

Most patients have **cervical lymphadenopathy**, and persistent generalised lymphadenopathy is a reactive follicular hyperplasia. Nodes that are asymmetric, rapidly growing, painful or matted, or that come with B symptoms, need FNA or excision. The causes then include *tuberculosis* (the commonest serious cause in endemic areas), atypical mycobacteria, histoplasmosis, toxoplasmosis, bacillary angiomatosis, Kaposi sarcoma and *lymphoma*. Deep neck abscesses are also more frequent.

IMAGING IN THE APP 1 case

Cervical lymphadenopathy in HIV.



Lymphadenopathy, HIV

Infection & inflammation · 1 image

3 CHAPTER THREE HIV-associated tumours

Kaposi, lymphoma and others.

3.1 Kaposi sarcoma

Kaposi sarcoma is an angioproliferative tumour caused by *human herpesvirus 8* (KSHV). Four variants are described: the classic form in elderly men of Mediterranean or Eastern European origin, the endemic African form, the

transplant-related form, and the AIDS-associated form, which is the commonest and most aggressive.

The head and neck are involved in most AIDS cases: the skin of the face, nose and ears; the oral cavity, especially the *hard palate*, gingiva and tongue; and the pharynx, larynx and nodes. The lesions are violaceous macules, plaques and nodules that do not blanch, and oral lesions may ulcerate, bleed, loosen teeth and obstruct.

Biopsy shows spindle cells, slit-like vascular channels, extravasated red cells and haemosiderin; the differential includes haemangioma, pyogenic granuloma and bacillary angiomatosis. Stage the disease by looking for visceral spread to the gut and lungs and by checking the CD4 count and viral load.

The foundation of treatment is *ART*, under which many lesions regress. Local lesions respond to radiotherapy, to which they are very sensitive, or to intralesional vinblastine, cryotherapy or laser. Extensive disease is treated with liposomal doxorubicin or paclitaxel, and surgery is used only to palliate or to secure the airway.

3.2 Lymphoma

Non-Hodgkin lymphoma is the second commonest malignancy in HIV. It is mostly of aggressive B-cell types: diffuse large B-cell, Burkitt and *plasmablastic*, the last often in the oral cavity. It is frequently EBV-driven, extranodal and advanced at presentation with B symptoms. In the head and neck it affects Waldeyer's ring, the sinonasal tract, oral cavity, orbit, salivary glands and nodes, and *primary CNS lymphoma* occurs when the CD4 count is very low. Diagnosis needs an excision or core biopsy, because FNA is usually not enough to classify it. Treatment is chemotherapy (R-CHOP, R-EPOCH) together with ART.

Hodgkin lymphoma is the commonest non-AIDS-defining cancer in HIV. It presents at a more advanced stage, usually as the mixed cellularity or lymphocyte-depleted types.

3.3 Other cancers

Basal and squamous cell carcinomas and melanoma of the skin are commoner and more aggressive, and HPV-related oropharyngeal, anal and cervical carcinomas are

more frequent. The practical rule is simple: keep a low threshold to biopsy any lesion that does not heal.

4 CHAPTER FOUR Protecting the team

Precautions and prophylaxis.

4.1 Risk of occupational infection

After a percutaneous injury from an HIV-positive source the risk of transmission is about 0.3%, or 1 in 300; after mucous membrane exposure it is about 0.09%. The risk is higher with a deep injury, a hollow-bore needle that has been in a vessel, visible blood on the device, and a source with a high viral load.

4.2 Universal (standard) precautions

Every patient's blood and body fluids are treated as potentially infectious. Wash hands before and after patient contact, wear gloves, and *double-glove* in surgery. An impermeable gown, mask and eye protection are needed wherever splashes are likely, as in tracheostomy, endoscopy, drilling, laser and oral surgery.

Sharps are passed in a dish through a "hands-free" zone, needles are never recapped, and sharps go straight into rigid containers; blunt suture needles are used where possible. Soiled linen and specimens are handled as infectious, and staff with weeping dermatitis should avoid direct patient care.

4.3 After an exposure

1. Wash the wound with soap and water and let it bleed. Do not scrub or squeeze. Irrigate mucous membranes and eyes with water.
2. Report the injury at once and assess the risk.
3. Test the source patient (with consent) for HIV, hepatitis B and hepatitis C.
4. Take baseline blood from the injured worker.
5. *Post-exposure prophylaxis (PEP)*: start as soon as possible, ideally within hours and no later than 72 hours. Take three drugs for 28 days, typically tenofovir with

emtricitabine (or lamivudine) plus dolutegravir or raltegravir.

6. Follow-up HIV testing at about 6 weeks and 3 to 4 months. Protect sexual partners and avoid blood donation meanwhile.

PEP REGIMENS

Older texts give two NRTIs (such as zidovudine and lamivudine) for low-risk exposures and add a third drug for high-risk ones. Current guidance uses a three-drug regimen built around an integrase inhibitor for every exposure that warrants PEP, because it is better tolerated and more often completed. Diagnosis has moved from ELISA and Western blot to fourth-generation antigen and antibody assays.

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The virion and its life cycle:** where the drugs act.
- **CD4 count and AIDS;** the stages of infection.
- **ENT clues to HIV:** oral hairy leukoplakia, persistent candida, necrotising gingivitis, bilateral parotid lymphoepithelial cysts, generalised lymphadenopathy, Kaposi sarcoma, and a unilateral effusion with adenoid hyperplasia.
- **Kaposi sarcoma:** cause, appearance, treatment.
- **Universal precautions and PEP.**

5.2 Common pitfalls

COMMON PITFALLS

- Treating a unilateral adult middle ear effusion with grommets without examining and biopsying the nasopharynx.
- Excising bilateral parotid cysts that would regress on ART.

- Relying on FNA alone to diagnose lymphoma.
- Delaying PEP beyond the first hours after a needlestick.

Written for this app from Oxford Handbook of ENT ch. 18, Dhingra ch. 75 and K. J. Lee ch. 11. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · NECK, THYROID & SALIVARY

Skin cancer and melanoma of the head and neck

The face, scalp and ears take the most sun, so the head and neck carry most skin cancers. Basal cell carcinoma is common and slow and almost never spreads, but it destroys locally. Cutaneous squamous carcinoma can metastasise to the parotid and neck and can track along nerves. Merkel cell carcinoma and melanoma are the dangerous minority. For all of them, biopsy first, clear margins before complex reconstruction, and think about the parotid and neck nodes.

AT A GLANCE

- Mohs surgery for high-risk and central-face BCC.
- cSCC of the scalp, temple and ear spreads to the parotid.
- Melanoma margins: 1 cm up to 1 mm thick, 2 cm over 2 mm.
- Sentinel node biopsy from T1b; checkpoint inhibitors for advanced disease.

1 CHAPTER ONE Overview

Who gets skin cancer.

1.1 The cancers

Basal cell carcinoma is the commonest human cancer and makes up about 70 to 75% of non-melanoma skin cancers; **cutaneous squamous cell carcinoma** accounts for about 20%. **Merkel cell carcinoma** is rare and aggressive. Between a fifth and a third of **melanomas** arise in the head and neck, and their incidence is rising. Rarer tumours include sebaceous carcinoma of the eyelid, adnexal carcinomas, dermatofibrosarcoma protuberans, angiosarcoma of the scalp in the elderly, and atypical fibroxanthoma.

1.2 Risk factors

Ultraviolet radiation is the dominant cause, but in two patterns: cumulative exposure drives BCC and SCC, while intense intermittent exposure and *blistering*

sunburn in childhood drive melanoma. Tanning beds add to both. Fair skin of Fitzpatrick types I and II, red or blond hair, freckling and many or atypical naevi mark the susceptible patient.

Immunosuppression transforms the risk of SCC. Organ transplant recipients have up to a hundredfold increase, and their tumours are aggressive; HIV and chronic lymphocytic leukaemia have a similar effect. Reducing immunosuppression where possible, or switching to an mTOR inhibitor such as sirolimus, helps.

Radiation, arsenic, and chronic scars and ulcers also predispose; SCC arising in a burn scar is a *Marjolin ulcer*. Among the inherited syndromes, *xeroderma pigmentosum* impairs DNA repair and produces skin cancers in childhood; *naevoid BCC (Gorlin) syndrome*, a *PTCH1* mutation in the hedgehog pathway, gives multiple BCCs with jaw keratocysts, palmar pits and a calcified falx, and radiotherapy must be avoided; the *familial atypical mole and melanoma syndrome* is linked to *CDKN2A*; and albinism removes the protection of pigment. The *Merkel cell polyomavirus* is found in most Merkel cell carcinomas.

2 CHAPTER TWO Assessment

History, examination, biopsy.

2.1 Examination

Record the lesion's size, borders, colour, ulceration and fixation, and then examine the whole skin. Palpate the draining nodes: the *parotid nodes* for the anterior scalp, temple, forehead, cheek and ear, the *postauricular and suboccipital nodes* for the posterior scalp, and the neck. Test the cranial nerves for perineural spread, looking for facial numbness and weakness (V and VII), especially with SCC of the cheek, temple and forehead.

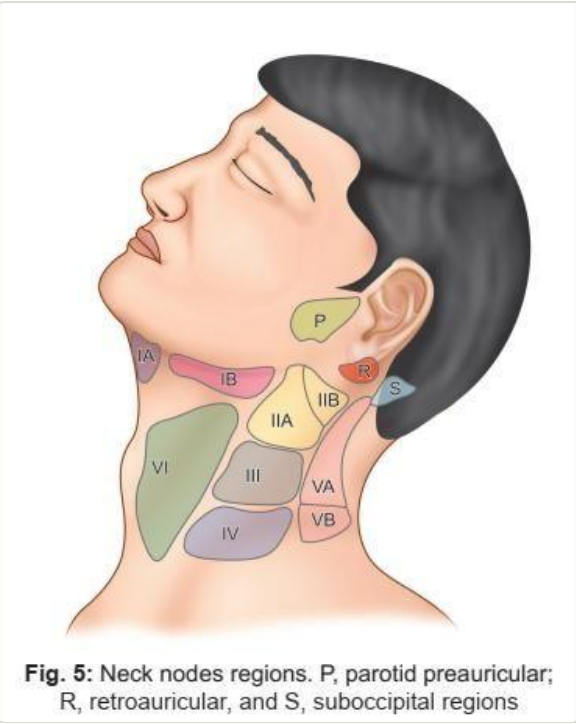


Fig. 5: Neck nodes regions. P, parotid preauricular; R, retroauricular, and S, suboccipital regions

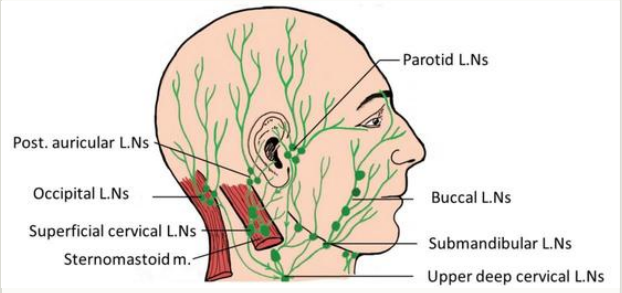


FIGURE 1 Lymph node regions of the head and neck, and the superficial nodes of the head (parotid, postauricular, occipital) that drain the scalp and face.

Anatomy atlas · ENT Steps p. 284, Doss p. 195

2.2 Biopsy

A shave biopsy suits superficial, non-pigmented lesions, and a punch or incisional biopsy through the thickest part suits larger ones. A small suspicious pigmented lesion is removed by excision biopsy with narrow margins of 1 to 3 mm. Two errors are avoided: a shave biopsy of a suspected melanoma, which loses the depth on which staging depends, and a wide excision before diagnosis, which distorts the lymphatics needed for sentinel node mapping and may be the wrong size.

2.3 Imaging

CT or MRI is used for large, recurrent or high-risk tumours, suspected bone invasion, nodes and perineural spread, with MRI following V and VII to the skull base. Nodes are assessed with ultrasound and FNA, and PET-CT stages advanced disease.

CHAPTER THREE

3 Basal cell carcinoma

Locally destructive, rarely spreading.

3.1 Features

The classic BCC is a pearly, translucent papule with rolled edges and telangiectasia that ulcerates centrally, the *rodent ulcer*. The **nodular** subtype is commonest; the **superficial** type is a red scaly patch, usually on the trunk; and **pigmented** BCC can mimic melanoma. The aggressive subtypes are the *morphoeic* (sclerosing) BCC, which looks like a scar with ill-defined edges, and the infiltrative, micronodular and basosquamous types.

BCC grows slowly but invades along the embryonic fusion planes of the nasal ala, inner canthus and preauricular region, and around cartilage and bone. Metastasis is very rare.

3.2 High-risk BCC

A BCC is high risk when it lies in the *H zone* of the central face (the nose, eyelids, lips, ear, and periauricular and temple areas), when it measures 10 mm or more on the cheek, forehead, scalp or neck, or when it has ill-defined borders, an aggressive subtype or perineural invasion. Recurrent tumours, tumours in previously irradiated skin and tumours in immunosuppressed patients are also high risk.

3.3 Treatment

Low-risk BCC is excised with margins of about *4 mm*, checked histologically; small low-risk lesions off the face can be treated by curettage and cautery. **High-risk, recurrent and H-zone BCC** is best treated by *Mohs micrographic surgery*, in which horizontal frozen sections examine the whole margin, giving the lowest recurrence rates with maximum preservation of tissue; the alternative is excision with formal margin assessment before reconstruction.

Radiotherapy suits patients unfit for surgery and positive margins that cannot be re-excised, but is avoided in Gorlin syndrome and connective-tissue disease. Superficial BCC responds to topical imiquimod or 5-fluorouracil, photodynamic therapy or cryotherapy. Locally advanced or metastatic BCC is treated with *hedgehog pathway inhibitors* such as vismodegib and sonidegib, whose side effects are muscle cramps, taste loss, alopecia and weight loss, and then with cemiplimab. Because new lesions are common, patients need annual skin checks and sun protection.

CHAPTER FOUR

4 Cutaneous squamous cell carcinoma

The one that spreads to the parotid.

4.1 Spectrum

SCC develops along a spectrum. *Actinic keratoses* are premalignant rough scaly patches on sun-damaged skin, and *Bowen's disease* is SCC in situ. The *keratoacanthoma* is a fast-growing crateriform nodule with a keratin plug that may regress, but it is treated as a well-differentiated SCC and excised. Invasive SCC is a keratotic nodule or ulcer that grows over months, most often on the lip, ear, scalp, temple and cheek.

4.2 High-risk features

An SCC is high risk if it is 2 cm or more across, *more than 2 mm deep* or invading beyond the subcutaneous fat, poorly differentiated or of the desmoplastic, acantholytic or adenosquamous variant, or shows *perineural invasion*, especially of named nerves, or lymphovascular invasion. Site on the ear or lip, recurrence, origin in a scar and *immunosuppression* all add to the risk.

4.3 Treatment of the primary

Field change and SCC in situ are treated with cryotherapy, topical 5-FU or imiquimod, photodynamic therapy or curettage. Low-risk invasive SCC is excised with *4 to 6 mm margins*. High-risk SCC needs wider margins of 6 to 10 mm or Mohs surgery with complete margin assessment, resection of underlying bone, cartilage

or parotid as required, and *adjuvant radiotherapy* for perineural invasion, close or positive margins, recurrence and T3 to T4 tumours.

4.4 Nodes and the parotid

About 5% of high-risk SCCs metastasise, mostly to the *parotid* from primaries on the scalp, forehead, temple, ear and cheek, and to the neck. Nodal disease is treated by superficial or total parotidectomy with dissection of the at-risk neck levels, followed by postoperative radiotherapy, with cisplatin or a trial for extranodal extension and positive margins. Primaries on the *posterior scalp and neck* need a *posterolateral neck dissection*, which takes levels II to V together with the postauricular and suboccipital nodes. Sentinel node biopsy may be considered for selected high-risk tumours with a node-negative neck.

For advanced, unresectable or metastatic disease, the PD-1 inhibitor *cemiplimab*, and also pembrolizumab, give high response rates and have replaced platinum and EGFR inhibitors as first-line systemic treatment.

OPERATIVE GUIDES IN THEATRE 3 operations

Management of skin cancer, posterolateral neck dissection, and superficial parotidectomy.

Management of Skin Cancer of the Head and Neck
Skin cancer
Posterolateral Neck Dissection
Posterolateral neck dissection
Superficial Parotidectomy

4.5 Lip cancer

Most lip SCCs arise on the vermilion of the *lower lip* of outdoor workers; upper lip and commissure tumours are more aggressive. Lesions up to a third of the lip are removed by V or W wedge excision, and larger defects are closed with the Abbe, Estlander, Karapandzic or Bernard–Burow flaps. Actinic cheilitis is treated by vermilionectomy (a lip shave), and radiotherapy or brachytherapy is an alternative to surgery. Submental and submandibular nodes are treated when involved.

OPERATIVE GUIDES IN THEATRE 1 operation

Cancer of the lip.

Cancer of the Lip
Lip cancer

5 CHAPTER FIVE Merkel cell carcinoma

Fast and neuroendocrine.

5.1 Features and treatment

Merkel cell carcinoma affects the elderly, the sun-damaged and the immunosuppressed, and is associated with the *Merkel cell polyomavirus*. It is a rapidly growing, painless, firm, red-violet dome-shaped nodule, remembered by *AEIOU*: asymptomatic, expanding rapidly, immunosuppressed, older than 50, and on a UV-exposed site.

Histology shows small round blue cells that are *CK20 positive* in a perinuclear dot pattern and positive for neuroendocrine markers but *TTF-1 negative*, which distinguishes it from metastatic small-cell lung cancer. It spreads early to nodes, so that about a third of patients whose nodes feel normal have occult nodal disease, and it recurs locally and metastasises; five-year survival is about 30 to 65%.

Treatment is wide excision with 1 to 2 cm margins and *sentinel lymph node biopsy*, the most sensitive staging test, with neck dissection or radiotherapy for nodal disease. Adjuvant radiotherapy to the primary and nodal basin is common, and checkpoint inhibitors such as avelumab, pembrolizumab and nivolumab treat advanced disease.

6 CHAPTER SIX Melanoma

Depth is destiny.

6.1 Precursors and types

Up to a third of melanomas arise in a pre-existing naevus. Giant congenital naevi over 20 cm carry the highest risk, dysplastic naevi a lesser one, and *lentigo maligna* (Hutchinson's melanotic freckle) is melanoma in situ on sun-damaged facial skin.

Superficial spreading melanoma is the commonest, about 70%, growing first radially and then vertically. **Nodular** melanoma (10 to 15%) grows vertically from the start and is thick at diagnosis. **Lentigo maligna melanoma** appears on the face of the elderly after a long radial phase. **Acral lentiginous** melanoma of the palms, soles and nails is the commonest type in dark-skinned populations and is not sun-related. **Desmoplastic** melanoma is often amelanotic, a scar-like nodule on the head and neck that is strongly *neurotropic*, so its problem is local recurrence rather than nodes; it needs wide margins and adjuvant radiotherapy. Mucosal melanoma is considered separately below.

6.2 Recognition

The ABCDE rule describes a suspicious lesion: **A**symmetry, **B**order irregularity, **C**olour variation, **D**iameter over 6 mm, and **E**volution, meaning change in size, shape or colour, itching or bleeding. The "ugly duckling" sign, a mole unlike the patient's others, and dermoscopy add to clinical judgement.

Head and neck melanomas carry a worse prognosis than those of the trunk and limbs, the scalp being worst, then the ear, then the face and neck. The mimics are seborrhoeic keratosis, pigmented BCC, blue naevus, Spitz naevus and naevus of Ota.

6.3 Pathology and staging

Breslow thickness, the depth from the granular layer to the deepest tumour cell, is together with *ulceration* the key prognostic factor and defines the T stage. Clark's levels of anatomical invasion are now historical. The report also gives the mitotic rate, microsatellites, lymphovascular and perineural invasion and the margins. Immunohistochemistry uses S-100, SOX10, HMB-45 and Melan-A, and tumours are tested for *BRAF V600* (about half of cutaneous melanomas), NRAS, and c-KIT in mucosal and acral types.

T STAGE OF CUTANEOUS MELANOMA (AJCC 8TH EDITION)

T	BRESLOW THICKNESS	ULCERATION
Tis	In situ	—
T1a	Under 0.8 mm	Absent
T1b	Under 0.8 mm with ulceration, or 0.8 to 1.0 mm	Either
T2	Over 1.0 to 2.0 mm	a: absent; b: present
T3	Over 2.0 to 4.0 mm	a: absent; b: present
T4	Over 4.0 mm	a: absent; b: present

Stages I and II are localised disease, stage III is nodal, satellite or in-transit disease, and stage IV is distant metastasis, with the LDH forming part of the M stage. Thin melanomas need no imaging; thick or ulcerated tumours and stage III and IV disease are staged with CT or PET-CT and brain MRI, with an LDH.

6.4 Treatment of the primary

Wide local excision margins depend on thickness: 0.5 to 1 cm for melanoma in situ, 1 cm up to 1 mm thick, 1 to 2 cm for 1 to 2 mm, and 2 cm beyond 2 mm, often modified on the face to preserve function. The depth of excision depends on site: down to the periosteum on the scalp, or the outer table if it is involved; a partial or total auriculectomy on the ear, or a lateral temporal bone resection if the canal is involved; deep to the mimetic muscles on the face if needed; and to the parotid fascia, or including a superficial parotidectomy, over the parotid. Lentigo maligna is

excised in stages with margin control, by the "spaghetti" technique or Mohs with immunostains, or treated with imiquimod or radiotherapy in the frail.

DO NOT RECONSTRUCT BEFORE THE MARGINS ARE KNOWN

Complex flaps should wait until the final pathology confirms clear margins. Otherwise residual tumour is buried under a flap. Use a temporary dressing or skin graft and reconstruct afterwards (*delayed reconstruction*).

6.5 The neck

Sentinel lymph node biopsy is offered for melanomas thicker than 0.8 mm (T1b and above) with a clinically negative neck. It is the strongest predictor of survival and guides adjuvant therapy.

1. Preoperative *lymphoscintigraphy* with technetium-99m colloid, with SPECT-CT to localise nodes in the complex drainage of the head and neck, including parotid, postauricular and contralateral sites.
2. An intraoperative gamma probe, with or without blue dye.
3. Facial nerve monitoring in the parotid.

A positive sentinel node is now usually followed by *ultrasound surveillance* rather than completion dissection, because MSLT-II showed no survival benefit from completion dissection, only better regional control. Clinically or radiologically involved nodes need a therapeutic neck dissection, adding a superficial parotidectomy for primaries of the anterior scalp, temple, forehead, ear and cheek, and a posterolateral dissection with postauricular and suboccipital nodes for the posterior scalp. Adjuvant radiotherapy, often hypofractionated, improves regional control after extranodal extension or multiple or large nodes.

OPERATIVE GUIDES IN THEATRE 1 operation

Sentinel lymph node biopsy.

Sentinel Lymph Node Biopsy
Sentinel node biopsy

6.6 Systemic therapy

Checkpoint inhibitors, anti-PD-1 agents (nivolumab, pembrolizumab) alone or with the anti-CTLA-4 agent ipilimumab, are first-line treatment for unresectable and metastatic melanoma and give durable responses in about half of patients, at the cost of immune side effects: colitis, hepatitis, pneumonitis, thyroiditis and hypophysitis. **BRAF with MEK inhibitors**, dabrafenib with trametinib or encorafenib with binimetinib, give rapid responses in BRAF-mutant disease. After resection of stage III and high-risk stage II melanoma, adjuvant anti-PD-1 therapy, or BRAF/MEK inhibitors for BRAF-mutant tumours, reduces relapse. Intralesional oncolytic virus (T-VEC) treats in-transit disease, while dacarbazine and interferon are now historical.

THE NECK AFTER A POSITIVE SENTINEL NODE, AND ADJUVANT IMMUNOTHERAPY

Older practice was completion neck dissection for every positive sentinel node, with interferon as the only adjuvant. Since MSLT-II and DeCOG-SLT, most patients are followed with nodal ultrasound instead. Adjuvant anti-PD-1 or BRAF/MEK therapy is now standard for resected stage III disease.

6.7 Mucosal melanoma

Mucosal melanoma makes up only about 1% of melanomas, but the head and neck is its commonest site: the nasal septum and turbinates, the sinuses, the palate and gingiva, and the nasopharynx. It is often amelanotic and presents late with unilateral obstruction, epistaxis or a pigmented palatal lesion. Breslow depth does not apply, and every mucosal melanoma is at least T3.

Treatment is complete endoscopic or open resection with postoperative radiotherapy for local control, and therapeutic rather than elective neck dissection. The prognosis is poor, with five-year survival of 20 to 30% because of distant metastasis, and immunotherapy and KIT inhibitors have a role.

6.8 Melanoma of unknown primary

Melanoma may present in the parotid or neck nodes with no primary, which has perhaps regressed. Examine the skin, scalp, mucosa and eyes, then treat it as stage

III disease with dissection, adjuvant radiotherapy and systemic therapy. The prognosis is similar to, or better than, that of stage III disease with a known primary.

7 CHAPTER SEVEN Reconstruction

After excision.

7.1 The reconstructive ladder

Defects are closed by the simplest reliable method on the *reconstructive ladder*:

1. Healing by secondary intention, which works well in concave sites such as the conchal bowl and medial canthus.
2. Primary closure along relaxed skin tension lines.
3. Skin grafts: full-thickness grafts from postauricular or supraclavicular skin for the nose and eyelids, and split-thickness grafts for the scalp over periosteum.
4. Local flaps: advancement, rotation, transposition (rhomboid, bilobed), and interpolated flaps (paramedian forehead, nasolabial).
5. Regional flaps: cervicofacial, deltopectoral, pectoralis major, trapezius.
6. Free tissue transfer.

Scalp defects can also be closed by tissue expansion.

OPERATIVE GUIDES IN THEATRE 5 operations

Skin grafting, cervicofacial and paramedian forehead flaps, scalp reconstruction, and free flaps.

Skin Grafting
Skin grafting
Cervicofacial Transposition Flaps
Paramedian Forehead Flaps
Forehead flap
Scalp Reconstruction: Role of Tissue Expansion and Flap Reconstruction
Microvascular Reconstruction of the Head and Neck

8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

8.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **BCC:** features, high-risk features, Mohs surgery and its rationale, hedgehog inhibitors.
- **cSCC:** high-risk features, parotid metastasis, perineural spread, cemiplimab.
- **Merkel cell carcinoma:** CK20 and TTF-1; sentinel node biopsy.
- **Melanoma:** ABCDE, Breslow and ulceration, margins by thickness, sentinel node indications and lymphoscintigraphy, MSLT-II, checkpoint and BRAF/MEK inhibitors.
- **Drainage:** of the anterior versus posterior scalp, and when to add parotidectomy.
- **Mucosal melanoma.**

8.2 Common pitfalls

COMMON PITFALLS

- Shave biopsy of a pigmented lesion.
- Wide excision before diagnosis, which ruins sentinel node mapping.
- Flap reconstruction before margins are clear.
- Missing facial numbness or weakness as perineural spread from a "treated" facial SCC.
- Radiotherapy in Gorlin syndrome.

Written for this app from K. J. Lee ch. 42, 43. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · NECK, THYROID & SALIVARY

Cysts and tumours of the jaws

Most jaw lesions are found by chance on a dental radiograph, and the radiograph does most of the diagnostic work. Is the lesion radiolucent or radio-opaque? Single or multiple? Well defined or ragged? Unilocular or multilocular? Related to a tooth crown, a root apex, or neither? A periapical cyst on a dead tooth needs a root canal treatment. A keratocyst in a young adult raises the question of Gorlin syndrome. A multilocular "soap-bubble" lesion at the angle is an ameloblastoma until biopsy says otherwise.

AT A GLANCE

- Radicular cyst: the only cyst with a dead tooth.
- Dentigerous cyst: around an unerupted crown.
- Multiple keratocysts: Gorlin syndrome.
- Ameloblastoma: soap bubble at the angle; resect with margins.

1 CHAPTER ONE Approach

Reading the jaw.

1.1 Clinical assessment

Most lesions present as a painless bony expansion of the jaw. When the cortex has thinned it gives a characteristic "egg-shell crackling" on pressure, and the teeth over it may be displaced, loose or, if they never erupted, missing. Pain and infection usually mean that a benign lesion has become secondarily infected, but *numbness of the lower lip* from involvement of the inferior alveolar nerve points towards malignancy or osteomyelitis and should never be dismissed.

Test the vitality of the related teeth with cold or an electric pulp tester. The rule is simple and useful: *every jaw cyst except the periapical (radicular) cyst is associated with vital teeth.*

Aspirate before any incisional biopsy. Blood suggests a vascular lesion such as an arteriovenous malformation or aneurysmal bone cyst, and *a pulsatile or blood-filled*

lesion must never be biopsied blind, because torrential bleeding may follow. Straw-coloured fluid suggests a dentigerous or radicular cyst, cheesy keratin an odontogenic keratocyst, and a dry tap a solid tumour or a traumatic bone cavity.

1.2 Imaging

A panoramic radiograph (OPG) with intraoral films is the first step. CT or cone-beam CT then defines size, any cortical breach, the relation to the inferior alveolar canal and the maxillary sinus, and features of malignancy, while MRI shows soft-tissue extension.

RADIOLOGICAL GROUPS

PATTERN	EXAMPLES
Radiolucent, unilocular	Radicular, residual, dentigerous, lateral periodontal and nasopalatine cysts; odontogenic keratocyst; Stafne defect; unicystic ameloblastoma; adenomatoid odontogenic tumour
Radiolucent, multilocular ("soap bubble")	Ameloblastoma, odontogenic keratocyst, odontogenic myxoma, central giant cell granuloma, aneurysmal bone cyst, glandular odontogenic cyst, Pindborg tumour
Radiolucent, multiple	Keratocysts in Gorlin syndrome, Langerhans cell histiocytosis, cherubism, myeloma, metastases
Radiolucent, ill-defined	Osteomyelitis, osteoradionecrosis, MRONJ, malignancy, metastases, Langerhans cell histiocytosis ("floating teeth")
Radio-opaque, well defined	Odontoma, cementoblastoma, osteoma, focal cemento-osseous dysplasia
Radio-opaque, ill-defined	Fibrous dysplasia ("ground glass"), osteosarcoma ("sunburst"), chondrosarcoma, Paget disease ("cotton wool")
Mixed	Ossifying fibroma, calcifying odontogenic cyst, adenomatoid tumour, cemento-osseous dysplasia

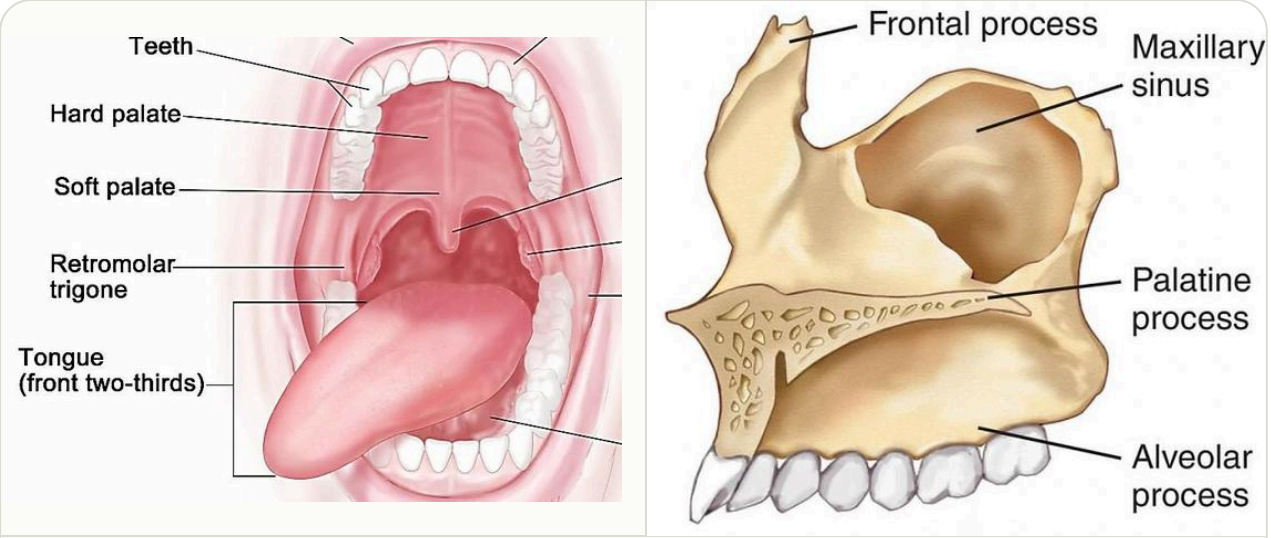


FIGURE 1 The teeth, palate and retromolar trigone, and the maxilla with its alveolar process.

Anatomy atlas · ENT Steps p. 172, ENT Steps p. 129

2 CHAPTER TWO Cysts of the jaws

Odontogenic and non-odontogenic.

A true cyst has an epithelial lining. Odontogenic cysts arise from the remnants of tooth-forming epithelium, and each remnant gives its own cyst: the *rests of Malassez* give radicular cysts, the *reduced enamel epithelium* gives dentigerous cysts, and the *dental lamina (rests of Serres)* gives keratocysts and lateral periodontal cysts.

2.1 Radicular (periapical) cyst

The radicular cyst is the *commonest jaw cyst*. It follows death of the pulp of a tooth, which leads to a periapical granuloma that then becomes cystic. On the radiograph it is a well-defined, round lucency *at the root apex* of the non-vital tooth, and it may announce itself with a sinus onto the gum or skin. It is treated by root canal treatment, or by extraction with enucleation of the cyst. A cyst that persists after the tooth has gone is a **residual cyst**, and is enucleated.

2.2 Dentigerous (follicular) cyst

The dentigerous cyst is the second commonest. It forms around the *crown of an unerupted tooth*, attached at the amelocemental junction, usually a lower third molar

or an upper canine, in patients aged 10 to 30. It expands painlessly and may push the tooth a long way, into the ramus or the antrum.

Most are enucleated with the tooth. Large cysts in children are better *marsupialised* or decompressed, which lets the cyst shrink and the tooth erupt. Rarely an ameloblastoma or carcinoma arises in the wall, which is why the lining is always sent for histology. Its soft-tissue counterpart in a child, the **eruption cyst**, is a bluish swelling of the gum over an erupting tooth that resolves as the tooth comes through.

2.3 Odontogenic keratocyst (OKC)

The keratocyst was renamed a "keratocystic odontogenic tumour" in 2005 and reclassified as a cyst in 2017, but whatever its label it behaves aggressively. It usually occupies the *posterior body and ramus of the mandible* of a young adult, and because it extends along the medulla with little expansion it is often large by the time it is found.

Histology shows a thin, *parakeratinised* lining with a palisaded, hyperchromatic basal layer and *daughter cysts* in the wall, and the aspirate is cheesy keratin. The friable lining and satellite cysts explain recurrence rates of up to 30%. Large cysts are first decompressed, then enucleated with a *peripheral ostectomy* or Carnoy solution; multiple recurrences need resection, and follow-up is long.

Multiple keratocysts, especially in a child, mean **Gorlin syndrome** (naevoid basal cell carcinoma syndrome), an autosomal dominant condition caused by a *PTCH1* mutation. Look for multiple basal cell carcinomas, palmar and plantar pits, a calcified falx, bifid ribs, frontal bossing with hypertelorism, and medulloblastoma, and avoid radiation in these patients. The **orthokeratinised odontogenic cyst** looks similar but has an orthokeratinised lining without palisading, rarely recurs and carries no syndrome.

2.4 Other odontogenic cysts

The **lateral periodontal cyst** is a small lucency between the roots of vital lower premolars and canines in adults. The **glandular odontogenic cyst** of the anterior mandible can be confused with central mucoepidermoid carcinoma, recurs, and may

need resection. The **calcifying odontogenic (Gorlin) cyst** contains "ghost cells" with calcification, and the **paradental (buccal bifurcation) cyst** lies buccal to an erupting lower molar in a child.

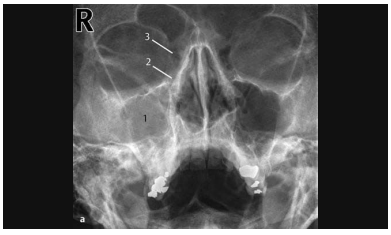
2.5 Non-odontogenic cysts

The **nasopalatine (incisive canal) cyst** is the commonest non-odontogenic cyst. It appears as a *heart-shaped* lucency between vital upper central incisors, with a swelling behind them, and is enucleated. The **nasolabial (nasoalveolar) cyst** lies in soft tissue beneath the ala, filling the nasolabial fold and lifting the nasal floor, mostly in women; it is lined by respiratory epithelium and is removed through a sublabial approach or marsupialised endoscopically into the nose.

Three lesions are pseudocysts, without an epithelial lining. The **traumatic (simple) bone cavity** is an empty space in a teenager that scallops between the roots and heals after surgical exploration. The **aneurysmal bone cyst** is a blood-filled expansile "soap bubble" treated by curettage. The **Stafne defect** is a lingual depression below the inferior alveolar canal at the angle, filled by submandibular gland, and needs no treatment.

IMAGING IN THE APP 1 case

Odontogenic disease extending into the maxillary sinus.



Odontogenic Sinusitis

Infection & inflammation · 3 images

3 CHAPTER THREE Odontogenic tumours

Most are benign.

3.1 Ameloblastoma

The ameloblastoma is the commonest clinically significant odontogenic tumour: benign, but *locally aggressive*. It affects adults in the third to fifth decades, and about 80% arise in the mandible, mostly in the *molar–ramus region*. It grows slowly and painlessly, displacing and loosening teeth and resorbing their roots.

The radiograph shows a *multilocular "soap-bubble"* or honeycomb lucency with cortical expansion and *knife-edge root resorption*. Histologically there are islands of odontogenic epithelium whose peripheral columnar cells palisade with *reverse polarisation* of their nuclei away from the basement membrane, in follicular, plexiform, acanthomatous or desmoplastic patterns. Most mandibular tumours carry *BRAF V600E*, and BRAF-targeted therapy is now used for unresectable or recurrent disease.

Several variants matter. The **unicystic** ameloblastoma of young patients often mimics a dentigerous cyst; its luminal type can be enucleated with close follow-up, but the mural type behaves like a solid tumour. The **peripheral** form is a gum mass. The **metastasising ameloblastoma** has benign histology yet spreads to lung or nodes, and **ameloblastic carcinoma** is frankly malignant.

Treatment is *resection with about 1 to 1.5 cm margins beyond the radiological edge*, including a soft-tissue margin wherever the cortex has been breached: a marginal or segmental mandibulectomy, or a partial maxillectomy, with reconstruction by a fibula free flap and dental implants. Enucleation and curettage give recurrence rates of 50 to 90%. Maxillary tumours are the more dangerous because they can reach the skull base, and follow-up is lifelong because recurrence may appear after decades.

OPERATIVE GUIDES IN THEATRE 3 operations

Marginal and segmental mandibulectomy, and microvascular reconstruction.

Marginal Mandibulectomy
Marginal mandibulectomy
Segmental Mandibulectomy
Segmental mandibulectomy
Microvascular Reconstruction of the Head and Neck
Free flap reconstruction

3.2 Other odontogenic tumours

The **odontoma** is the commonest odontogenic tumour overall, though really a hamartoma. *Compound* odontomas in the anterior maxilla contain small tooth-like structures, and *complex* odontomas in the posterior mandible are a disorganised mass of enamel and dentine. Both are radio-opaque with a lucent rim, often stop a tooth from erupting, and are enucleated.

The **adenomatoid odontogenic tumour** occurs in teenage girls in the anterior maxilla around an unerupted canine, as a lucency with fine calcifications, and is enucleated. The **calcifying epithelial odontogenic (Pindborg) tumour** of adults in the posterior mandible contains amyloid with concentric *Liesegang ring* calcifications and is resected with a margin. The **ameloblastic fibroma** of young patients is enucleated. The **odontogenic myxoma**, with its "soap bubble" or "tennis racket" lucency of stellate cells in myxoid stroma, is unencapsulated and infiltrative, and recurs in about a quarter even after resection. The **cementoblastoma** is a radio-opaque mass fused to a lower molar root with a lucent halo, removed with the tooth.

4 CHAPTER FOUR Non-odontogenic lesions

Bone, fibro-osseous, systemic.

4.1 Fibro-osseous lesions

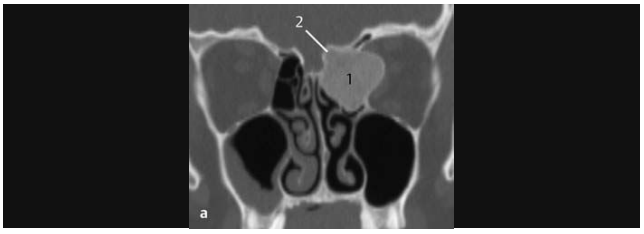
In **fibrous dysplasia** bone is replaced by fibrous tissue and woven bone as a result of a *GNAS* mutation. About 80% is monostotic; polyostotic disease with café-au-lait macules and endocrine overactivity such as precocious puberty is *McCune–Albright syndrome*. It causes painless facial asymmetry in a child or teenager, the maxilla more often than the mandible, and growth usually slows after puberty. The

radiograph shows an ill-defined *ground-glass* expansion blending into normal bone. Most patients are observed, recontouring is done for deformity once growth stops, and the optic canal is decompressed if vision is threatened.

The **ossifying fibroma**, by contrast, is a true, well-demarcated neoplasm of mixed lucency and opacity, typically in the mandible of young women; its juvenile trabecular and psammomatoid forms are aggressive in children and in the sinuses. It is enucleated or resected. **Cemento-osseous dysplasia**, periapical, focal or florid, occurs in middle-aged women of African or Asian descent and is best left alone, because extraction is followed by infection.

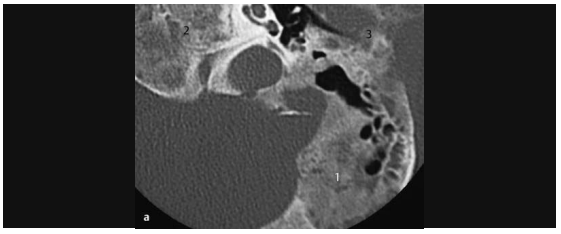
IMAGING IN THE APP 2 cases

Fibrous dysplasia on imaging.



Fibrous Dysplasia of the Sinuses (Monostotic and Polyostotic)

Bone disease · 5 images



Fibrous Dysplasia of the Temporal Bone and Skull Base

Bone disease · 7 images

4.2 Giant cell lesions

The **central giant cell granuloma** affects patients under 30 in the anterior mandible and may cross the midline. Its multinucleated giant cells are indistinguishable from those of a *brown tumour of hyperparathyroidism*, so calcium and PTH must always be checked. It is treated by curettage or with intralesional steroids, calcitonin or denosumab.

Cherubism is autosomal dominant and produces painless *bilateral* swelling of the mandibular angles, and often the maxillae, from the age of two to five, giving the "cherub" face with upturned eyes. The radiograph shows multilocular lucencies, and the condition regresses after puberty.

4.3 Bone tumours

An **osteoma** is a dense, well-defined mass. Multiple osteomas with colonic polyposis and soft-tissue tumours make up *Gardner syndrome* (an APC mutation), and colonoscopy is mandatory because the polyps become malignant. The **osteoblastoma** is painful, and any cartilaginous jaw tumour, even one called a chondroma, should be treated as a possible chondrosarcoma.

Osteosarcoma of the jaw occurs a decade or two later than in the limbs, in the third and fourth decades, and also follows radiotherapy and Paget disease. It presents with swelling, pain, loose teeth and lip numbness, and the radiograph shows a *sunburst* periosteal reaction and *symmetrical widening of the periodontal ligament space* (Garrington sign). It needs radical resection with clear margins and chemotherapy. **Chondrosarcoma** affects older adults, the anterior maxilla more than the mandible, and needs radical resection because it resists radiotherapy and chemotherapy.

Metastases from breast, lung, prostate, kidney and thyroid usually land in the posterior mandible and present with pain, lip numbness or an extraction socket that will not heal; the "*numb chin*" can be the first sign of cancer. Myeloma and lymphoma give punched-out lesions, and **Langerhans cell histiocytosis** in children produces punched-out lucencies around the roots so that the teeth seem to float ("*floating teeth*"); it is treated by curettage, steroids, low-dose radiotherapy or chemotherapy. Finally, **squamous carcinoma** of the overlying mucosa invades the mandible through the occlusal ridge or the tooth sockets.

4.4 Paget disease

Paget disease is disordered bone remodelling in older people. The maxilla and skull enlarge with "*cotton-wool*" radiodensities, the roots show hypercementosis, the alkaline phosphatase rises, and bony overgrowth can compress cranial nerves and cause hearing loss. It is treated with bisphosphonates.

5 CHAPTER FIVE Osteonecrosis and infection

Dead bone in the jaw.

5.1 Osteomyelitis

Osteomyelitis usually affects the mandible after dental infection or a fracture, and diabetes and immunosuppression predispose to it. It causes pain, swelling, pus and sequestra, and numbness of the lower lip (*Vincent sign*). Treatment is debridement, sequestrectomy and prolonged antibiotics.

5.2 Osteoradionecrosis

Osteoradionecrosis follows radiotherapy, especially above 60 Gy. It is prevented by dental treatment before radiotherapy, and treated along a ladder from debridement and pentoxifylline–tocopherol to resection with a free fibula flap.

5.3 Medication-related osteonecrosis (MRONJ)

MRONJ is exposed bone for more than eight weeks in a patient taking antiresorptive drugs such as bisphosphonates or denosumab, or antiangiogenic drugs, who has never had radiotherapy to the jaw. The risk is highest with intravenous treatment for cancer, and it usually follows an extraction, so dental treatment should be completed before the drugs are started. It is managed with chlorhexidine, antibiotics and conservative debridement, with resection for advanced disease.

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Systematic reading of a jaw radiograph,** and why aspiration comes before biopsy.
- **Radicular versus dentigerous cyst:** the tooth relationship and vitality.
- **Odontogenic keratocyst:** its lining, why it recurs, and Gorlin syndrome.
- **Ameloblastoma:** age, site, soap-bubble picture, reverse polarity, resection margins, BRAF.

- **Fibrous dysplasia** (ground glass) versus ossifying fibroma (well-defined).
- **Giant cell granuloma:** exclude hyperparathyroidism.
- **Osteosarcoma signs;** the numb chin.
- **Osteoradionecrosis and MRONJ.**

6.2 Common pitfalls

COMMON PITFALLS

- Biopsying a jaw lucency without aspirating first, and meeting an AVM.
- Curetting an ameloblastoma.
- Missing Gorlin syndrome in a young patient with several keratocysts.
- Calling a central giant cell lesion without checking calcium and PTH.
- Extracting teeth in an irradiated jaw or in a patient on intravenous bisphosphonates without precautions.

Written for this app from K. J. Lee ch. 39. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · NECK, THYROID & SALIVARY

Head and neck reconstruction and facial reanimation

A head and neck resection is only half the operation; the other half is putting back what was taken, and the patient lives with the second half. Reconstruction starts with a defect analysis: which tissues are missing (skin, lining, muscle, bone, nerve), what each one did, and what the bed is like. Choose the simplest method that restores both function and appearance. Today that is often a free flap at the same operation. Facial reanimation follows the same logic: what is missing, and for how long?

AT A GLANCE

- Concave sites heal well by secondary intention.
- Abbe for a third to a half of the lip; Karapandzic up to two-thirds.
- Fibula for the mandible; radial forearm for the tongue; check the Allen test.
- Facial muscle survives about 18 to 24 months of denervation.

1 CHAPTER ONE Principles

Defect analysis and the ladder.

1.1 What is missing

Every defect is analysed in the same way. First, which tissues have gone: skin, mucosal lining, muscle, bone, cartilage or nerve. Second, what each did, because the reconstruction must restore function as well as form: a skeleton for occlusion, contour and *dental implants*; cover for vital structures such as the carotid, dura and brain; oral competence and a lip sphincter; pliable, mobile tissue for tongue movement, speech and swallowing; and volume, sensation and secreting mucosa. Third, the state of the bed: irradiation, infection, a fistula and the vessels available, remembering that irradiated or infected beds need *vascularised* tissue. Finally, the patient's health, prognosis and wishes.

1.2 The reconstructive ladder (and the elevator)

1. **Healing by secondary intention:** good for *concave* sites (medial canthus, temple, conchal bowl, nasolabial groove, glabella); bad near *free margins* (eyelid, ala, lip), which contract and distort.
2. **Primary closure:** along relaxed skin tension lines, as a fusiform excision about three times as long as it is wide.
3. **Grafts:** skin, cartilage, bone, nerve and composite grafts.
4. **Local flaps.**
5. **Regional (pedicled) flaps.**
6. **Free tissue transfer.**
7. **Prostheses.**

Climbing the ladder rung by rung is not compulsory. The surgeon may take the "elevator" straight to the option that gives the best result, which for a major defect is often a free flap.

---

2 CHAPTER TWO Grafts

Tissue without its own blood supply.

2.1 Skin grafts

A skin graft survives for about 48 hours by plasmatic *imbibition*, and then by *inosculation* of its vessels with those of the bed and capillary ingrowth at three to five days. Anything that separates graft from bed in that time kills it: haematoma, seroma, *shearing* and infection, which are prevented with a bolster or tie-over dressing, quilting sutures, pie-crusting or negative pressure. Grafts do not take on bare bone or cartilage stripped of periosteum or perichondrium, nor on irradiated beds.

Split-thickness grafts, taken with a dermatome from the thigh or buttock, take more reliably but *contract more* and match colour poorly; they line the mouth and maxillectomy cavities and cover the scalp over periosteum and free flap donor sites. **Full-thickness grafts**, defatted and taken from postauricular, preauricular, upper

eyelid or supraclavicular skin, contract less and match better, so they are used on the nose, eyelids and ear.

2.2 Cartilage, bone and nerve grafts

Septal cartilage is harvested leaving an L-strut of 1 to 1.5 cm; curved **conchal cartilage** suits the ala and ear; and **costal cartilage** is plentiful for the dorsum and microtia but can warp. **Composite** skin–cartilage grafts from the ear repair alar rim and columella defects up to about 1 to 1.5 cm.

Cortical bone from split calvaria, rib or iliac crest rebuilds the orbital floor, nasal dorsum and facial buttresses, and **cancellous bone** from the iliac crest fills alveolar clefts and small mandibular gaps. Every free bone graft needs a well-vascularised, unirradiated bed and rigid fixation, and all resorb to some degree.

Nerve grafts come from the great auricular nerve, which lies in the operative field and gives up to about 10 cm; the sural nerve, up to 30 to 35 cm, used for cross-face grafts; and the medial antebrachial cutaneous nerve.

OPERATIVE GUIDES IN THEATRE 5 operations

Skin grafting, fascial and allografts, calvarial and rib grafts, and nerve grafting.

- Skin Grafting
- Autologous Fascial Grafts and Acellular Allografts
- Split Calvarial Graft
- Rib Graft
- Nerve Graft Techniques

3 CHAPTER THREE Local flaps

Like for like.

3.1 Principles

Random-pattern local flaps live on the subdermal plexus. On the well-vascularised face they tolerate a length-to-width ratio of up to about 3:1, more than on the trunk. They are designed to place scars in the relaxed skin tension lines and at the borders

of *aesthetic subunits*, to borrow skin from areas of laxity such as the cheek, glabella and neck, and to close the secondary defect without distorting a free margin. Smoking, tension, irradiation and haematoma are their enemies.

3.2 Types

Advancement flaps include the V-Y and the unipedicle and bipedicle flaps of the forehead and of the lip with a crescentic perialar excision. **Rotation** flaps include the cervicofacial and scalp flaps. **Transposition** flaps include the *rhomboid (Limberg)* flap, the *bilobed (Zitelli)* flap for defects up to about 1.5 cm on the lower third of the nose, the *nasolabial (melolabial)* flap for the ala, and the note flap.

Interpolated flaps pass over or under intact skin and are divided later. The *paramedian forehead flap*, on the *supratrochlear artery* and divided at three weeks, is the workhorse for large or full-thickness nasal defects; the interpolated nasolabial flap and the *Abbe* cross-lip flap are the others. A full-thickness nasal defect needs all *three layers* rebuilt: lining from local mucosal flaps or a turnover flap, a framework of cartilage grafts, and cover from the forehead flap.

OPERATIVE GUIDES IN THEATRE 4 operations

Paramedian forehead, cervicofacial and scalp flaps, and nasal reconstruction.

Paramedian Forehead Flaps
Forehead flap
Cervicofacial Transposition Flaps
Scalp Reconstruction: Role of Tissue Expansion and Flap Reconstruction
Nasal Reconstruction

3.3 Lip reconstruction

FULL-THICKNESS LIP DEFECTS

DEFECT	RECONSTRUCTION
Under a third (up to about a half of the lower lip)	Primary (V or W) closure
A third to a half	Abbe cross-lip flap (on the labial artery, divided at about three weeks); Estlander flap if the commissure is involved

DEFECT	RECONSTRUCTION
A half to two-thirds	Karapandzic (neurovascular, sensate, innervated rotation–advancement); bilateral advancement
Over two-thirds to total	Bernard–Burow (cheek advancement), Gillies fan flap, or a free radial forearm flap with a palmaris longus sling

What matters most is restoring the *orbicularis sphincter*, the vermilion line and the commissure. *Microstomia* is the price paid for large local flaps.

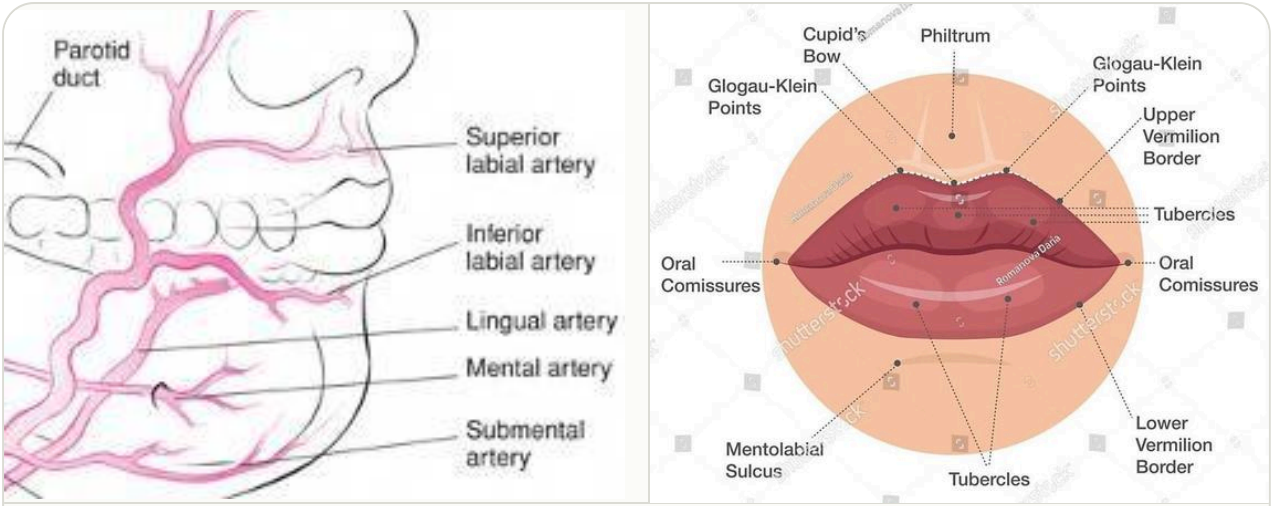


FIGURE 1 The arterial supply of the lips, and the lip landmarks.
Anatomy atlas · ENT Steps p. 168, ENT Steps p. 166

4 CHAPTER FOUR Regional flaps

Pedicled from nearby.

REGIONAL FLAPS

FLAP	PEDICLE	USES AND NOTES
Pectoralis major myocutaneous	Pectoral branch of the *thoracoacromial artery*	Workhorse salvage flap: oral, pharyngeal and neck defects, carotid cover; reaches about to the zygomatic arch; bulky, can sag
Deltopectoral (fasciocutaneous)	Perforators of the internal thoracic (mammary) artery, 2nd to 4th intercostal spaces	Neck skin, pharyngostome; often needs delay and a second stage

FLAP	PEDICLE	USES AND NOTES
Temporalis muscle	Deep temporal arteries	Orbit, maxillectomy, skull base, facial reanimation
Temporoparietal fascia	Superficial temporal artery	Ear, orbit and skull base; thin and vascular; can be free
Supraclavicular island	Supraclavicular branch of the transverse cervical artery	Thin, pliable neck and face skin; pharynx
Submental island	Submental branch of the facial artery	Oral cavity and face (beware nodes in oral cancer)
Sternocleidomastoid	Occipital, superior thyroid, transverse cervical	Small buttress; parotid contour
Trapezius (lateral, lower island)	Transverse cervical and dorsal scapular arteries	Posterior neck, scalp
Latissimus dorsi (pedicled)	Thoracodorsal artery	Large defects of neck and scalp
Nasoseptal, turbinate, pericranial	Septal artery, sphenopalatine branches, supraorbital and supratrochlear	Skull base
Palatal island	Greater palatine artery	Palate and nasopharynx

OPERATIVE GUIDES IN THEATRE 5 operations

Pectoralis major, deltopectoral, trapezius and latissimus, temporoparietal fascia and temporalis flaps.

Pectoralis Major Musculocutaneous Flaps in Head and Neck Surgery
Pectoralis major
Deltopectoral Flap
Trapezius and Latissimus Dorsi Regional Flaps
Temporoparietal Fascial Flap
Temporalis Muscle Transposition

5 CHAPTER FIVE Free flaps

Microvascular transfer.

5.1 Principles

Free flaps are joined to recipient arteries such as the facial, superior thyroid, lingual or transverse cervical, and to the internal jugular vein end-to-side or the external jugular or facial veins; a previously treated neck may need a contralateral approach or vein grafts. Two teams work together, ischaemia time is kept short (bowel is the most sensitive, then muscle, then bone), and the pedicle is laid out without kinking, twisting or tension. Success rates are about 95 to 99%.

Most failures occur within 72 hours, usually from venous thrombosis, so the flap is checked hourly at first for colour, capillary refill, temperature and pinprick bleeding, with a hand-held or implantable Doppler. The patient is kept warm, well filled and normotensive, vasoconstrictors are avoided where possible, and aspirin or heparin is given according to unit policy. **Venous congestion** makes the flap purple and swollen with brisk dark bleeding; **arterial insufficiency** makes it pale and cool with no bleeding. Either means an urgent return to theatre, because salvage is far more likely within hours.

5.2 The workhorse flaps

COMMON FREE FLAPS

FLAP	PEDICLE	TISSUE	TYPICAL USE
Radial forearm	Radial artery and venae comitantes, cephalic vein	Thin, pliable fasciocutaneous; sensate; can carry radius	Tongue, floor of mouth, palate, pharynx (tubed), lip; check the *Allen test*
Anterolateral thigh	Descending branch of the lateral circumflex femoral artery (perforators)	Skin, fat, fascia, vastus lateralis; adjustable bulk	Tongue (total glossectomy), cheek, skull base, pharynx; hidden donor site; variable perforators
Fibula	Peroneal artery	Up to about 25 cm of straight bone, segmented by osteotomy; skin paddle	*Mandible*, maxilla; implant-capable (double-barrel for height); image leg vessels first (peronea arteria magna)

FLAP	PEDICLE	TISSUE	TYPICAL USE
Scapula and parascapular (subscapular system)	Circumflex scapular, thoracodorsal (angular branch to tip)	Bone, skin, latissimus, serratus in combinations ("chimeric")	Complex 3D maxillary and mandibular defects; elderly with poor leg vessels
Iliac crest	Deep circumflex iliac artery	Thick curved bone with internal oblique	Mandible, maxilla; donor pain and hernia
Rectus abdominis	Deep inferior epigastric artery	Muscle and skin; bulky	Skull base, maxillectomy, orbit, total glossectomy
Latissimus dorsi	Thoracodorsal artery	Large thin muscle	Scalp, large surface defects
Jejunum	Mesenteric arcade	Tubed, secretory mucosa	Circumferential pharyngo-oesophageal defects; peristalsis and voice quality are drawbacks
Gracilis	Medial circumflex femoral artery; anterior branch of the obturator nerve	Functioning muscle	*Facial reanimation*

5.3 Choosing the flap

In the **mandible**, a lateral segmental defect is usually rebuilt with a fibula, though a plate with a soft-tissue flap may suit a patient with a poor prognosis. An anterior arch defect must always be rebuilt with bone, or the "*Andy Gump*" deformity follows: a collapsed chin, drooling and loss of oral competence. The **maxilla** is restored with an obturator prosthesis or a free flap (fibula, scapula or anterolateral thigh), chosen by the size of the defect and the involvement of the palate and orbital floor.

The **tongue** needs thin, mobile tissue after partial glossectomy, so a radial forearm flap is used, but bulk after total glossectomy, so an anterolateral thigh or rectus flap fills the mouth and reaches the palate. In the **pharynx**, partial defects take a patch of radial forearm or anterolateral thigh, while circumferential defects need a tubed anterolateral thigh or radial forearm flap, jejunum, or a gastric pull-up when the oesophagus is removed.

OPERATIVE GUIDES IN THEATRE 3 operations

Microvascular reconstruction, and marginal and segmental mandibulectomy.

Microvascular Reconstruction of the Head and Neck
Free flaps
Marginal Mandibulectomy
Segmental Mandibulectomy

CHAPTER SIX

6 Facial reanimation

Time decides the method.

6.1 Questions to ask

1. What caused the palsy, and is the nerve in continuity?
2. Is the proximal stump available?
3. *How long has the face been paralysed?* Facial muscles survive denervation for about 18 to 24 months, after which motor end plates are lost and muscle atrophies.
4. Which regions are affected, and what does the patient want: eye protection, symmetry at rest, a smile?

6.2 Protect the eye first

Whatever else is planned, the cornea is protected first: lubricating drops by day and ointment at night, taping and a moisture chamber. A *gold or platinum weight* in the upper lid helps it close, a lateral tarsal strip or canthoplasty tightens a lax lower lid, a brow lift corrects brow ptosis, and a tarsorrhaphy is done if the cornea is at risk. Botulinum toxin to the levator produces a temporary protective ptosis.

6.3 Restoring the nerve (within about 18 to 24 months)

A nerve cut at the time of injury is repaired at once by tension-free epineurial **neurorrhaphy**, and any transection in the temporal bone or at parotidectomy is repaired immediately. Where the gap is 1 cm or more, an **interposition graft** of great auricular or sural nerve is used; function returns at 6 to 12 months, usually with

some *synkinesis*. Branches medial to a vertical line through the lateral canthus are usually not repaired, because of cross-innervation.

When the proximal stump is unavailable but the distal nerve and muscle are still viable, a **nerve transfer** is used. The *masseteric-to-facial* (V to VII) transfer is now the commonest: fast, strong and with little morbidity, though the smile is driven by biting and is usually not spontaneous. The *hypoglossal-to-facial* (XII to VII) transfer, done as a partial split or jump-graft technique to limit tongue atrophy, gives good tone but synkinesis. *Cross-face nerve grafting* carries axons from the contralateral buccal branches through a sural graft, with a second stage once Tinel's sign shows they have crossed at about 8 to 12 months. It is the only method that gives a *spontaneous, emotional smile*, but it is weaker, so it is often combined with a masseteric transfer acting as a "*babysitter*" to keep the muscles alive.

6.4 Long-standing palsy (muscle lost)

Once the facial muscles are lost, a new muscle must be brought in. The **free functional gracilis transfer**, powered by the masseteric nerve in one stage or by a cross-face graft in two, is the gold standard for smile reanimation and is also used in *Möbius syndrome*. The **temporalis transfer**, as an orthodromic lengthening temporalis myoplasty from the coronoid or as the older turnover over the arch, is the alternative. **Static procedures** add support: fascia lata or acellular dermal slings to lift the commissure and support the nasal valve, a brow lift, and lower lid tightening.

6.5 Synkinesis and hyperkinesis

Aberrant regeneration after Bell's palsy or nerve repair produces involuntary co-contractions, such as the eye closing on smiling, and hypertonicity. These are treated with *neuromuscular retraining* by facial physiotherapy, *botulinum toxin* to the overactive muscles on the affected side and to the normal side for symmetry, and selective neurectomy or myectomy.

OPERATIVE GUIDES IN THEATRE 2 operations

Facial reanimation, and eyelid reanimation.

Facial Reanimation
Facial reanimation

Eyelid Reanimation in Patients With Facial Paralysis
Eyelid reanimation

7 CHAPTER SEVEN Other options

Prostheses and beyond.

7.1 Beyond tissue transfer

Auricular, nasal and orbital **prostheses** retained by *osseointegrated implants* or adhesives, and palatal *obturators* after maxillectomy, suit the elderly and the irradiated, follow failed reconstruction, and allow the cavity to be inspected for recurrence. **Tissue expansion** recruits skin for the scalp and neck. **Virtual surgical planning** uses CT-based 3D models, cutting guides and custom plates for mandibular and maxillary reconstruction. **Facial allotransplantation** of composite tissue has been done for devastating defects, but lifelong immunosuppression, rejection and ethical questions still limit it.

8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

8.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The reconstructive ladder;** sites that heal well by secondary intention.
- **Split- versus full-thickness grafts:** take, contraction, uses.
- **Bilobed, rhomboid and paramedian forehead flaps:** blood supply, three-layer nasal repair.
- **Lip reconstruction by size of defect:** Abbe, Estlander, Karapandzic.
- **The pectoralis major flap:** its pedicle and reach.
- **Radial forearm, anterolateral thigh and fibula flaps:** pedicles, uses, and pre-operative checks (Allen test, leg vessel imaging).

- **Free flap monitoring** and the failing flap.
- **Facial reanimation by time since injury;** masseteric versus hypoglossal versus cross-face grafting; the gracilis flap; the eye first.

8.2 Common pitfalls

COMMON PITFALLS

- Skin grafting onto irradiated tissue or bare bone.
- Leaving an anterior mandibular defect unreconstructed.
- Raising a radial forearm flap without an Allen test.
- Delaying return to theatre for a congested free flap.
- Waiting beyond two years before trying to reinnervate the face.

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CLINICAL · NECK, THYROID & SALIVARY

Head and neck cancer: biology, staging and principles

Head and neck squamous carcinoma is several diseases that share a microscope slide. One is tobacco- and alcohol-driven cancer of the mouth, larynx and hypopharynx in older smokers. Another is HPV-driven cancer of the tonsil and tongue base in younger non-smokers, which carries a far better prognosis. A third is EBV-driven nasopharyngeal carcinoma. The principles here apply across sites: biology, staging by the eighth edition of the AJCC system, the work-up, the multidisciplinary decision, and life after treatment.

AT A GLANCE

- Tobacco and alcohol act synergistically.
- HPV: E6 degrades p53, E7 inactivates Rb, p16 rises.
- AJCC 8 added depth of invasion, extranodal extension and p16 staging.
- Treat the N0 neck when occult risk exceeds about 15 to 20%.

1 CHAPTER ONE Who and why

Epidemiology and causes.

1.1 Burden

Head and neck cancer is about the seventh or eighth commonest cancer worldwide. In India it is among the leading cancers in men, mostly *oral cavity* cancer linked to smokeless tobacco and betel quid. More than 90% of tumours of the mucosa of the upper aerodigestive tract are squamous cell carcinomas, and in many low-resource settings more than 70% of patients present at stage III or IV.

1.2 Risk factors

Tobacco is the principal cause, whether smoked as cigarettes, bidis or hookah, or used in smokeless forms: chewing tobacco, gutka, *betel quid with areca nut* and snuff. *Reverse smoking*, with the lit end inside the mouth, causes cancer of the palate.

Alcohol acts *synergistically* with tobacco, so that the combined risk is multiplicative rather than additive.

HPV-16 causes oropharyngeal cancer, especially of the tonsil and tongue base; it is linked to oral sex and the number of partners, and its incidence is rising in high-income countries. **Epstein–Barr virus** causes undifferentiated nasopharyngeal carcinoma, together with salted fish and preserved foods and genetic susceptibility in southern China.

Other factors include poor oral hygiene and chronic dental trauma, submucous fibrosis from areca nut, Plummer–Vinson syndrome for postcricoid cancer, immunosuppression, radiation, occupational exposure to wood dust, leather and nickel for sinonasal adenocarcinoma and squamous carcinoma, Fanconi anaemia and dyskeratosis congenita, and diets poor in fruit and vegetables.

1.3 Field cancerisation and second primaries

Carcinogens bathe the whole mucosa, so genetically altered fields surround a cancer and give rise to new ones. Second primary tumours develop at about 3 to 5% a year, in the head and neck, lung and oesophagus, and they are a leading cause of death after successful treatment of an early cancer. This is why every patient has a panendoscopy at diagnosis, is helped to stop smoking, and is followed for life.

1.4 Premalignant lesions

Leukoplakia and erythroplakia, of which erythroplakia carries the highest risk, submucous fibrosis, erosive lichen planus and laryngeal dysplasia are the recognised premalignant states. The grade of dysplasia, from mild through moderate and severe to carcinoma in situ, predicts the risk of invasion.

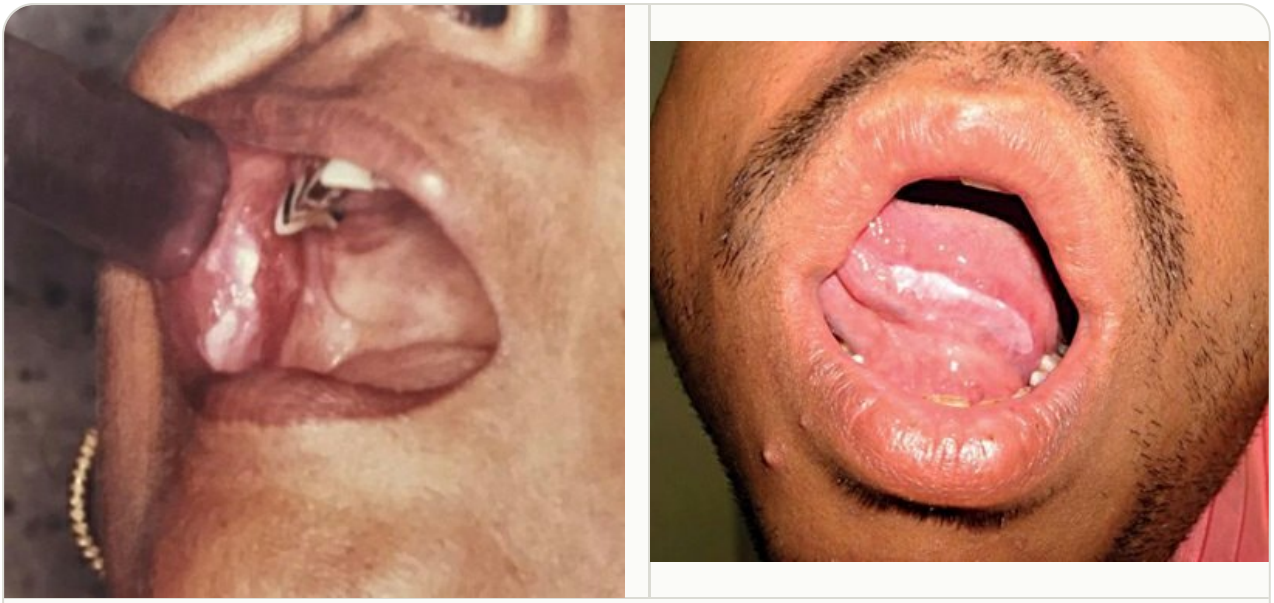


FIGURE 1 Leukoplakia of the cheek and of the lateral tongue.
Dhingra, figs 44.7 and 44.8

CHAPTER TWO

2 Tumour biology

Two diseases.

2.1 HPV-negative carcinoma

HPV-negative cancer is driven by tobacco carcinogens that cause accumulated mutations. Tumour suppressors are inactivated: *TP53* is mutated in about 80%, *CDKN2A* (p16) is lost by mutation, deletion of 9p21 or methylation, and *FAT1*, *NOTCH1* and *CASP8* are often affected. Oncogenes are activated: *cyclin D1* by 11q13 amplification, *PIK3CA* at 3q26, *EGFR*, which is overexpressed in about 90% and amplified in some, and *FGFR1*, *MYC* and *HRAS*. Califano's progression model describes the sequence: loss of 9p21 (p16), then 3p, then 17p (p53), then further changes, as normal mucosa moves through dysplasia to invasive cancer.

2.2 HPV-positive carcinoma

In HPV-positive cancer the viral oncoprotein *E6* degrades *p53* and *E7* inactivates *Rb*. With Rb gone, *p16 is strongly over-expressed*, which is why *p16 immunohistochemistry* serves as the surrogate marker for HPV in oropharyngeal cancer. p53 is usually *wild-*

type, the mutational burden is lower, *PIK3CA* activation is frequent, and changes in *TRAF3* and *CYLD* are linked to immune escape.

Clinically the patients are younger and often non-smokers, with small primaries in the tonsil or tongue base and early, often *cystic*, nodes. The tumours respond much better to radiotherapy and chemotherapy, with about 80% three-year survival against 55% for HPV-negative tumours in RTOG 0129, although smoking more than 10 pack-years worsens the outlook.

2.3 Prognostic factors

The tumour factors that matter are stage, especially nodal disease and *extranodal extension*; *depth of invasion* in oral cancer; perineural and lymphovascular invasion; the margins; and HPV status. Comorbidity, performance status and continued smoking are the patient factors, and total treatment time and gaps in radiotherapy the treatment factors.

3 CHAPTER THREE Staging

AJCC eighth edition.

3.1 Principles

The **T** stage describes the primary, mostly by size, with important exceptions: vocal fold mobility and subsite spread in the larynx, *depth of invasion* in the oral cavity, and extension in the nasopharynx and sinuses. The **N** stage describes regional nodes by size, number and side and now by *extranodal extension* (ENE), and the **M** stage distant metastases. Clinical staging (cTNM) comes from examination and imaging, pathological staging (pTNM) from the resected specimen.

T4 is subdivided into *T4a*, moderately advanced and resectable, and *T4b*, very advanced and usually unresectable, involving for example the masticator space, pterygoid plates, skull base, prevertebral fascia or mediastinum, or encasing the carotid.

T STAGE: ORAL CAVITY (INCLUDING LIP)

T	DEFINITION
T1	2 cm or less, depth of invasion (DOI) 5 mm or less
T2	2 cm or less with DOI over 5 mm, or over 2 to 4 cm with DOI 10 mm or less
T3	Over 4 cm, or DOI over 10 mm
T4a	Invades through cortical bone of mandible or maxilla, the maxillary sinus or facial skin (lip: through bone, inferior alveolar nerve, floor of mouth or skin)
T4b	Masticator space, pterygoid plates, skull base, or encases the internal carotid

T STAGE: LARYNX (GLOTTIS)

T	DEFINITION
T1a / T1b	One vocal fold / both vocal folds, normal mobility
T2	Extends to supraglottis or subglottis, or impaired mobility
T3	Fixed vocal fold, paraglottic space or inner cortex of thyroid cartilage
T4a	Through the outer cortex of the thyroid cartilage or beyond the larynx (trachea, strap muscles, thyroid, oesophagus, deep tongue muscles)
T4b	Prevertebral space, encases carotid, or mediastinum

The other sites follow the same logic. In the **supraglottis**, T1 involves one subsite, T2 more than one subsite or the glottis or adjacent mucosa, and T3 a fixed fold or the postcricoid area, pre-epiglottic or paraglottic space or minor cartilage erosion, with T4 as for the glottis. In the **hypopharynx**, T1 is 2 cm or less in one subsite, T2 over 2 to 4 cm or more than one subsite, T3 over 4 cm or with a fixed hemilarynx or oesophageal mucosal spread, and T4a invades cartilage, hyoid, thyroid, oesophageal muscle or the central compartment. In the **oropharynx**, for both HPV types, T1 is 2 cm or less, T2 is 2 to 4 cm and T3 is over 4 cm or reaches the lingual surface of the epiglottis, and HPV-positive tumours have no T4b. In the **nasopharynx**, T1 is confined to the nasopharynx or extends to the oropharynx or nasal cavity, T2 reaches the parapharyngeal space or adjacent muscles, T3 involves the bone of the skull base, cervical vertebrae, pterygoid structures or sinuses, and

T4 extends intracranially or to the cranial nerves, hypopharynx, orbit, parotid or beyond the lateral pterygoid.

3.2 Nodal staging

N STAGE: CLINICAL (ORAL CAVITY, LARYNX, HYPOPHARYNX, HPV-NEGATIVE OROPHARYNX)

N	DEFINITION
N1	Single ipsilateral node 3 cm or less, ENE negative
N2a	Single ipsilateral node over 3 to 6 cm, ENE negative
N2b	Multiple ipsilateral nodes, none over 6 cm, ENE negative
N2c	Bilateral or contralateral nodes, none over 6 cm, ENE negative
N3a	Any node over 6 cm, ENE negative
N3b	Clinically overt ENE (skin involvement, fixation, cranial nerve or muscle invasion)

Pathologically, a single ipsilateral node of 3 cm or less *with* ENE becomes pN2a, and larger or multiple nodes with ENE become pN3b. In the **HPV-positive oropharynx**, clinical N1 is ipsilateral nodes of 6 cm or less, N2 contralateral or bilateral nodes of 6 cm or less, and N3 any node over 6 cm, while pathologically *pN1 is one to four nodes and pN2 five or more*. In the **nasopharynx**, N1 is unilateral cervical, or unilateral or bilateral retropharyngeal nodes, of 6 cm or less above the cricoid; N2 is bilateral cervical nodes of 6 cm or less above the cricoid; and N3 is any node over 6 cm or below the caudal border of the cricoid.

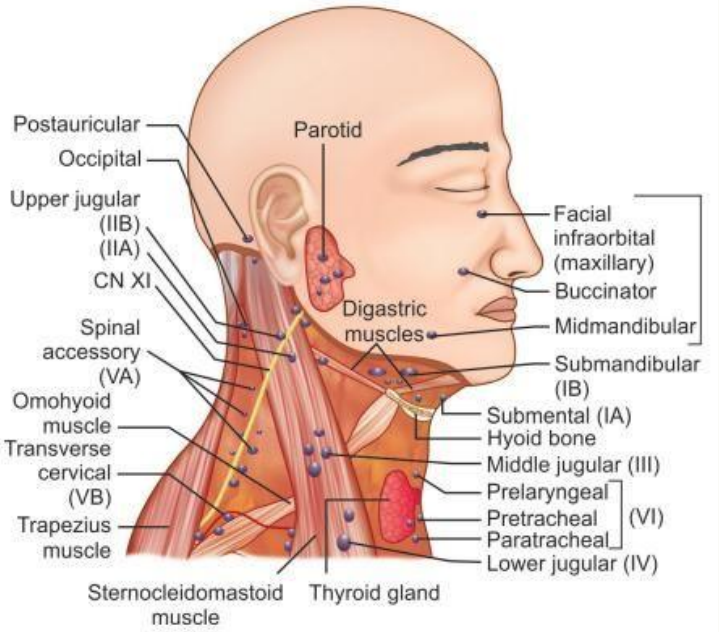


Fig. 6: Lymph nodes of head and neck. Neck nodes regions (levels) are mentioned in parenthesis

FIGURE 2 Lymph node levels of the neck.

Anatomy atlas · ENT Steps p. 285

3.3 Stage groups

STAGE GROUPS (ORAL CAVITY, LARYNX, HYPOPHARYNX, HPV-NEGATIVE OROPHARYNX, SINUSES, SALIVARY)

STAGE	TNM
I	T1 N0 M0
II	T2 N0 M0
III	T3 N0, or T1 to T3 N1, M0
IVA	T4a N0 to N1, or T1 to T4a N2, M0
IVB	Any T N3, or T4b any N, M0
IVC	Any M1

HPV-positive oropharyngeal cancer has its own clinical groups: stage I is T0 to T2 with N0 to N1, stage II is T0 to T2 with N2 or T3 with N0 to N2, stage III is T4 or N3, and stage IV is M1. Many patients are downstaged compared with the seventh edition, which reflects their better prognosis.

THE EIGHTH EDITION CHANGES

The 2017 AJCC eighth edition introduced:

- *Depth of invasion* for oral cancer.
- *Extranodal extension* in nodal staging.
- A separate staging system for *p16-positive oropharyngeal cancer*.
- Separate staging for unknown primaries by EBV and HPV status.

Older texts and exam questions based on the seventh edition will give different stages for the same tumour.

4 CHAPTER FOUR Work-up

Before the decision.

4.1 Assessment

The history covers symptoms, risk factors, comorbidity, performance status (ECOG or Karnofsky), nutrition and social support. The examination covers the whole head and neck, with palpation of the tongue base and floor of mouth, flexible nasendoscopy, the cranial nerves, and the neck by level. Tissue comes from biopsy of the primary and FNA of nodes, with p16/HPV testing for the oropharynx and unknown primaries and EBV for the nasopharynx.

4.2 Imaging

CT and MRI of the primary and neck are complementary: MRI is better for the tongue, nasopharynx, skull base and perineural spread, CT for bone and the larynx. Distant staging uses CT of the chest for advanced disease and *PET-CT* for stage III and IV disease, unknown primaries and recurrence, and the neck is assessed with ultrasound and FNA.

4.3 Panendoscopy

Examination under anaesthesia, with laryngoscopy, pharyngoscopy, oesophagoscopy and sometimes bronchoscopy, maps the tumour's extent into the tongue base, vallecula, postcricoid area and subglottis, takes biopsies, and searches for synchronous second primaries. For unknown primaries it includes tonsillectomy and tongue base mucosectomy.

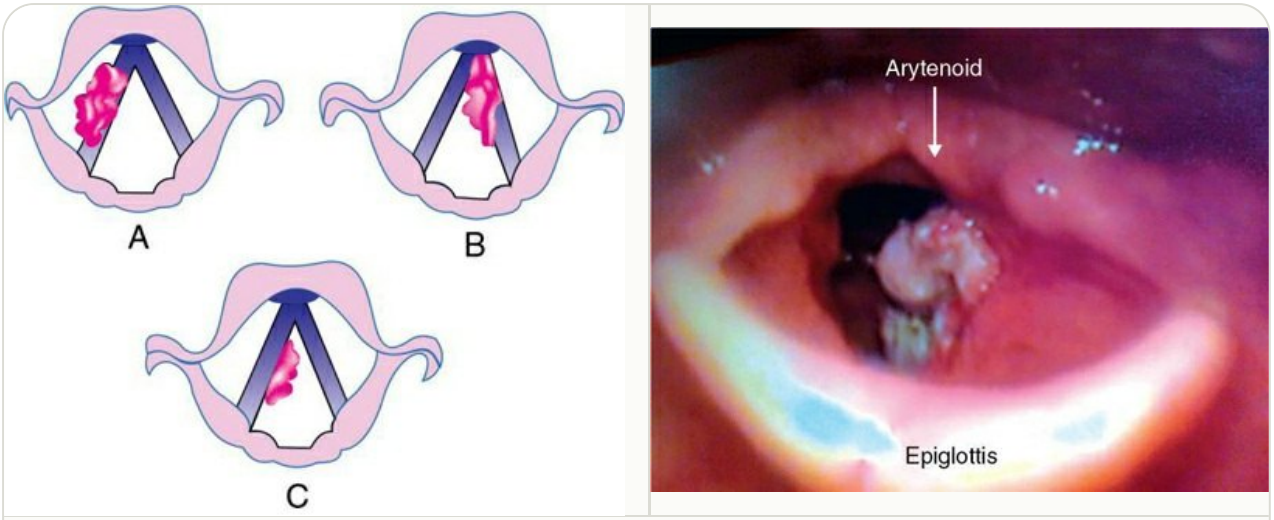


FIGURE 3 Laryngeal cancer by subsite: supraglottic, glottic and subglottic.
Dhingra, figs 62.4 and 62.5

4.4 Before treatment

Before treatment begins, teeth at risk are extracted ahead of radiotherapy, a dietitian reviews nutrition and a prophylactic gastrostomy is considered, and a speech and swallowing therapist makes a baseline assessment and starts *prehabilitation*. Smoking and alcohol cessation are supported, the anaesthetist plans the airway (some patients need a tracheostomy under local anaesthesia first), and the clinical nurse specialist begins psychosocial support.

OPERATIVE GUIDES IN THEATRE 2 operations

Management of the unknown primary, and panendoscopy.

Management of the Unknown Primary Cancer
Unknown primary

Esophagoscopy and Hypopharyngoscopy
Panendoscopy

5 CHAPTER FIVE Treatment

The multidisciplinary decision.

5.1 The team

Every patient is discussed at a *multidisciplinary team (tumour board)* of head and neck and reconstructive surgeons, clinical and medical oncologists, radiologist, pathologist, restorative dentist, speech and language therapist, dietitian, specialist nurse and palliative care. The patient's own priorities are central: cure, voice, swallowing, appearance and time.

5.2 General principles

Early disease (stages I and II) is treated with a single modality, surgery *or* radiotherapy, which give similar cure rates. The choice is driven by function, access, morbidity, the patient's wishes and the wish to keep options open for second primaries: transoral laser surgery or radiotherapy for a T1 glottic cancer, for example, or surgery for a small oral tongue cancer.

Advanced disease (stages III and IV) needs combined treatment. *Surgery followed by radiotherapy*, or chemoradiotherapy for positive margins or extranodal extension, is preferred for oral cavity cancer and for cancers destroying cartilage or bone, such as T4a larynx and mandibular invasion. *Primary concurrent chemoradiotherapy* with surgical salvage is the organ-preservation approach for the larynx, hypopharynx and oropharynx, and the standard for the nasopharynx.

Unresectable T4b disease is treated with chemoradiotherapy or palliation.
Recurrent or metastatic disease is treated by salvage surgery or re-irradiation for local and regional recurrence, systemic therapy with pembrolizumab with or without platinum and 5-FU or cetuximab-based regimens, and palliative care.

5.3 The neck

A **clinically node-negative neck** is treated electively, by selective neck dissection or elective radiotherapy, when the risk of occult nodes exceeds about 15 to 20%. In the oral cavity this means a depth of invasion over 3 to 4 mm, and the D'Cruz trial showed a survival advantage for elective neck dissection in early oral cancer; sentinel node biopsy is an alternative. A **clinically positive neck** is treated by therapeutic neck dissection, modified radical or comprehensive, or by chemoradiotherapy with a PET-CT at 12 weeks deciding whether dissection is needed. Tumours of the tongue base, soft palate, supraglottis and floor of mouth crossing the midline drain to *both* sides, and both necks are treated.

5.4 Surgery

The aim of surgery is complete resection with clear margins, generally *5 mm or more* on histology, a margin under 5 mm being close; frozen sections help. The approaches range from transoral laser microsurgery and transoral robotic surgery to the lip-split mandibulotomy, lateral pharyngotomy, total laryngectomy and pharyngolaryngectomy. Reconstruction usually takes place at the same operation, often with free flaps, and a tracheostomy protects the airway after major oral and pharyngeal resections.

6 CHAPTER SIX After treatment

Survival and survivorship.

6.1 Follow-up

Most recurrences occur within the first *two years*, so patients are seen every 1 to 3 months in the first year, every 2 to 4 months in the second, every 3 to 6 months in years three to five, and yearly thereafter. Each visit includes examination and nasendoscopy, with a PET-CT about 12 weeks after radiotherapy and further imaging as indicated. *TSH* is checked every 6 to 12 months after neck radiotherapy,

the teeth are reviewed, smoking status is recorded, and the whole upper aerodigestive tract and chest are watched for second primaries.

6.2 Rehabilitation and quality of life

Swallowing is rehabilitated with exercises, dilatation of strictures, and a gastrostomy for those who cannot eat safely. After laryngectomy, voice is restored with a *tracheo-oesophageal valve*, an electrolarynx or oesophageal speech. Shoulder dysfunction after accessory nerve injury needs physiotherapy and lymphoedema manual drainage, and trismus, xerostomia, hearing loss and osteoradionecrosis are the other late effects to manage.

Depression, fear of recurrence, alcohol relapse and suicide are all common in this group, so patients are screened and referred. Palliative care controls symptoms, manages a fungating tumour, and plans for *carotid blow-out*, airway obstruction and bleeding, with a crisis plan of dark towels and sedation.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Risk factors:** tobacco (including smokeless) and alcohol synergy; HPV; EBV.
- **Field cancerisation** and the second primary rate.
- **HPV-positive versus HPV-negative cancer:** E6 degrades p53, E7 inactivates Rb so p16 is over-expressed; prognosis; a separate AJCC staging system.
- **The eighth edition changes:** depth of invasion, ENE, p16 staging.
- **Oral cavity and glottic T staging;** the clinical N stages; the stage groups.
- **The work-up:** imaging, PET-CT, panendoscopy, the dental and nutrition review.
- **Early versus advanced disease:** single versus combined modality; when to treat the N0 neck; margins.

- **Follow-up schedule** and late effects.

7.2 Common pitfalls

COMMON PITFALLS

- Staging an oropharyngeal cancer without p16.
- Using seventh-edition definitions in an eighth-edition world.
- Forgetting the contralateral neck in midline tumours.
- Neglecting dental assessment and smoking cessation before treatment.
- Discharging patients without thyroid and second primary surveillance.

Written for this app from Oxford Textbook of Otolaryngology ch. 68, 69, Oxford Specialist Handbook ch. 4, 12, Oxford Handbook of ENT ch. 14, K. J. Lee ch. 41, 48 and Dhingra ch. 73. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

Imaging for this region

The core imaging conditions of this region, with their key facts and a representative scan. For the normal anatomy and a search pattern, read Lesson 10, The neck by spaces; Lesson 13, Lymph nodes; Lesson 14, Thyroid and salivary glands in the imaging volume.

NECK: TRIANGLES, FASCIA & NODES · INFECTION & INFLAMMATION

Abscess of the Neck

IMAGING FINDINGS

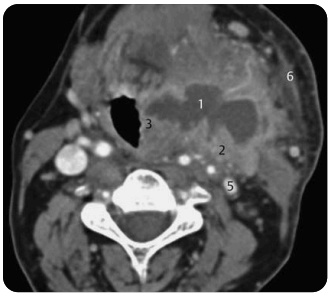
- Low-density collection with a well-formed enhancing rim
- Surrounding cellulitis and fat stranding
- The source: tooth, tonsil, node, stone or foreign body

TOP DIFFERENTIALS

- Phlegmon (no fluid centre)
- Necrotic metastatic node
- Infected congenital cyst

CHECKLIST

- Airway, mediastinum and jugular vein
- Which space: decides the drainage route



CT with intravenous contrast enhancement, axial.A large abscess (1) with a thick, contrast-enhanced capsule (2). The abscess is extendin…

NECK: TRIANGLES, FASCIA & NODES · INFECTION & INFLAMMATION

Parapharyngeal Abscess

IMAGING FINDINGS

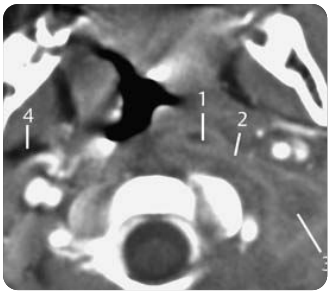
- Collection in the parapharyngeal fat, usually from the tonsil
- Medial bulge of the pharyngeal wall
- Trismus from medial pterygoid involvement

TOP DIFFERENTIALS

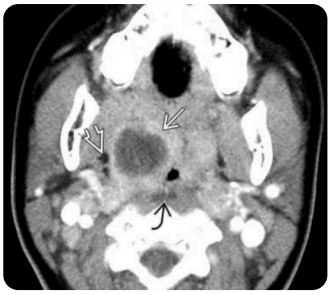
- Peritonsillar abscess
- Retropharyngeal abscess
- Deep lobe parotid infection

CHECKLIST

- Carotid space and jugular vein (Lemierre)
- Spread to the mediastinum



CT with intravenous contrast enhancement, axial. Note the bulging left pharyngeal wall, as a result of a parapharyngeal abscess (1) which is…



Axial CECT shows a large, low-density tonsillar abscess. No extension through the capsule into PPS is present. The left tonsil is prominent…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 28.

NECK: TRIANGLES, FASCIA & NODES · INFECTION & INFLAMMATION

Cervical Lymphadenitis

IMAGING FINDINGS

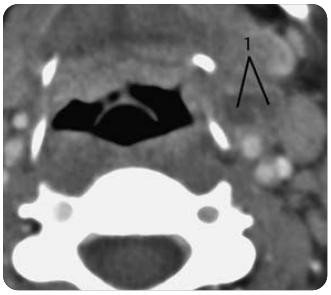
- Enlarged nodes with preserved fatty hilum (reactive)
- Suppurative node: thick rim around a fluid centre with surrounding inflammation
- Matted necrotic nodes in tuberculosis

TOP DIFFERENTIALS

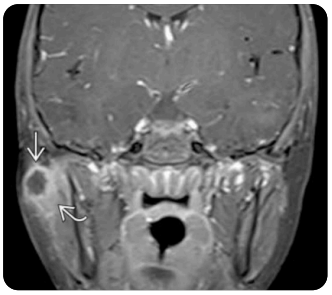
- Metastatic squamous carcinoma
- Second branchial cleft cyst
- Atypical mycobacteria

CHECKLIST

- Ultrasound first in children
- Progression to abscess



CT with intravenous contrast enhancement, axial. Various enlarged level III lymph nodes are present near the internal carotid artery and jugular…



Coronal T1 C+ FS MR in an 8-year-old boy with right cheek swelling shows a peripherally enhancing, centrally necrotic intraparotid mass with…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 314.

NECK: TRIANGLES, FASCIA & NODES · TUMOURS

Lymphoma, Non-Hodgkin

IMAGING FINDINGS

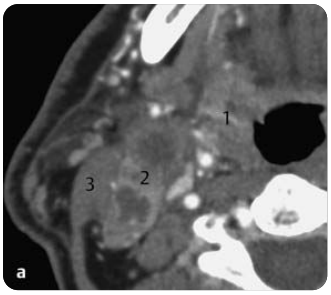
- Large, homogeneous, non-necrotic nodes in many chains
- Waldeyer ring involvement
- Strong diffusion restriction

TOP DIFFERENTIALS

- Reactive nodes
- Tuberculosis
- Metastatic carcinoma

CHECKLIST

- Extranodal sites: tonsils, sinuses, orbit, thyroid
- Core or excision biopsy, not FNA alone



CT with intravenous contrast enhancement, axial. On this slice, an area of different structure with enhancement is seen at the base of…



Axial CECT shows multiple large, round, solid lymph nodes in every chain of the suprahyoid neck. Bilateral nodes are evident in levels IIA, IIB, and…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 336.

NECK VESSELS & NERVES · TUMOURS

Carotid Body Paraganglioma

IMAGING FINDINGS

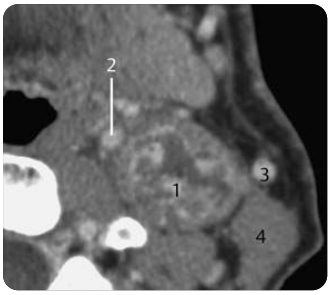
- Avidly enhancing mass in the carotid bifurcation
- Splaying of the internal and external carotid arteries
- Flow voids ("salt and pepper") on MRI

TOP DIFFERENTIALS

- Vagal paraganglioma (pushes ICA forwards)
- Carotid space schwannoma
- Jugulodigastric node; pseudoaneurysm

CHECKLIST

- Shamblin class: degree of carotid encasement
- Look for multiple paragangliomas



Patient with a slowly progressive lesion in the neck.

Dunnebier, Imaging for Otolaryngologists, Fig. 9.10 a



Axial CECT shows a classic carotid body paraganglioma with avid, fairly uniform enhancement. Notice the clear definition of the tumor sitting in the…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 140.

NECK: TRIANGLES, FASCIA & NODES · CONGENITAL

Branchial Cleft Cyst

IMAGING FINDINGS

- Cyst at the angle of the mandible in front of the sternocleidomastoid
- Lateral to the carotid sheath, behind the submandibular gland
- Thick wall when infected

TOP DIFFERENTIALS

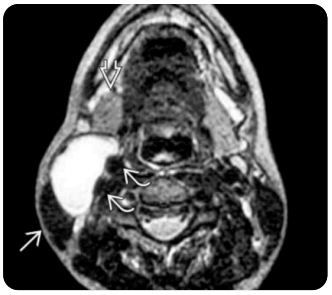
- Cystic nodal metastasis in an adult
- Lymphatic malformation
- Thymic cyst

CHECKLIST

- In adults: exclude HPV oropharyngeal cancer
- Tract towards the tonsillar fossa



CT with intravenous contrast enhancement, coronal (a) and axial (b).
Dunnebier, Imaging for…



Axial T2WI MR shows a well-defined hyperintense mass anterior to the right SCM, posterolateral to the right submandibular gland, and…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 602.

NECK: TRIANGLES, FASCIA & NODES · CONGENITAL

Lymphatic Malformation (Cystic Hygroma, Lymphangioma)

IMAGING FINDINGS

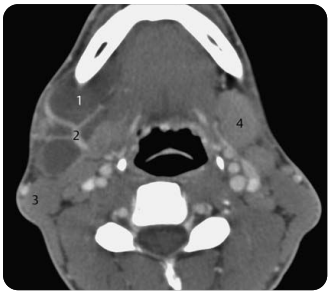
- Multicystic, septated, transspatial mass
- Fluid–fluid levels after bleeding
- Insinuates between structures without mass effect

TOP DIFFERENTIALS

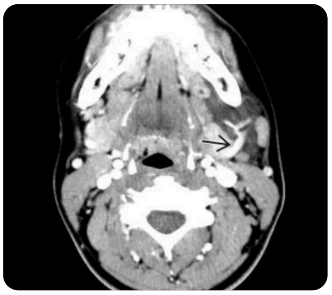
- Infantile haemangioma
- Venous malformation (phleboliths)
- Abscess

CHECKLIST

- Airway and mediastinal extent
- Macrocystic versus microcystic (sclerotherapy)



CT with intravenous contrast enhancement, axial. There is hypodense multilobular cystic structure (1) at the level of the…



Axial CECT in a 22-yearold woman with an angle of a mandible mass reveals a low-density submandibular space
Koch et al., Diagnostic Imaging: Head…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 580.

THYROID & PARATHYROID · CONGENITAL

Thyroglossal Duct Cyst

IMAGING FINDINGS

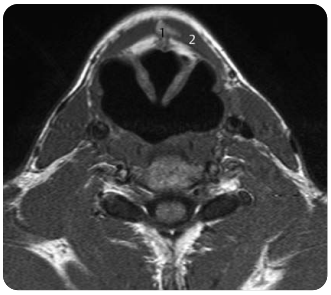
- Midline or paramedian cyst at or below the hyoid
- Infrahyoid cysts embedded in the strap muscles
- Thick wall or enhancement if infected

TOP DIFFERENTIALS

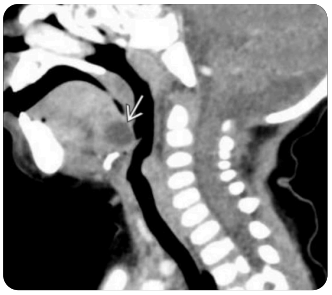
- Dermoid or epidermoid
- Lymphatic malformation
- Lingual thyroid

CHECKLIST

- Normal thyroid in the lower neck on ultrasound
- Solid nodule in the wall in an adult: carcinoma



MRI, T1-weighted, axial. Small lesion (1) with high signal intensity on T1-weighted images, probably due to presence of protein-rich fluid, situate…



Sagittal CECT shows a well-defined cysticappearing mass at the midline base of the tongue. This was incidentally found on a CT performed…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 590.

THYROID & PARATHYROID · INFECTION & INFLAMMATION

Hypertrophy of the Thyroid Gland, Thyroid Goiter

IMAGING FINDINGS

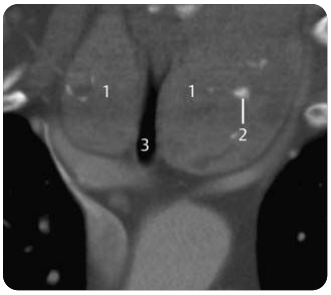
- Enlarged, heterogeneous gland with nodules, cysts and coarse calcification
- Tracheal deviation and narrowing
- Retrosternal extension

TOP DIFFERENTIALS

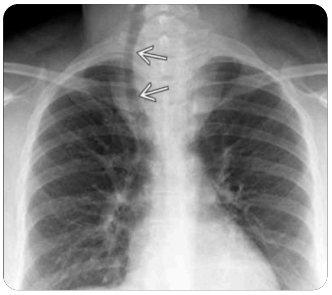
- Thyroid adenoma
- Differentiated or anaplastic carcinoma
- Lymphoma (rapid growth, Hashimoto)

CHECKLIST

- Lowest extent relative to the aortic arch and brachiocephalic vein
- Tracheal calibre for anaesthesia



CT with intravenous contrast enhancement, coronal. This slice at the level of the lower neck/upper thoracic cavity, shows enlarged thyroi…



Posteroanterior chest radiograph demonstrates marked displacement of cervical and thoracic trachea by a large, left-sided neck & superior…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 226.

SALIVARY GLANDS · INFECTION & INFLAMMATION

Sialadenitis

IMAGING FINDINGS

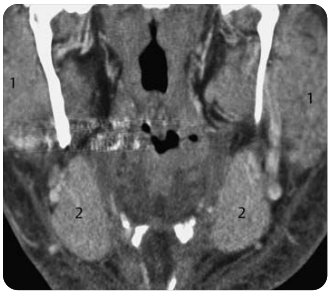
- Enlarged, enhancing gland with stranding of the surrounding fat
- Dilated duct, often with a stone
- Abscess: rim-enhancing collection

TOP DIFFERENTIALS

- Dental infection
- Nodal metastasis
- Gland tumour

CHECKLIST

- Look for a stone along the duct
- Chronic: small, fibrotic gland



Patient with chronic intermittent enlargement of the salivary glands and diffuse pain.
Dunnebier, Imaging for…



Axial CECT demonstrates an asymmetrically enhancing, enlarged left SMG compared to the right. Calculus is evident in the distal…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 406.

SALIVARY GLANDS · INFECTION & INFLAMMATION

Sialolith of the Submandibular Gland

IMAGING FINDINGS

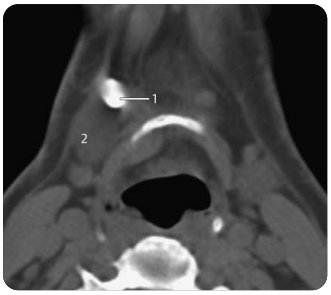
- Dense calculus in the Wharton duct or hilum on non-contrast CT
- Dilated duct proximal to the stone
- Enlarged inflamed or atrophic gland

TOP DIFFERENTIALS

- Phlebolith (venous malformation)
- Calcified node
- Dental fragment

CHECKLIST

- Position relative to the mylohyoid and hilum
- Ultrasound shows stone and duct without radiation



CT, axial bone window.A hyperdensity (1), with similar density as bone, suggestive of a sialolith (stone, calculus), probably obstructin outflo…



Axial CECT demonstrates an asymmetrically enhancing, enlarged left SMG compared to the right. Calculus is evident in the distal…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 406.

SALIVARY GLANDS · INFECTION & INFLAMMATION

Ranula of the Sublingual Gland

IMAGING FINDINGS

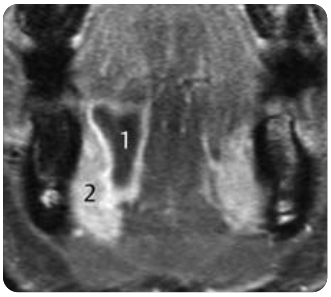
- Simple ranula: thin-walled cyst in the sublingual space
- Diving ranula: passes behind or through the mylohyoid into the submandibular space
- Tail sign pointing back to the sublingual space

TOP DIFFERENTIALS

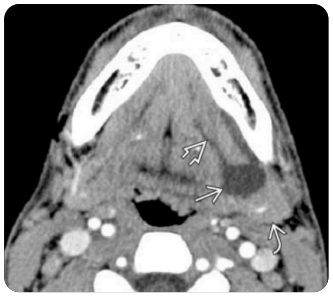
- Lymphatic malformation
- Dermoid or epidermoid
- Second branchial or thyroglossal cyst

CHECKLIST

- Relation to the mylohyoid decides the approach
- Remove the sublingual gland to prevent recurrence



MRI, T1-weighted with gadolinium enhancement and fat suppression, coronal. There is a cystic lesion suggestive of a ranula (1), originating…



Axial CECT demonstrates a rounded cystic mass in the submandibular space abutting the posterior margin of the mylohyoid muscle and displacing…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 400.

SALIVARY GLANDS · TUMOURS

Pleomorphic Adenoma of the Parotid Gland

IMAGING FINDINGS

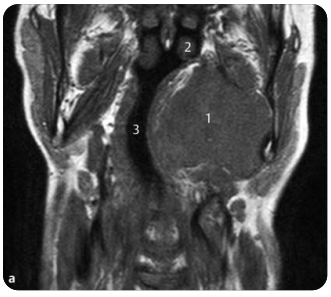
- Well-defined, lobulated mass with a capsule
- Very bright on T2
- Deep lobe: widens the stylomandibular gap, pushes parapharyngeal fat medially

TOP DIFFERENTIALS

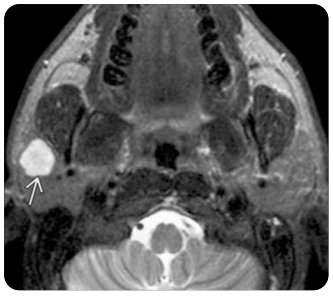
- Warthin tumour (tail, multiple, smokers)
- Low-grade mucoepidermoid carcinoma
- Intraparotid node

CHECKLIST

- Superficial or deep to the retromandibular vein
- Never enucleate: recurrence



MRI, T1-weighted, coronal. Note the extensive, isointense, smooth-bordered lesion (1) in the hypopharyngeal area: from the level o…



Axial T2 FS MR reveals a sharply circumscribed, high-signal BMT in the superficial lobe of the parotid gland. This tumor at the very least abuts, if…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 88.

SALIVARY GLANDS · TUMOURS

Malignancies of the Parotid Gland

IMAGING FINDINGS

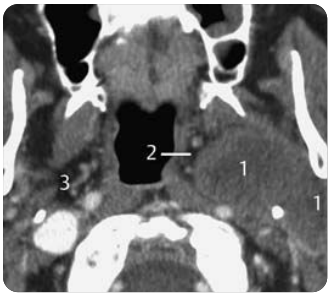
- Ill-defined or infiltrative margins
- Low or intermediate T2, low ADC
- Facial nerve: enhancement through the stylomastoid foramen

TOP DIFFERENTIALS

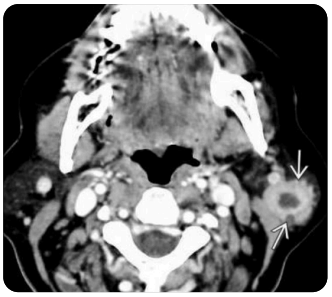
- Pleomorphic adenoma
- Warthin tumour
- Metastatic node from skin cancer; lymphoma

CHECKLIST

- Perineural spread to the mastoid segment and beyond
- Nodes in the parotid and level II



Slight swelling of the left parotid region and deep pain.
Dunnebier, Imaging for Otolaryngologists, Fig. 9.30 a



Axial CECT reveals a well-defined mass in the superficial lobe of the parotid gland. It has a thick rind of peripheral enhancement and is centrally cystic …

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 98.

THYROID & PARATHYROID · TUMOURS

Thyroid cancer

IMAGING FINDINGS

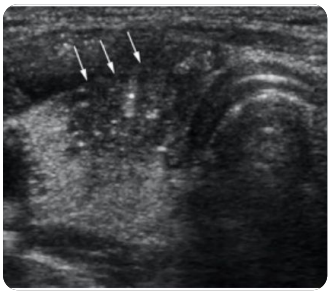
- Solid hypoechoic nodule with microcalcifications, taller than wide
- Irregular margins or extrathyroidal extension
- Cystic or calcified nodes in levels III, IV and VI

TOP DIFFERENTIALS

- Benign adenoma or colloid nodule
- Multinodular goitre
- Medullary or anaplastic carcinoma

CHECKLIST

- Trachea, oesophagus, larynx and recurrent nerve
- Iodinated contrast delays radioactive iodine



16-year-old lady presenting with hard right neck lump confirmed to be papillary ca on cytology. US shows strongly hypoechoic lesion with…



Axial NECT shows a well-defined mass arising from the left thyroid lobe with fine, speckled microcalcifications centrally. Small calcifications such as…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 238.